

# THE POPULAR SCIENCE MONTHLY

EDITED BY  
J. MCKEEN CATTELL

VOL. LVII  
MAY TO OCTOBER, 1900

NEW YORK AND LONDON  
MCCLURE, PHILLIPS AND COMPANY  
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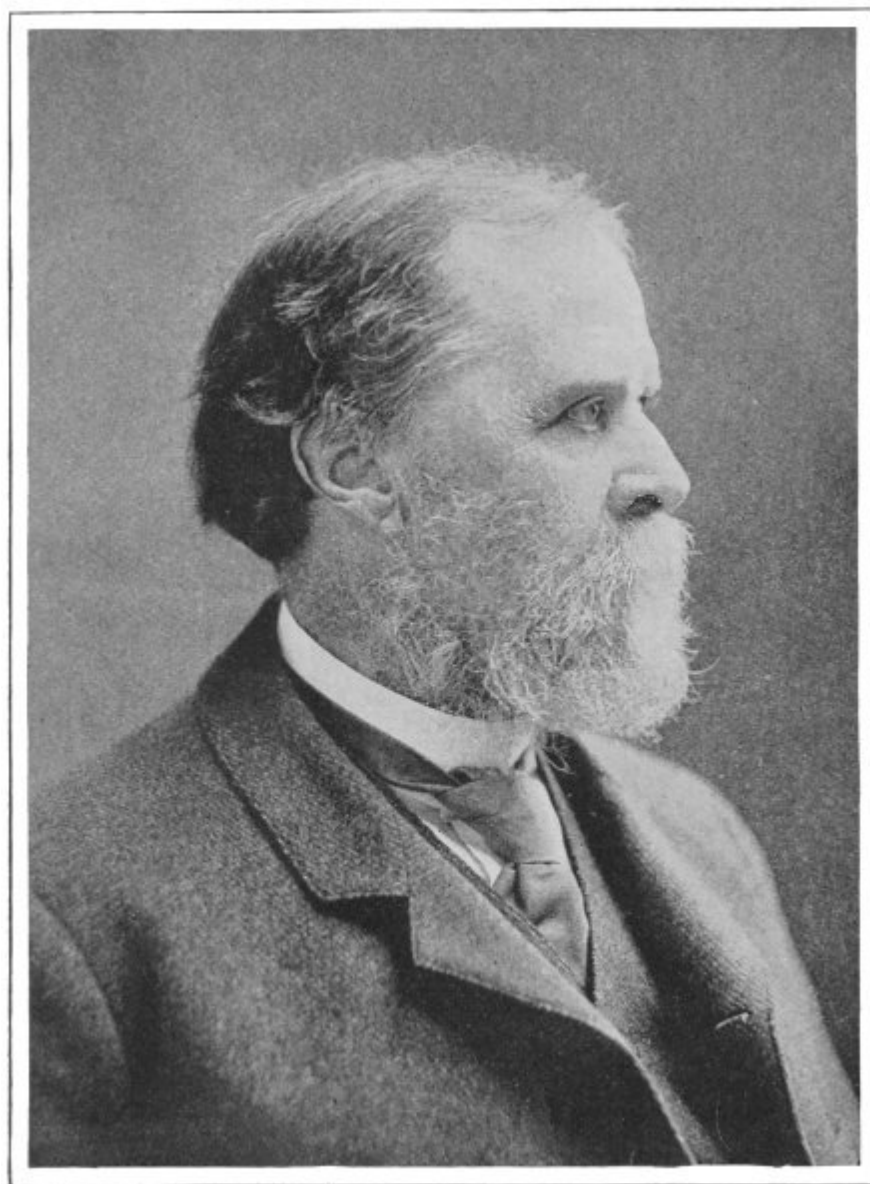
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**PROFESSOR WOLCOTT GIBBS,**  
**PRESIDENT OF THE NATIONAL ACADEMY OF SCIENCES,**  
**EMERITUS RUMFORD PROFESSOR AND LECTURER ON**  
**THE APPLICATIONS OF SCIENCE TO THE USEFUL ARTS**  
**AT HARVARD UNIVERSITY.**

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**JUNE, 1900.**

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# PREVENTIVE INOCULATION. (I.)

BY DR. W. M. HAFFKINE.

DIRECTOR-IN-CHIEF, GOVERNMENT PLAGUE RESEARCH LABORATORY, BOMBAY.

**I**T was due to certain particularly favorable circumstances that the first ideas on preventive inoculation were gathered from observations on smallpox patients. Such circumstances were presumably the following:

*a.* It is a disease which attacks epidemically, in a short time and within a small area, large numbers of people, thus permitting of easy comparisons and suggesting conclusions from the facts observed.

*b.* Its fatality is comparatively small, so that after each outbreak a large number of convalescent persons remain alive to serve as objects for future observation and comparison.

*c.* These convalescents are marked and are thus easily distinguishable from the rest of the population who have not been attacked, and even the severity of the disease they have gone through is, so to say, written down on their faces and bodies.

*d.* The disease is easily communicable, owing to the infectious matter appearing on the surface of the patient's body in the pustules.

It was easy, therefore, to notice in this case, as was indeed very early done in the East, that a person who has gone through one attack, as shown by his pitted face, very rarely suffers even during severe subsequent epidemics. Smallpox, like other epidemic diseases, breaks out in some years in very fatal, in others in milder forms. It is admissible that by a mixed process of thought and faith an impression insensibly gained ground that it was *lucky* to have been touched by the smallpox deity—of course, not in years when that deity appeared in terrifying mortality. Accordingly, in times of *mild* outbreaks people would not be very careful in avoiding infected persons, and would even seek their company so as to get infected from

them. The practice of intentionally rubbing one's skin with a pustule, or with bits of it, from an attacked person, must have been a subsequent stage.

Such or a similarly gradual development of ideas may explain why it is impossible to fix a date or place for this discovery, which indeed goes back to the darkness of antiquity. Research points to its practice among the Chinese and Hindus in very ancient times. The Chinamen induced a mild attack by inserting a crust from a smallpox pustule into the nostrils. The Hindus, on the contrary, used the fluid pus, which they inoculated under the skin of the arm. In either case, in the course of a week, the inoculated was attacked by some slight preliminary symptoms followed by an eruption, sometimes profuse, sometimes scanty, and then the disease would run its ordinary course. The only difference between an attack caused by inoculation and that caused by natural infection was, as a rule, the milder nature of the former, especially when the matter for inoculation was taken from a notoriously mild case. The result, however, was by no means certain. A mild form of an infectious disease may be due either to the virus being of a weak nature; and then such a virus would be the desired one for inoculating persons seeking artificial protection; or else the mildness of the case may be due to the patient himself being of a resistant organization, in which case, though exhibiting mild symptoms himself, he may be harboring an intense form of contagion, apt to cause a severe outbreak when transferred to other less resistant persons. Many plans were consequently adopted to secure with more certainty a mild artificial infection. Some of these were directed to the treatment of the patient preparatory to inoculation, others to the preparation of the infectious matter in order to attenuate its virulence. The Brahmans, who were the operators in India, in addition to selecting material from patients with a mild form of the disease, were accustomed not to employ the pus at once, but to keep it wrapped up in cotton wool for a period of about twelve months, and thus to weaken its power. They inoculated in the early part of the year, at the time when smallpox prevailed, and the practice they used was to moisten with water a bit of cotton wool prepared in the previous outbreak, to place it on the arm of the person to be inoculated, and to prick the arm, through the wool, over an area of about the size of a twenty-five cent piece. In a few days a vesicle would appear at the seat of the inoculation, which later on developed into a pustule and eruption. Notwithstanding these precautions, great variation in

the results was observed, and many succumbed to the operation; but those that passed through it safely were proof against further attacks.

Besides the personal risk to the inoculated, the illness produced in them was infectious to others, and unprotected persons coming in contact with the inoculated were likely to get infected from them. The latter result was largely avoided by the practice adopted by the Brahmans of inoculating all the inhabitants of a family or village at the same time. The benefits secured under the above precautions were considered far to outweigh the risks of inoculation.

With the extension of smallpox westward the system of artificial protection spread toward Europe through the intermediary of travelers and merchants. The Arabs and Turks appreciated its benefits at an early date. The slave dealers supplying the bazaars and harems of Constantinople adopted the system to protect against disfigurement their Circassian and other live stock. In the early part of the eighteenth century the method was made known to the English practitioners by Lady Mary Wortley Montagu, the wife of the English ambassador at Constantinople, who had her two children inoculated according to the Turkish system. Curiously enough, it was soon afterward discovered that a similar method was in practice among the peasants of some of the districts in Wales and the Highlands of Scotland, and had long been known there as 'buying the smallpox.' When inoculation was given a more extensive trial it was found, in England as in the East, that the effect of it was decidedly beneficial, but fraught with danger. At first one in every fifty of those operated upon succumbed to the consequences of inoculation. By improved methods the mortality was gradually reduced to one in a thousand; but the most serious danger lay in the spread of infection to healthy persons. The precaution of inoculating whole groups of inhabitants at one time, or of keeping the inoculated apart from the healthy, as had been practiced by the Brahmans ages ago, was overlooked, and the result was often disastrous to the community.

\* \* \* \* \*

It was at this time that Jenner achieved great progress and threw a vast amount of new light on the question. As is well known, he started from a belief that existed in the west of England, that cowpox was a bovine form of smallpox, and that the milkers who attended on cows suffering from that

disease and who became infected with the eruptions on the teats and udders, passed through a mild illness, which rendered them immune against smallpox. Jenner determined to put this tradition to the test, and succeeded in establishing, by a few accurate and well-planned experiments, a series of most important facts.

He showed, first, that cowpox could be artificially given to the cow by infecting it with virus from a smallpox patient, and that the disease thus produced was transferable by inoculation from cow to cow.

He showed further, that by having been bred in the tissues of the cow, the virus lost its intense infective properties for man. When the matter from an artificially infected animal was transferred by inoculation to a human being, it produced at the seat of its insertion a discrete vesicle, which was not followed by a general eruption, as would often be the case with the original smallpox virus.

Though the illness thus induced was not infectious in the sense that it would not be communicated spontaneously from person to person, it could be so transferred artificially by inoculating patients with the lymph from a ripe human vesicle.

When transferred from cow to cow or from man to man the matter preserved unchanged the same property of producing the mild inoculation vesicle, harmless to the patient and to his surroundings; and thus a matter for inoculation was obtained of *invariable strength*, what was called later on, by Pasteur, ‘virus fixe.’

The last and the most essential property which Jenner demonstrated to belong to the substance in question was the following: A man who had been inoculated with that substance could afterward be with impunity infected with a virus taken direct from a smallpox patient; the inoculation would be either abortive altogether or the effect much milder than in a man not so prepared. Jenner concluded from this most striking result that the inoculation with the matter cultivated by him in the cow would protect a man forever against contamination with smallpox, and he called that matter ‘vaccine,’ or cow lymph.

Jenner’s experiments produced an immense impression throughout the world, and inoculation according to his system, which was called

‘vaccination,’ was rapidly applied to large numbers of people. When outbreaks of smallpox occurred in the midst of vaccinated communities, observations began to come in as to the actual effectiveness of the method in protecting against the disease.

These observations proved that the system possessed an undoubted and exceedingly high beneficial effect, though the following two restrictions had to be imposed upon the originally conceived expectations:

1. The protection was not absolute. In every outbreak of smallpox a number of patients were and are met with who are attacked, generally mildly, but also in some cases fatally, though they had undergone a successful vaccination, some even at a comparatively recent date before the attack. Only the proportion of such patients to the whole of the vaccinated community is very markedly smaller than the proportion of attacks in the non-vaccinated; and also the severity of the attack, as well as the proportion of deaths to attacks, is in the vaccinated much smaller.

2. This favorable difference between the outbreaks among vaccinated and non-vaccinated is maintained not for life, but for a limited number of years, and disappears gradually, and at length altogether, unless the individuals be revaccinated. Observation has shown that the period during which the protective effect of vaccination lasts extends over from three to seven years.

Vaccination very rapidly displaced inoculation, and spread to every part of the civilized world. The results have been dwelt upon in innumerable books and pamphlets. At present great outbreaks of smallpox have become very rare, at least in the civilized part of the world, and there is a tendency to forget or ignore the devastations they used to cause.

\* \* \* \* \*

The first successful attempt in extending the system of inoculation to other diseases was made only after the discovery of the fact that ‘infection’ is caused by living animal or vegetable parasites, capable in the majority of cases of being cultivated and bred in artificial media outside the animal body. Pasteur found that he was able to effect protection against disease similar to vaccination against smallpox by the use of such artificially bred micro-organisms.

It may be interesting to relate that this important discovery was made unintentionally, and represents one of those happy 'accidents' which occur to those who diligently search. Pasteur had been working with cultures of chicken-cholera microbes, an extremely fatal form of virus when it is introduced into fowls and small birds. It so happened that one of his cultures was left forgotten in the incubator when work was stopped for the vacation. On the return of Pasteur and his assistants the experiments were continued. When the bottle was discovered, thinking that the microbes might have been exhausted or dead from long starvation, Pasteur tried to make what is called a fresh culture of them, by inseminating a sample from the old bottle into a freshly prepared nutritious broth. The microbes were not dead, and multiplied and grew luxuriantly; but when they were injected into a fowl they caused only a transient and non-fatal disease. To make a fresh start, Pasteur took some old blood, which he had drawn a long time previously from a chicken-cholera fowl and preserved in a cupboard in the laboratory in a sealed-up tube, and made a culture with the material that was in that tube. The culture thus obtained killed fresh fowls as usual, but when it was injected into the bird that had resisted the first culture it resisted this injection also. Pasteur, who excelled all men I ever knew in his ability of quickly analyzing and discerning true connections between facts, required no further hints. Others might perhaps have dwelt on the peculiarity of the fowl that happened to resist the injections, or on some other circumstances. Pasteur relinquished this and other suggestions at once, and thought of the microbe. The fact that old specimens of microbes may become impotent when injected into animals was known to him, and was readily explained by the vitality of such microbes being lowered or exhausted by starvation. But, then, such a microbe when transferred into a fresh medium, if not dead, generally regains its vigor, and after that, when inoculated into an animal, it produces its usual effect. The remarkable circumstance about the culture left in the incubator was that even when it was transferred into a fresh medium and its vitality renewed, it remained still impotent. Pasteur concluded from this that an infectious microbe possesses two distinct properties: one, which it shares with any other living being—viz., *vitality*—which may be weakened or strengthened according to the conditions of life and food; and another, which he considered as its 'virulence,' its power of causing diseases, which may be also weakened or strengthened by special means, but which is quite independent of 'vitality.'

The lucidity of thought of which Pasteur made proof on this occasion was magnificent. Later researches confirmed and explained these facts with a singular completeness, and now the idea, as is always the case, looks simple and self-evident. One must remember that at that time Pasteur had every reason to believe that disease is caused by the mere fact of a foreign micro-organism of a given species penetrating and settling down to live in the system of a man or animal. Its capability of *living* there, *i. e.*, its vital properties, seemed all that was necessary for causing disease. It was only later that it was found that pathogenic microbes cause diseases by producing so-called toxins or poisonous substances distinct from their own bodies and separable from them. The process may be illustrated by a comparison, for instance, with a cobra or any other animal producing a special venom. By starvation or some other treatment the vitality of the cobra may be temporarily weakened. When it obtains fresh food again and gets generally in good condition, it recovers, without its ability of producing venom having been in any way impaired. On the other hand, a snake may be by an operation deprived of its fangs and power of secreting poison without its health and strength being in the least affected. Pasteur at once asserted that in a similar way it was possible by starvation to weaken a breed of microbes without their virulence being diminished, and, on the other hand, to deprive them of their power of producing disease without impairing their vitality, though what the above power consisted in he did not know. He called the latter result *attenuation* of a virus. An attenuated virus in his meaning is therefore a special breed of pathogenic microbes which can be maintained, by suitable breeding, in best conditions of health, but which has lost either partially or entirely its power of producing poison and disease.

Pasteur extracted from the few experiments related above a further most-important conclusion—viz., that such an attenuation was due to and could be produced artificially by the effect of oxidation. This he deduced from the fact that the microbes in the sealed-up tube had not lost their virulence, while those forgotten in the open bottle in the incubator and exposed to the access of air had done so. Oxidation proved indeed to be one of the most general methods of artificially producing attenuated virus, to which method later on were added others—the effect of light, of chemicals, of passage through certain animals, etc.



And, of course, the last and crowning conclusion was that an ordinary, susceptible fowl that has undergone the injection with an attenuated culture becomes immune against a culture which kills other fowls; and that conclusion, in the particular circumstances under which Pasteur was working, proved to be true.

Pursuing the new line of research, Pasteur demonstrated that a protection similar to that obtained against smallpox and chicken cholera could be secured also against *anthrax*, a disease which, by the destruction it caused among sheep and cattle, was entailing heavy loss on the farmers of France. After a long series of experiments he prepared two specimens of virus, different in strength, but both weaker than the natural contagion, and worked out the proportions in which sheep, horses and cows could be safely injected first with the weakest virus and then with the virus of the somewhat greater strength, after which they became capable of withstanding the strongest anthrax infection.

In honor of Jenner, who was the first to discover the way of preparing a virus of a fixed strength safe to be used for the preventive treatment of men, Pasteur proposed that all such artificially bred, so to say, domesticated forms of microbes be called *vaccines*, while the word *virus* be reserved for a contagion growing in nature in a natural condition, or taken direct from an infected individual. The French distinguish between ‘vaccin,’ which is used as a generic term in Pasteur’s sense, and ‘vaccine,’ which name they reserve for smallpox *vaccinia lymph*. The word ‘vaccination’ has been also extended to designate inoculation with artificially vaccinized virus, while the word ‘inoculation’ is used for the injection of a natural, not vaccinated virus, taken direct from a patient. The latter distinction is, however, not yet strictly maintained in English literature, nor in the subsequent pages of this paper.

Pasteur gave a memorable demonstration of the efficiency of his method of anthrax vaccination. At Pouilly-le-Fort, in the midst of an assemblage of scientists, delegates of agricultural societies, government officials, landlords, farmers and representatives of the press, he performed the following experiment: Sixty sheep were taken; ten of these were put aside, twenty-five were vaccinated with the two attenuated anthrax vaccines at an interval of twelve days, and twenty-five were left untouched. Twelve days afterward the two groups of twenty-five sheep were inoculated with

virulent anthrax; and the result was that at the next visit the twenty-five unvaccinated and one vaccinated pregnant female were found dead, while twenty-four out of the twenty-five that had been vaccinated were perfectly well, and exhibited during the whole time they were kept under observation the same degree of health as the ten sheep that had been put aside for comparison.

\* \* \* \* \*

An impetus was given by these discoveries to researches having for their object the protection of *men* against infectious diseases. The most important of these researches was Pasteur's own into the nature of hydrophobia and rabies, and the way of inoculating against that disease. This was followed a few years later by the preparation of a prophylactic against cholera.

Inoculation against hydrophobia was rendered possible by the discovery of the fact that the rabies or hydrophobia virus is found in a pure condition, free from other microbes, in the nervous centers of animals. The material for inoculation is prepared from such nervous matter, the virulence of which is rendered *fixe*, as will be mentioned below.

The cholera microbe, which was subsequently named *comma bacillus*, was discovered by Koch in 1883, in the intestinal contents of cholera patients. Two years later cholera broke out in Spain, and Dr. James Ferran, a Spanish physician, began inoculating men with living cultures of comma bacillus taken from patients attacked with the disease. The procedure in its essential features corresponded to the pre-Jenner method of inoculation. The failure to fix the strength of the virus used for treatment rendered the method subject to the same uncertainty as that which was connected with inoculation with smallpox virus taken direct from patients. It was impossible to predict the effect of the injections. Comma bacilli taken from cholera patients may, under cultivation, show themselves extremely virulent, or, on the contrary, extremely mild. There are specimens which, when injected into a Guinea pig, even in an insignificant dose, will prove fatal to it, and there are others which will appear harmless when given in a dose seventy times greater. The immediate effect, and the protection caused by the inoculation, must, of course, vary accordingly. The attempt made by Ferran caused great interest, and a number of scientific commissions were

sent to Spain from different countries of Europe to study the results of his work. They could, however, come to no conclusion, and the treatment speedily lost its position. Only some seven years later a method was found of fixing the strength of the cholera virus. I was connected with this stage of the work, and it may perhaps present some interest to the reader to relate the way in which the problem was solved, and to show how gradual is the development of ideas by which results in laboratory investigation are arrived at.

It has been mentioned already that the virulence of microbes changes under the influence of different agents in Nature—heat, light, chemicals. When a virus is first obtained from a patient or outside a patient its preceding history, its antecedents, the conditions under which it lived before, are extremely variable. Jenner's method of cultivating the smallpox virus by transferring it from calf to calf secured for that virus uniform conditions of life, and its strength could thus be maintained unchanged for an indefinite length of time. Pasteur, in the preparation of hydrophobia vaccine, followed the same plan, and found in the successive inoculation from rabbit to rabbit a method of propagating the hydrophobia virus in a uniform condition. But attempts made to cultivate in a similar way the comma bacillus by transferring it from animal to animal failed.

The most susceptible animals for the cholera microbe are Guinea pigs. There are two principal methods of ingrafting upon them the virus: Koch's method of administering it through the mouth and leaving it to develop in the intestines of the animal, and Pfeiffer's of injecting it, not into the intestines, but into the abdominal or peritoneal cavity, where the intestines are lodged, by introducing there the virus with a hypodermic needle not allowed to penetrate into the intestines themselves. But by neither of these methods could the microbe be cultivated in an unbroken series of animals, as it became gradually weakened and soon lost its power of affecting such animals. For the purpose in question, cultivation in the peritoneal cavity had the advantage that in a healthy individual the peritoneum is free from other microbes, whereas in the intestines there are always present a large number of micro-organisms which interfere in variable ways with the growth of the particular microbes.

But when one inoculates the peritoneal cavity of a Guinea pig with a dose of cholera microbes sufficient to cause a fatal disease, it is found, when

the animal dies, that the microbes have died also. Thus, the attempt to ingraft the virus from a first animal to a subsequent one is checked at the very beginning. This initial difficulty was overcome by merely giving to the first animal a dose larger than was necessary to cause a fatal effect. The animal then succumbs more rapidly, and the microbes have no time to disappear. At the post-mortem examination there is found, in the peritoneal cavity, a small amount of exudate liquid which contains large numbers of those microbes alive. When, however, that exudate is injected into the peritoneal cavity of a second animal that animal does not succumb to the infection, or even if it succumbs one finds that the microbes have again disappeared in this second animal. By starting with, a still larger initial dose one may have three, perhaps four, successive animals affected by the virus, but it invariably ends by being weakened, and finally dies out.

In trying to obviate this result I found, perhaps contrary to expectations, that the exudation liquid should be exposed to the air for a few hours before it is injected into a subsequent animal. This result was contradictory to the effect which Pasteur had found to be exercised by atmospheric oxygen on the virulence of microbes, and it requires at least some provisional explanation. The microbes of cholera differ from a certain number of other microbes in that they stand in need of a free and abundant access of air for growing and multiplying quite satisfactorily. They are deprived of this condition in the peritoneal cavity of an animal. It is possible, therefore, that a certain opposition between the maintenance or development of virulence on the one hand, and a lowering in vitality on the other, takes place while they are cultivated there, and a respite must be given them between each successive 'passage' through the Guinea pig by leaving them for a time in the free atmosphere. Be that explanation true or not, the result is that under such conditions the successive animals inoculated with the virus do succumb, and even in a shorter and shorter time, after the inoculation, the microbe apparently undergoing under such a treatment a progressive increase in virulence. A similar development up to a certain stage was observed by Pasteur when transferring the rabies virus from rabbit to rabbit. The last difficulty that presented itself was the following: The exudation liquid which is found in the peritoneal cavity post mortem varies in quantity; sometimes it is inconveniently large and diluted; sometimes, on the contrary, so scanty that it becomes difficult to collect and transfer it to another animal. I found that this variation stands in connection with the size

of the animal, so that a diluted exudation fluid can be concentrated by injecting it into a small animal, while a too much concentrated exudate is rendered more dilute by transferring it to an animal of a larger size.

Thus, by the initial use of more than a fatal dose, by alternating cultivation in an animal with exposure to air, and by attention to the size of the animal employed, a material was obtained which, as mentioned, increased in intensity from the first and proved fatal to animals in a shorter and shorter time after inoculation. Later the virus reached a stage when it killed a Guinea pig of three hundred and fifty grammes weight in eight hours. After that, in each further inoculation the time of eight hours remained stationary, showing that the virus has reached the condition of a 'virus fixe.' These experiments were conducted by me in the Pasteur Institute, in 1889 to 1893, simultaneously on the cholera microbe and on the bacillus of typhoid. The two exhibited a number of common features in their nature, and the results as above detailed for the cholera microbe were found valid for the typhoid bacillus also.

Starting from the 'virus fixe' obtained as above, a method of double inoculation was worked out, one with an attenuated virus prepared from the 'virus fixe,' and another with the latter itself. The two 'vaccines,' when inoculated successively into Guinea pigs, protected them against all possible forms of cholera infection. The vaccines were cultivated on a solid medium, and when the crop of microbes was ready at the end of some twenty-four hours, they were washed off the surface of the medium and used as a kind of medicinal plant. It was found that the substances contained in the microbes preserved to a great extent their immunizing properties even when the microbes were killed by some delicate processes not affecting considerably their chemical constitution. The washings could, therefore, be prepared in dilute solutions of carbolic acid, and employed in the form of preserved vaccines. In 1892 and in the beginning of 1893 I made a series of experiments in Paris, in Netley, in London, in Cambridge, and in Calcutta, with these carbolized cholera vaccines, which had been preserved in sealed tubes for a period of six to seven months, and it was possible to show the protective effect of the method on animals as conclusively as Pasteur had done in the demonstration at Pouilly-le-Fort with anthrax. For the inoculation in man, however, I decided to use at first only unaltered living

vaccines, as much more promising than the dead ones, especially from the point of view of the durability of the effect.

[*To be concluded.*]

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## PROFESSOR EWART'S PENYCUIK EXPERIMENTS.<sup>A</sup>

<sup>A</sup> Abstract from an article in the Quarterly Review discussing Professor Ewart's 'Experimental Investigations on Telegony,' read before the Royal Society last year, and his book, 'The Penycuik Experiments,' published by Messrs. A. & D. Black.

THE views and works of Darwin have influenced in an unexpected way the nature of the work carried on by biological investigators during the latter end of this fast-dying century. To a great extent, while generally holding the doctrines he held, they have forsaken his methods of inquiry.

If animals and plants have arrived at their present state by descent with modification from simpler forms, it ought to be possible by careful searching to trace the line of ancestry; and it is this fascinating but frequently futile pursuit which has dominated the minds of many of our ablest zoölogists for the last thirty years. To such an extent has this pedigree hunting been carried that there is scarcely a group of invertebrates from which the vertebrates have not been theoretically derived; and to-day one of the ablest of our physiologists is using his great powers in the attempt to trace the origin of the backboneed animals from a spiderlike creature, and is exercising his ingenuity in a plausible but unconvincing effort to equate the organs of a king-crab with those of a lamprey. This appeal to comparative anatomy and the consequent neglect of living animals and their habits are no doubt partly due to the influence of Huxley, Darwin's most brilliant follower and exponent. He had the engineer's way of looking at the world, and his influence was paramount in many schools. The trend which biology has taken since Darwin's time is also partly due to a fervent belief in the

recapitulation theory, according to which an animal in developing from the egg passes through phases which resemble certain stages in the past history of the ancestors of the animal. For example, there is no doubt that both birds and mammals are descended from some fishlike animal that lived in the water and breathed by gills borne on slits in the gullet, and every bird and mammal passes through a stage in which these gill-slits are present, though their function is lost and they soon close up and disappear. In the hope, which has been but partially realized, that a knowledge of the stages through which an animal passes on its path from the ovum to the adult would throw light on the origin of the race, the attention of zoölogists has been largely concentrated on details of embryology, and a mass of facts has already been accumulated which threatens to overwhelm the worker.

The two chief factors which play a part in the origin of species are heredity and variation, and until we know more about the laws which govern these factors we can not hope to arrive at any satisfactory criteria by which we can estimate the importance of the data accumulated for us by comparative anatomists and embryologists. Signs are not wanting that this view is beginning to be appreciated. The publication of 'Materials for the Study of Variation,' by Mr. Bateson, a few years ago, shows that there exists a small but active school of workers in this field; and the recent congress on hybridization, held in London under the auspices of the Royal Horticultural Society, is evidence that in America, on the Continent and in Great Britain one of the most important sides of heredity is being minutely and extensively explored. Prof. Cossar Ewart's experiments, which we shall attempt to summarize, deal with heredity and cognate matters, and, although they are so far from complete that the results hitherto obtained can not be regarded as final, they mark an important stage in the history of the subject.

Five years ago Professor Ewart began to collect material for the study of the embryology of the horse, about which, owing to the costliness of the necessary investigations, very little is at present known. At the same time he determined to inquire into certain theories of heredity which have for centuries influenced the breeders of horses and cattle, and the belief in which has played a large part in the production of our more highly bred domestic animals. Foremost among these is the view widely held among breeders that a sire influences all the later progeny of a dam which has once produced a foal to him. This belief in the 'infection of the germ,' or



‘throwing-back’ to a previous sire, is probably an old one—possibly as old as the similar faith in maternal impressions which led Jacob to place peeled wands before the cattle and sheep of his father-in-law Laban. The phenomenon has recently been endowed with a new name—Telegony. Since the publication of Lord Morton’s letter to Dr. W. H. Wollaston, President of the Royal Society, in 1820, it has attracted the attention not only of practical breeders, but of theoretical men of science. The supporters of telegony, when pressed by opponents, having almost always fallen back on Lord Morton’s mare, it will be well to recall the chief incidents in the history of this classic animal.

It appears that early in this century Lord Morton was desirous of domesticating the quagga. He succeeded in obtaining a male, but, failing to procure a female, he put him to a young chestnut mare, of seven eighths Arab blood, which had never been bred from before. The result was the production of a female hybrid apparently intermediate in character between the sire and the dam. A short time afterward Lord Morton sold his mare to Sir Gore Ouseley, who bred from her by a fine black Arabian horse. The offspring of this union, which were examined by Lord Morton, were a two-year-old filly and a year-old colt. He describes them as having “the character of the Arabian breed as decidedly as can be expected where fifteen sixteenths of the blood are Arabian, and they are fine specimens of that breed; but both in their color and in the hair of their manes they have a striking resemblance to the quagga.” The description of the stripes visible on their coats is careful and circumstantial, but the evidence of the nature of the mane is less convincing: “Both their manes are black; that of the filly is short, stiff, and stands upright, and Sir Gore Ouseley’s stud groom alleged that it never was otherwise. That of the colt is long, but so stiff as to arch upward and to hang clear of the sides of the neck, in which circumstance it resembles that of the hybrid.”

This is the classical, we might almost say the test, case of telegony: the offspring resembled not so much the sire as an earlier mate of the dam. The facts related tended to confirm the popular view, and that view is widely spread. Arab breeders act on the belief, and it is so strongly implanted in the minds of certain English breeders that they make a point of mating their mares first with stallions having a good pedigree, so that their subsequent

progeny may benefit by its influence, even though poorly bred sires are subsequently resorted to.

The evidence of Lord Morton's mare convinced Darwin of the existence of telephony; after a careful review of the case he says "there can be no doubt that the quagga affected the character of the offspring subsequently got by the black Arabian horse." Darwin, however, latterly came to the conclusion that telephony only occurred rarely, and some years before his death expressed the opinion that it was "a very occasional phenomenon." Agassiz believed in telephony. He was strongly of the opinion "that the act of fecundation is not an act which is limited in its effects, but that it is an act which affects the whole system, the sexual system especially; and in the sexual system the ovary to be impregnated hereafter is so modified by the first act that later impregnations do not efface that first impression."

Romanes also believed that telephony occasionally occurred. He paid a good deal of attention to the matter, commenced experiments in the hope of settling the question, and corresponded at length on this subject with professional and amateur breeders and fanciers. The result of his investigations led him to the conclusion "that the phenomenon is of much less frequent occurrence than is generally supposed. Indeed, it is so rare that I doubt whether it takes place in more than one or two per cent of cases." A recent controversy in the *Contemporary Review* shows us that Mr. Herbert Spencer is a firm upholder of telephony, and that he has a theory of his own as to the mode in which it is brought about. He suggests that some 'germ-plasm' passes from the embryo into the mother and becomes a permanent part of her body, and that this is diffused throughout her whole structure until it affects, among other organs, the reproductive glands. This view, which in some respects recalls the pangenesis of Darwin, is intermediate between the saturation and the infection hypothesis. Professor Ewart refers to it as indirect infection.

Weismann, to whom we owe the term telephony, came to consider the facts for and against its existence in connection with his well-known inquiry into the inheritance of acquired characters. If telephony be true, there is no need to look further for a clear case of the inheritance of a character which has been acquired during the lifetime of the parent. The quagga-ness—if one may be permitted to use such an expression—of Lord Morton's mare

was acquired when she was put to the quagga or shortly afterward, and was transmitted to her foals. A clearer case of a character acquired during lifetime and transmitted to offspring could not be imagined. Weismann does not absolutely deny the possibility of the existence of telegony, but he would like more evidence. In the *Contemporary Review* he writes, "I must say that to this day, and in spite of the additional cases brought forward by Spencer and Romanes, I do not consider that telegony has been proved." And further: "I should accept a case like that of Lord Morton's mare as satisfactory evidence if it were quite certainly beyond doubt. But that is by no means the case, as Settegast has abundantly proved." He would, in fact, refer the case to reversion, and quotes Settegast to the effect that every horse breeder is well aware that the cases are not rare when colts are born with stripes which recall the markings of a quagga or zebra. We shall return to this point later.

A considerable number of German breeders support the contention of Weismann that telegony is as yet unproved, and it may be pointed out that in Germany, on the whole, breeders have had a more scientific education than in England, and that in that country science is regarded with less aversion or contempt than is usually the case among so-called practical men in England. We may mention one more case of an experienced breeder who was equally skeptical—the late Sir Everett Millais, who was, as is well known, an authority of great weight in the matter of dog-breeding. He writes as follows, in a lecture entitled *Two Problems of Reproduction*: "I may further adduce the fact that in a breeding experience of nearly thirty years' standing, during which I have made all sorts of experiments with pure-blood dams and wild sires, and returned them afterward to pure sires of their own breeds, I have never seen a case of telegony, nor has my breeding stock suffered. I may further adduce the fact that I have made over fifty experiments for Professor Romanes to induce a case of telegony in a variety of animals—dogs, ducks, hens, pigeons, etc.—but I have hopelessly failed, as has every single experimenter who has tried to produce the phenomenon."

It is thus evident that there is a considerable body of opinion, both practical and theoretical, for and against telegony; and that a reinvestigation of the subject is urgently needed. Such a reinvestigation has been begun by Professor Ewart at Penycuik. Since the clearest and most definite evidence

of this throwing back to a previous sire is derived from the crossing of different species of the *Equidae*, it was desirable to repeat the experiment of Lord Morton. This is now unfortunately impossible, because the quagga is extinct. The zebra is, however, still with us, and the mating of a zebra stallion with every variety of horse, pony, and ass, and subsequently putting the dam to pure-bred sires, has been the more important part of the numerous experiments carried on in the Midlothian village some ten miles southwest of Edinburgh.



**MATOPPO.**

Before considering in detail the result of the experiments it will be necessary to say a few words on the question of the various species of zebra; and since, like Weismann, Professor Ewart explains certain of the phenomena attributed to telegony by reversion, it will be as well to inquire how far reversion is known among the *Equidae*, and what evidence we have that the ancestor of the horse was striped.

Matopo, the zebra stallion from which Professor Ewart had up to last midsummer bred eleven zebra-hybrids from mares of various breeds and sizes, belongs to the widely distributed group of Burchell's zebras. Many subspecies or varieties are included in this group, which, as regards the pattern of the stripes, passes—in certain varieties found in Nyassaland—into the second species, the mountain zebra, once common in South Africa. The third species is the Grévy's zebra of Shoa and Somaliland; it is probably this species which attracted so much attention in the Roman amphitheaters during the third century of our era. A pair of Somali zebras has recently been presented to the queen by the Emperor Menelik, and is now lodged in the Zoological Gardens, Regent's Park. The species measures about fifteen hands high, is profusely striped, and stands well apart from the other two groups. It is important to note that, in Professor Ewart's opinion, it is the most primitive of all the existing striped horses.

There is no direct evidence that the ancestors of horses were striped. Certain observers think that some of the scratches on the lifelike etchings on bone, left us by our palæolithic cave-dwelling ancestors, indicate such stripes, but little reliance can be placed on this. On the other hand, there is much indirect evidence. Every one who has an eye for a horse, and who has traveled in Norway, is sure to have noticed the stripings, often quite conspicuous, on the dun-colored Norwegian ponies. Colonel Poole assured Darwin that the Kattiawar horses had frequently "stripes on the cheeks and sides of the nose." Breeders are well aware that foals are often born with stripes, usually on the shoulders or legs, less frequently on the face. Such stripes, as a rule, disappear as the colt grows up, but can often be detected in later life for a short time after the coat has been shed; they are sometimes only visible in certain lights, and then produce somewhat the same impression as a watered silk. From the facts that more or less striped horses are found all over the Old World; that in Mexico and other parts of America the descendants of horses which were introduced by the Spaniards and which afterward ran wild are frequently dun-colored and show stripes; that foals are frequently striped; and that mules not uncommonly have leg and shoulder stripes, the inference is largely justified that the ancestors of all our horses were striped.

We now pass to the experiments made at Penycuik in crossing the zebra Matopo with various mares of different breeds: 1. Matopo was first mated

with Mulatto, one of Lord Arthur Cecil's black West Highland ponies. The result was the hybrid Romulus (see p. 132), which on the whole, both in mental disposition and bodily form, takes more after his father than his mother. His striping is even more marked than that of his sire. He has a semi-erect mane which has been shed annually. The pattern of the markings, on both body and face, resembles the stripes on a Somali zebra—which, as we have seen, is regarded by Professor Ewart as the most primitive type—more than they resemble that of any of Burchell's zebras. The profuse striping is a point of difference between this hybrid and Lord Morton's. The quagga-hybrid was less striped than many dun-colored horses.

The mother Mulatto was next mated with a highly bred gray Arab horse, Benazrek. The offspring agrees in all respects with ordinary foals; it had, however, a certain number of indistinct stripes, which could only be detected in certain lights. The stripes were not nearly so clear as in a foal bred by Mr. Darwin from a cross-bred bay mare and a thoroughbred, and they disappeared entirely in about five months.

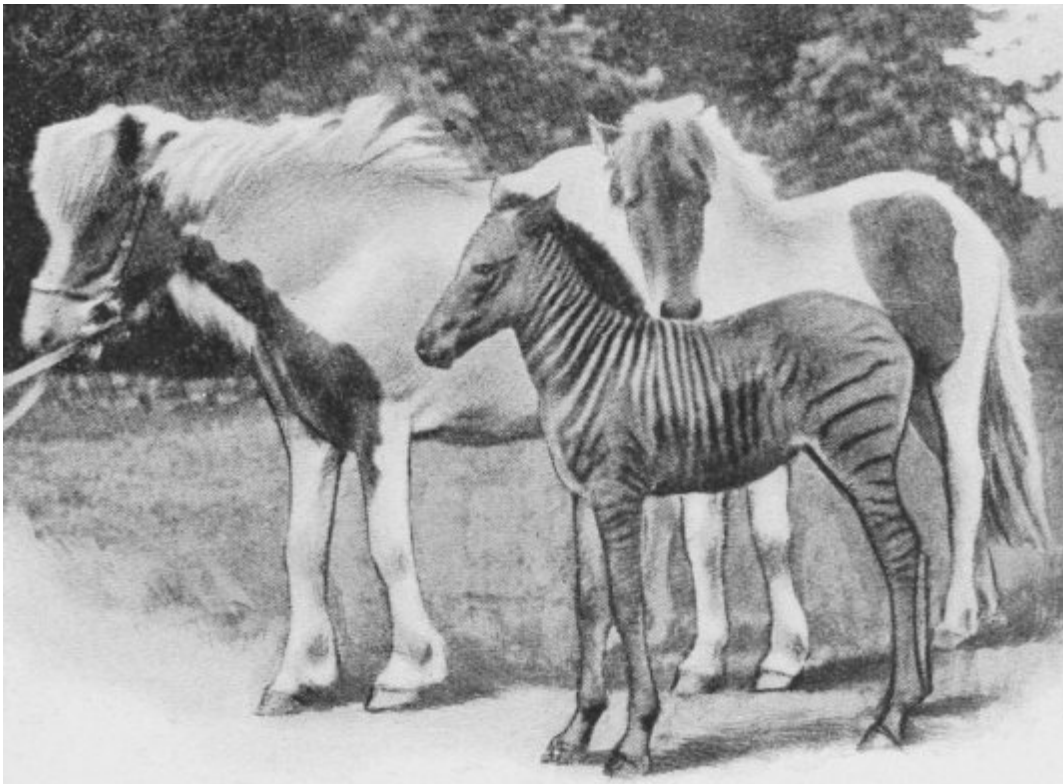


**ROMULUS, TWENTY-SEVEN DAYS OLD.**

Recently Mulatto has produced a third foal to Loch Corrie, a sire belonging to the Isle of Rum group of West Highland ponies, and closely resembling his mate. This foal was about as much striped as its immediate predecessor. In both cases the pattern of the stripe differed not only from that of Matopo, the previous sire, but from that of the hybrid Romulus. These two foals seem to lend some support to telegency; but the evidence which might be drawn from the second of them is destroyed by the fact that the sire Loch Corrie has produced foals from two West Highland mares, one brown and one black, and each of these foals has as many and as well marked stripes as the foal of Mulatto.

2. Four attempts were made to cross the zebra with Shetland ponies; only one succeeded. The hybrid was a smaller edition of Romulus. The dam Nora had been bred from before, and had produced, by a black Shetland pony, a foal of a dun color which was markedly striped. After the birth of the hybrid she was put to a bay Welsh pony; the resulting foal had only the faintest indication of stripes, which soon disappeared. It is a remarkable fact that Nora's foals were more striped before she had been mated with the zebra than afterward.

3. Five Icelandic ponies were mated with Matopo, of whom one produced, in 1897, a dark-colored hybrid. The dam, Tundra, was a yellow and white skewbald which had previously produced a light bay foal to a stallion of its own breed. Her third foal (1898) was fathered by a bay Shetland pony, and in coloration closely resembled its dam. There was no hint of infection in this case. This year Professor Ewart has bred from this mare, by Matopo, a zebra-hybrid of a creamy-fawn color, and so primitive in its markings that he believes it to stand in much the same relation to horses, zebras, and asses as the blue-rock does to the various breeds of pigeons (see illustration).



**TUNDRA (AN ICELAND PONY), HER FOAL, CIRCUS GIRL (BORN 1898),  
AND HER HYBRID FOAL, SIR JOHN (BY MATOPO), WHEN A MONTH  
OLD (BORN 1899).**

4. Two Irish mares, both bays, produced hybrids by Matopo, and subsequently bore pure-bred foals. One of the latter was by a thoroughbred horse, the other by a hackney pony. The foals were without stripes, and showed no kind of indication that their mother had ever been mated with a zebra.

5. Although Professor Ewart experimented with seven English thoroughbred mares and an Arab, he only succeeded in one case. The mare produced twin hybrids, one of which unfortunately died immediately after birth. This summer the same mare has produced a foal to a thoroughbred chestnut; “neither in make, color, nor action” does it in any way resemble a zebra or a zebra-hybrid.

6. A bay mare which had been in foal to Matopo for some months miscarried. Here—if there is anything in the direct infection theory—the



unused germ-cells of the zebra had a better chance than usual of reaching the ova from which future offspring are to arise, yet neither of the two foals which this mare subsequently produced to a thoroughbred horse “in any way suggests a zebra.”

The above is the record of the successful experiments which have been tried at Penycuik, with a view of throwing light on the existence of telegony in the *Equidae*. Experiments have also been made with other animals, such as rabbits, dogs, pigeons, fowls, and ducks. Space allows us to quote but one. Six white doe rabbits, all of which had borne pure white offspring to white bucks, were crossed with wild brown rabbits. The result was forty-two young rabbits, all of a bluish-black color, which in a very short time turned to a brown. These, at the time of writing, were about half grown, and Professor Ewart tells us that it is almost impossible to distinguish them from a full-blooded wild rabbit kept in the same inclosure. The half-breeds, however, were tamer and slightly lighter in color. The mother does next bred with white bucks again, and in every case bred true. The pure white young showed no trace of throwing back to a previous sire.

A phenomenon somewhat similar to telegony, and one which seems at present quite unexplained, is that a hen which has been crossed with a cock of another breed often lays eggs whose shell is no longer like that of its own breed, but in color, and frequently in texture, resembles that of the breed with which it has been crossed. When one calls to mind that the shell is deposited by a special shell-gland which is in no way connected with the ovary, but is a part of the quite distinct oviduct, and that the change in the color of the eggshell must be caused by some change brought about in this gland by cross-fertilization, we begin to recognize how mysterious and inexplicable are many of the problems which affect breeding.

Throughout his account of his experiments Professor Ewart is extremely cautious in claiming to prove anything, but we think he has justified his claim to have shown that telegony by no means always occurs, as many breeders believe. His experiments so far support the view of Continental mule breeders, that telegony, if it takes place, occurs very seldom. But the experiments are not complete, and it is much to be hoped that they may be continued. If it should subsequently appear that out of fifty pure-bred foals from dams which have been previously mated with the zebra no single instance of telegony be found, the doctrine may surely be

neglected by breeders; and if in the experiments which are now being carried out with various other mammals and birds telegony does not occur, the doctrine may be relegated to the 'dumping-ground' of old superstitions. The present state of the matter may be summed up in the professor's own words: "The experiments, as far as they have gone, afford no evidence in support of the telegony hypothesis." Nothing has occurred which is not explicable on the theory of reversion.

There is no factor in breeding of more importance than prepotency, and none which is more difficult to estimate. The term is necessarily a relative one, and, further, it may affect some characters and not others. Often it must go undetected, as in the case of the leader of a herd of wild cattle, who may be highly prepotent, but whose prepotency, unless he is mated with members of another herd displaying different characters, may pass unnoticed. Breeders claim to be able to produce cattle so prepotent that they will produce their like, however mated. A well-known dealer in highly-bred ponies used to boast that he had a filly so prepotent that, though she were sent to the best Clydesdale stallion in Scotland, she would throw a colt showing no cart-horse blood. Prepotency is usually obtained by inbreeding, which up to a certain point fixes the character of a race, and in all cases tends to check variation and reversion—the Jews, for instance, as a race are strongly prepotent—but there is no doubt that it may also arise as a sport, and this is probably its more usual origin in a state of Nature. Professor Ewart, however, believes that inbreeding is much commoner among wild animals than has usually been conceded, and he holds the opinion that the prepotency so induced has played a considerable part in the origin of species. This, if true, would to some extent take the place of Romanes' 'physiological selection'; for Romanes also thought that, though of great importance, variation and natural selection were insufficient to account for the origin of species without some factor which would help to mitigate the swamping effect of intercrossing—some such agency as the fences of modern farms and cattle ranches—without which the famous cattle breeds of the world would soon disappear in a general 'regression toward mediocrity.'

In inbreeding the great difficulty of the breeder is to know when to stop. Carried too far it undoubtedly leads to degeneracy. In the 'Domesticated Animals of Great Britain,' Low records the case of a gentleman who inbred

foxhounds to such an extent that “the race actually became monstrous and perished.” Hogs, if too closely inbred, grow hair instead of bristles; their legs become short and unable to support the body; and not only is their fertility diminished, but the mothers can not nourish the young.

So far as is known, no direct investigations have been made to test how far inbreeding may be carried in the *Equidae*; but, on the other hand, the breeding of race-horses may perhaps be looked upon as a gigantic experiment in this direction. Our English thoroughbreds can be traced back to a few imported sires—the Byerly Turk, imported in 1689; the Darley Arabian, in 1710; and the Godolphin Arabian, in 1730. Since then, by careful breeding and nutrition, they have increased on an average some eight or nine inches in height. There is, however, a widely spread impression that at present there is a marked deterioration in the staying power and in the general ‘fitness’ of the racer. The falling off is further shown by a fact commented on by Sir Walter Gilbey—viz., “the smallness of the percentage of even tolerably successful horses out of a prodigious number bred at an enormous outlay.” In support of this he quotes a sentence from the *Times* (December 27, 1897), referring to a sale in which thirty-two yearlings had been sold for 51,250 guineas: “These thirty-two yearlings” (said the *Times*) “are represented by two winners of five races, Florio Rubattino and La Reine, who have contributed about 2,000*l.* to the total cost; and there is not, so far as can be known, a single one of the thirty others with any prospect of making a race-horse.”

If, then, it is true that the English race-horse is on the down grade, what steps should be taken to arrest this descent? Sir Everett Millais restored a pack of basset hounds by crossing them with a bloodhound, the original forefather of bassets. The resulting pups were bassets in form, but not quite bassets in color; when, however, these cross-breeds were mated with bassets the majority of the pups turned out to be perfect bassets both in shape and coloration. This indicates that one way to rejuvenate the race-horse would be to have recourse to a new importation of the best Arab mares that the plains of Arabia can produce. Breeders hesitate to adopt this course, because their present breed is not only larger but, over very short distances, fleeter than its forefathers. The shortening of the course in recent years is probably a further sign of the degeneracy of our present racers. Were new blood introduced and more three- or four-mile races instituted, we should

doubtless soon have a return to the champion form of bygone days. Another method would be to import some of the racers of Australia or New Zealand and cross them with the home product. Different surroundings, food, etc., soon influence the constitution, and this being so, it would be advisable to select those horses of pure descent which have been longest subjected to these altered conditions. Thus the chance of reversion occurring would be increased.

It has been noticed more than once in the preceding pages that a young animal showing reversion is strong and vigorous. It is the belief of dog breeders that those members of an inbred litter which show reversion are the strongest and best. Similarly, experience shows that if an inbred sire and dam produce a dun-colored striped foal it almost always turns out well. Reversion is accompanied by a rejuvenescence; it is as though the young animal had appeared at an earlier period in the life history of the race, before the race had undergone those changes in the way of deterioration which so often accompany inbreeding.

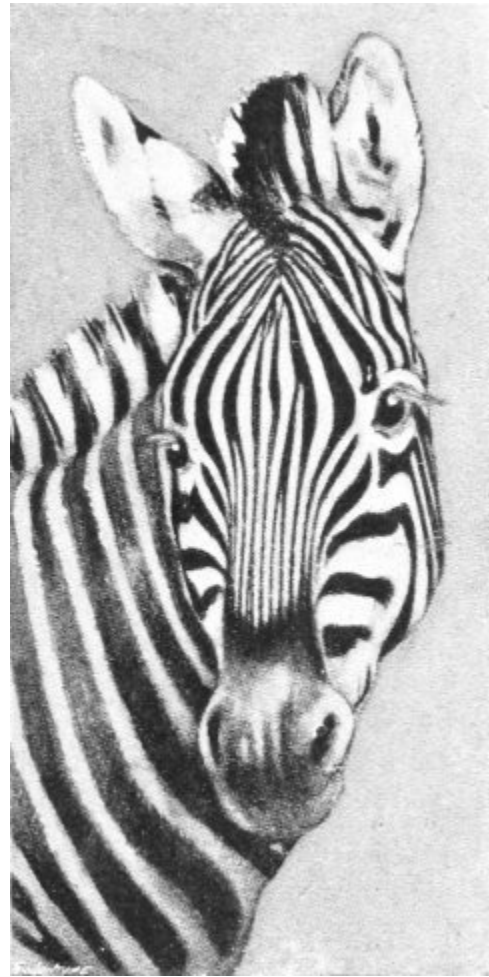
Wild animals are frequently thought to be prepotent over tame ones, but of the eleven zebra-hybrids bred at Penycuik only two took markedly after their sire, the zebra Matopo.<sup>B</sup> There are other experiments recounted which tell the other way, and at present this matter remains in a state of considerable uncertainty.

<sup>B</sup> The illustration shows the difference between the facial marks of the zebra and those of the hybrid. The latter, in this respect, bears much the same relation to the former as a blue-rock pigeon does to a fancy type.

This article must not close without a word or two more about the zebra-hybrids. It is mentioned above that only two out of the eleven which have already been born took strongly after their father.



**ROMULUS.**



**MATOPPO.**

Those who have seen the young hybrids playing about in the fields at Penycuik must agree that they are the most charming and compactly built little animals possible. Of Romulus, the eldest of the herd, Professor Ewart says: "When a few days old [he] was the most attractive little creature I have ever seen. He seemed to combine all the grace and beauty of an antelope and a well-bred Arab foal.... What has struck me from the first has been his alertness and the expedition with which he escapes from suspicious or unfamiliar objects. When quite young, if caught napping in the paddock, the facility with which he, as it were, rolled on to his feet and darted off was wonderful."

The writer can fully confirm all the praise Professor Ewart lavishes on his pets; in truth, Romulus has been well described as a "bonnie colt with

rare quality of bone ... and with the dainty step and the dignity of the zebra.” Remus, the offspring of the Irish mare, has been from the first more friendly than his half-brother; he objected less to the process of weaning, and, if he survives, promises to be the handsomest and fleetest of the existing hybrids.

On the whole, the hybrids are unusually hardy; only two have been lost—one, a twin, which died almost as soon as it was born, and another which lived some three months and then succumbed. It is only fair to say that the dam of the latter, who was only three years old when the hybrid was born, had been much weakened by attacks of the strongylus worm, and that she was the victim of close inbreeding. Both the zebras and the hybrids which have been under observation at Penycuik show a remarkable capacity for recovering from wounds. Accidental injuries heal with great rapidity. On one occasion the surviving twin was discovered with a flap of skin some five inches long hanging down over the front of the left fetlock. The skin was stitched into its place again, during which operation the little hybrid fought desperately and cried piteously; but it soon recovered, the wound healed, and now scarcely a scar remains. There was no lameness and no swelling either at the fetlock or above the knee. About a year ago four hybrid colts and three ordinary foals were attacked by that scourge of the stable, the strongylus worm. One of the latter died and another was reduced almost to a skeleton; the hybrids, though obviously affected, suffered much less than the others and soon recovered. It is further noticeable that the hybrids suffer less from colds and other slight ailments than the mares and horses among which they live.

There is no doubt that it is a comparatively easy matter to breed these hybrids, and that they are not only extremely attractive animals to the eye, but hardy and vigorous, possessed of great staying powers, and promising to be capable of severe work.

From what we have said, it is evident that the Penycuik experiments are of the highest interest, both practical and theoretical, and the public spirit and self-devotion shown by the Edinburgh professor in carrying them out can not be too widely recognized.

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# COLONIES AND THE MOTHER COUNTRY. (I.)

BY JAMES COLLIER.

WE may conceive a country, with its colonies and dependencies grouped round it at unequal distances and in different directions, as a giant organism, which has its laws of growth and development, its phases of expansion, activity and decay, like all other organisms. We see it enlarging in mass, but to the last remaining amorphous, or assimilable to no known forms. We observe the heart and brain fostering, helping and sometimes hindering, directing, controlling and guarding the evolution of the nearer or remoter portions. We perceive on scrutiny threads of relationship being woven, new nervous, muscular and circulatory systems being developed, which connect the extremities with the center and unite both into an organic whole. We are struck at times with the rupture of the mass and the permanent separation of parts of it; at times we are impressed by an unexpected augmentation in previously unknown areas, as if to repair the loss of the old. We witness the extremities reacting on the original nucleus, to some extent remodeling the heart and brain and thus creating a type of organism unprecedented in Nature. And we find that of a limited number of such colossal types, ever battling for predominance, one or another gains an ascendancy and the rest are reduced to a secondary rank, or, being lopped of their colonial extensions, cease to be world-wide organisms and shrink into the merely national organisms from which they sprang.

Snails put out their feelers as they go. The bolder insects and the more adventurous birds fly small or great distances in search of a feeding ground; some are carried out to sea, and become involuntary 'discoverers' of new lands.<sup>C</sup> The social organism puts out its feelers and extends in mass. The



community pushes out its scouts, and a portion of it, at longer or shorter intervals, follows their lead. Thus the mother country discovers many of the territories it colonizes. Cadiz was unknown to the Eastern world till a Phœnician merchant ship was blown thither. The West African coast and the mouth of the Rhone were discovered by the Carthaginians. Libya (west of Egypt) was a *terra incognita* to the Greeks till a Greek sailor who had been driven on its inhospitable coast informed the emigrant Theræans of its existence. The Portuguese discovered the Azores. The Spaniards discovered the West India Islands, Mexico, Peru and Florida. France discovered the country lying along the basin of the St. Lawrence and the valley of the Mississippi. Holland, through an English navigator, discovered the Hudson River and the future site of New York. England, through another alien, discovered the New England coast and that of Virginia; it discovered, or rediscovered, vast Australia, New Zealand, Tasmania and other South Sea islands; in quite recent years it discovered the sources of the Nile. All these countries have been or are about to be colonized by the peoples that discovered them.

C Examples of insects are given in Darwin's 'Journal,' and of birds in Wallace's 'Malay Archipelago.'

Discovery is chiefly the work of private enterprise. It was Phœnician and Greek traders who explored the northern and western coasts of Africa, of Spain, of Gaul, and of Britain. Scandinavian mariners, Norman and English fishermen, discovered North America. Spanish adventurers found the Canaries. The host of travelers have explored on their private account. Yet there are animals, like Mr. Thompson's Lobo, the wolf, and Spot, the crow, able generals and leaders of large bands, who seem to direct exploratory movements. So after a while governments lend their aid when they have ends of their own or their aid is needed. The two most memorable exploring expeditions of modern times, and the most momentous in their results, were either in great part or wholly equipped by their respective governments. Two of the vessels of Columbus were impressed ships, and the equipment proceeded from the Castilian treasury, the third being fitted out by merchant mariners of Palos. The expedition of Captain Cook, which

practically added a new continent to the globe, was altogether a state enterprise, and its celebrated commander was not, like Columbus, its designer and organizer, but only its director. The Portuguese discoveries of the Azores and the Cape were also state-aided. From this time forward Spanish and Portuguese adventurers received a royal license to discover, and the South American continent, with Mexico and Peru as its brightest jewels, was discovered by just such adventurers. Where a government refuses itself to discover, it may, like the States-General of Holland, assure to the enterprising a terminable monopoly of trade with newly discovered lands, and to this assurance the exploration of New York and its neighborhood and the discovery of Connecticut were due. Merchant companies have naturally a keen eye to the main chance, but those English and Dutch merchants can not be accused of timidity who chartered Cabot, Gilbert, Hudson and other daring mariners to seek a northwest passage to the East. Kings, in their private capacity, newspaper proprietors and rich individuals, from generous motives, sometimes equip and support explorers like Stanley and Winwood Reade.

Geographical, like scientific, discovery is often accidental. Phœnician and Greek traders, Spanish adventurers, Norman and English fishermen were blown by a succession of gales to Cadiz and Cyrene, the Canaries, Mexico and Newfoundland. Diaz was storm-carried southward to the Cape, where two shipwrecked mariners long afterward induced the Dutch to settle. Columbus, Cabot and Hudson sought a passage to India or China. The day comes, however, when chance gives way to a systematic art of discovery. The voyage of Columbus was the first where the end was deliberately aimed at and patiently worked up to. Under Ferdinand the Catholic maritime discovery was raised to an art. A board of eminent Spanish navigators, with Vespucci at its head, sat to construct charts and trace out routes for projected voyages. The primary object of Cook's first voyage was astronomical, and he was scientifically equipped for discovery on that, as of course also on the two later voyages, whose sole end was the one so gloriously gained.

Prior discovery confers an indefeasible title to occupy as against any other colonizing power. Misled by a false statement, a British man-of-war entered the Mississippi presumably to take possession of Louisiana, but turned aside on being informed of the earlier French occupation. In the thirties two naval expeditions were exploring at the same time in Spencer

Gulf, South Australia. Though the French gracefully yielded the *pas* to the prior English ship, they left a mark on a number of points that still bear French names. There seems to be now no doubt that Brazil had been discovered and rediscovered by Spanish navigators before the Portuguese *carbajal* set foot on it, but, owing to an international agreement, the discoverers ceded their claim.

Discovery does not necessarily issue in colonization. The more or less mythical discoveries of the coasts of North America and Australia in the ninth and sixteenth centuries interest the antiquarian rather than the historian. They resemble the so-called anticipations of scientific discoveries—Cesalpino's, of Harvey; Vico's, of Wolf and Niebuhr; Swedenborg's, of Kant; and a host of guessers, of Darwin. As proof alone is discovery in science, so only exploration is discovery in geography. For lack of this essential element even well-certified discoveries are apt to be fruitless. Tasman's frightened glimpse of New Zealand and his more careful coasting of Tasmania left durable marks on both countries, but only in nomenclature. They led to nothing. No Dutch settlement seems ever to have been made south of New Guinea; no northern nationality is more conspicuously absent among the colonizers of the South Seas. The earlier Portuguese discovery of the Cape of Good Hope was regarded as that of a halfway house to a more distant goal; they stopped to recruit, then hurried off to rich Cathay. The French left their names to a dozen headlands and rivers on the coast of Western Australia, but, though they often excited the suspicions of New South Wales, they made no attempt to settle.

Discovery, to assure sovereignty over the discovered country, must be followed, among animals as among men, by effective occupation. The Portuguese were roused into warlike excitement a few years ago by the advance of the Chartered Company into Mashonaland, where their settlements had long ceased to exist. Their claims to the basin of the Congo were on the same ground equally disregarded—this time by all the powers. A bit of seacoast can more easily be kept, and Delagoa Bay was assured to them by the French arbitrator. Mere occupation has at various times given a valid right to a territory. The Puritans found several islands off the New England coast to be destitute of inhabitants, and the shores so thinned of Indians by an epidemic as to be practically uninhabited. Yet they were careful to assure their title by purchase. The Manowolko Islands of the

Malay Archipelago were without indigenes when the first settlers arrived. Pitcairn and Norfolk Islands were found by the mutineers of the *Bounty* and the convicts from New South Wales to answer to Defoe's notion of a desert island. The first English settlers in Australia and the first French settlers in New Caledonia met with no resistance from the blacks at the initial stages of occupation. When the Boers trekked across the Vaal they entered on a country that had been left, through exterminating native wars, to the beasts of the field and the forest. The situation is very different when a rival civilized power lays claims to the territory. When Great Britain forcibly took possession of West Griqualand in 1871 she had to salve, without satisfying, the claims of the Orange Free State to one of the richest diamond fields in the world by a payment differently stated at £90,000 and over £100,000. Having to deal with a European power, she was constrained to submit to arbitration her pretensions to the so very useful and convenient Delagoa Bay. In attempting to extend British Guiana, so as to gain command of the Orinoco, she came into collision with the mightiest of American peoples, which now guards the interests of all the others. The United States refused to acknowledge the doctrine of 'squatter sovereignty,' and one of the preliminaries of the Venezuela arbitration was the addition to international law of the rule that a period of fifty years' uninterrupted occupancy was required to constitute valid sovereignty. England has gone through the world, like Sir Tantalus's man with his iron flail, beating down the weak and robbing the helpless. Yet few countries can show an equal record of honorable renunciations. It long refused to annex New Zealand, now one of the finest of its colonies. It long refused Fiji and Natal. It refused Samoa. It refused Bechuanaland for a time. It refused Angra Pequena. It would not listen to the discoverer who called on it to occupy equatorial Africa. It disavowed the action of Queensland in annexing New Guinea. It surrendered the Ionian Islands. Its constant injunction to its high commissioner in South Africa was not to advance the line of conquest. It surrendered, in 1854, its sovereignty over the Boers of the Orange River. That surrender was condemned by British governors, is still condemned by historians, and was disliked by the wealthier and more intelligent Boers; how wise and just it was is shown by the jealousy with which the republic has since watched over its independence. To its eternal honor—or rather to that of Gladstone—it nobly gave back the Transvaal to its stalwart farmers. France long relinquished Algeria and Madagascar, which her missions and

commercial stations in the seventeenth century gave her a prescriptive right to occupy two centuries later. It refused to support De La Tours, and abandoned Labourdonnais and Dupleix. Through mere inertia Portugal has let slip from her hands a grand inheritance. The Dutch repressed the extension of their colony at the Cape. Java flourishes, but Dutch New Guinea lies rotting.

A species, extending beyond its original habitat, has often to battle with lower species already in possession of that portion of the earth or water. So, except in rare cases, occupation means the necessity of conquest. The Puritans, as they advanced into the interior, had to fight for the possession of New England. The nomadic Australian blacks offered no resistance to the earliest settlers, but as they were driven inward they disputed, and are still fiercely disputing, every foot of territory. As the indigenes rise in the scale, have clearings and cultivate the soil, the resistance increases. No savage peoples have cost the invaders so much in disturbance, blood and treasure as the Indians, Maoris, Kaffirs and Algerian Arabs. Mashonaland was occupied by the Chartered Company without firing a shot or losing a life, but it had soon to fight for possession. The incessant turmoil, though the waves of it spread to the remote mother country, affects the settlers mainly. The blood shed is both colonial and metropolitan. The North American settlers fought their own hard battles; though British troops engaged, to their cost, with the Indians, it was against these as allies of the French; in recent years the British garrison in Canada has been employed against the half-castes. In New Zealand colonial volunteers joined with the regular troops to defeat the Maoris, and the former were sometimes found the more efficient.

The most picturesque conquests in history were effected by private enterprise. Mexico was conquered by local recruits. Pizarro was authorized to conquer Peru in the name of the Spanish crown, and, besides various other encouragements, he received a modest sum from the Spanish treasury. But it was again by local recruits, not one of them furnished by the Spanish Government, that the conquest was made and maintained. Algeria has a very different story to tell. The troops employed in effecting a difficult conquest spread over thirty years were French from first to last. In general, it may be said that where there have been regular campaigns and pitched battles the metropolitan troops bear the brunt of the fighting. Where there is a guerrilla warfare, as with the Australian blacks, it is carried on by the colonial police

or by the settlers, sometimes with the aid of the natives themselves. The Carthaginians built up their empire by native auxiliaries. The French and English conquered Canada with the Hurons and Iroquois as auxiliaries. The English mastered New Zealand with Maoris for allies, and defeated the Kaffirs with the help of the Fingoes (a related variety of the Bantu race). Rhodesia was won by a force half Kaffir. Peruvians aided Pizarro. India has been made British by armies of which four fifths were Indian. A people, like a man, contributes to its own subjugation. The expense is likewise distributed. Fifty years of intermittent war with the Kaffirs cost Great Britain twelve million pounds, and it may safely be assumed that a no smaller sum was expended in New Zealand. The colonists honorably bear their share. The premier of the latter colony told a London audience in jubilee year that it was now cheerfully paying the interest on a debt of eleven millions incurred in "holding the colony for the empire." After a Kaffir war Cape Colony was saddled with a debt of three or four millions. Other losses fall more directly on the settlers. Probably none have borne such disasters and so much suffering as the early colonists of North America. The destruction of property in a single New Zealand campaign amounted to £150,000, and the farmers on the frontiers of Cape Colony have suffered far more severely, as those on the frontier of Queensland are suffering now. If blood and money, poured forth like water, can furnish conquest with a valid title to territory, not a few British and French colonies have been justly annexed.

The expansion of an organism or a species is determined also by its struggle with other equal organisms or species which conflict with it. The hardest fight is with individuals of the same or similar species. So are rival colonizing powers usually more formidable opponents to the acquisition of a country than its indigenes. The Carthaginians were robbed of some of their colonies by the more numerous Greeks, and the Greeks of many of theirs by the all-conquering Romans. The Swedes lost a colony to the Dutch. The short but decisive struggle between the Dutch and the English was followed by the loss of the Dutch colonies in North America and the West Indies. In the eighteenth century, after every great war a group of colonies fell into the hands of the victorious power. The West India Islands and those of the Indian Ocean were for many years tossed as in a game of battledore and shuttlecock between France and England. The possession of Canada was a bone of contention between the two countries for several

decades. Seeley even maintains that the hundred years' war ending in 1815 was a long rivalry between France and England for the New World and India. If so, it was marked by striking acts of generosity. Conquests made in Canada by England, with the efficient aid of the American colonies, were more than once given back to France. When all but two of the West India colonies were surrendered in 1814 the Foreign Minister explained that it was desired to open to France the means of peaceful expansion, and it was not the interest of England to make her a military and conquering power. The rivalry did not end with the Napoleonic wars. According to one historian, Australia was saved to the English in 1788 by six days, and for long afterward there was a constant jealousy of French occupation. Ships were sent by Australian governors to take possession of Van Diemen's Land, of southern, western and northern Australia when it was believed that the French had designs on them. An English war ship, sent by the governor of the North Island of New Zealand to annex the rich and fertile South Island, anticipated by only a few hours a French ship dispatched for the same purpose. The rising of the French Canadians in 1838 has been described as "the last convulsion of despair of a sinking nationality." The English, French, and now the Germans are still rivals in present and future colonizing grounds in Africa, China, and the South Seas. But no British colonist doubts that further pacific defeats (if only by being bought out of their possessions) await the French in different quarters of the globe, for it is the colonies that press forward. The North American colonies were at all times more aggressive than the mother state, as the Australasian are now. They are unconsciously on the way to become the suns of new systems.

Conquests may be made on various pretexts. The Cape was twice seized by the English to prevent it from falling into the hands of the French, and a few years later the Dutch were constrained to cede the colony to its temporary possessors. Gambetta schemed to annex and colonize the whole North African coast from Egypt to Morocco, and thus to create a *France nouvelle* along the northern shores of the Mediterranean in place of the New France lost in Canada more than a century before, or of that still older New France on the shores of the Bosphorus. In pursuance of this policy, the powers at the Berlin Conference in 1878 permitted France to occupy (not to annex) Tunis, prohibiting her, however, from fortifying its chief port. But no one doubts that the 'regency' there, as in Madagascar, will speedily give way to undisputed sovereignty, and Bizerta is already fortified. Writers are

said to be dreamers, and Locke's constitution for Carolina, Rousseau's for Corsica, Bentham's for Russia, with many another quixotic proposal, furnish proof of their simplicity or their wrong-headedness. It is nevertheless a fact that most of the new ideas that are being carried into effect are the suggestions of publicists—journalists who stand midway between men of thought and men of action. Sometimes the former contribute immediately practicable proposals. Russia, Germany, France, and Great Britain—the four grasping powers—along with Italy and Belgium, are now, after a digestive interval of thirty years, carrying out a suggestion made by Renan in 1871. The Chinese, who threw off the yoke of the conquering Mongols, belong (it appears) by ordination of Nature to the subject races. The people that produced Confucius and Lao-tze consist of laboring men who need direction and organization. China, therefore, calls for conquest. The powers have obeyed the call, and that vast and peaceful land is now undergoing dismemberment, with a view to final wholesale partition. It is a parallel to the conquest of Peru by the Spaniards. In both countries a more perfect civilization of a lower type was or will be superseded by a less perfect civilization of a higher type. In this alone lies its assumed justification. Evidently the argument may be carried far. It would justify Russia in occupying Turkey, the United States in conquering not only Spanish colonies but Spain itself. It was the moral justification of the Gothic invasions of the early Christian centuries. It is a sentence of death or (it may be) of new life to all moribund nationalities.

Other modes of acquiring colonies are by cession and by purchase. The former is often disguised conquest, like that of the Cape to England. The latter may be so as well, like that of the African diamond fields by England. Colbert bought, for less than a million francs, certain of the West India Islands and the Antilles. The United States has bought her last colonies from dying Spain for four million pounds. At no distant time the Australian colonies will probably buy France and Germany out of the Pacific, and Holland and Germany out of New Guinea. It will be none the less a moral conquest. The right of the stronger, or the more fit to colonize, will still be, as it ever has been, the sole title to possess.

By these various means habitats have been found for future colonies, spheres for future colonial expansion.



[*To be continued.*]



# THE FUTURE OF THE NEGRO IN THE SOUTHERN STATES.

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WHATEVER danger there may be of serious conflict between the negroes and whites in the Southern States—at most but slight—is likely to arise from the fact that the old class of slaveholders, men accustomed to hold a caretaking relation to the lower race, is passing away. Already the greater number of the white people know the blacks only as they are known by the Northern folk. Race prejudice, which in the days of slavery was hardly more than formal, finding expression mainly in certain rules as to the behavior of the inferior class, is likely to increase in proportion as the two peoples become parted from one another in interests. If the present movement to disfranchise the negroes should lead to their general and permanent separation from political life, or if in elections they should again array themselves as they did immediately after the war—under the lead of white adventurers against the property interests of the commonwealth—then there may be disaster. The aim of the statesman—of every citizen in his quality of a statesman—should be to make the present political separation of the races, as far as possible, temporary. Their effort should be to develop in the blacks the qualities which may make them safe holders of the franchise, and to give that trust to all who become worthy of it. We may at once put aside all the futile expedients for other dispositions of the negroes than the simple plan of adopting them into our national life. The ancient project of returning them to Africa, the suggestions that they should be deported to some part of the American tropics, or be segregated in some one of the Southern States, are all too impracticable to deserve a moment's attention. They must be dismissed, if for no other reason, because the labor of the negroes is needed where they now dwell. Their exodus

would mean the commercial ruin of half a dozen great States. It is hardly necessary to suggest that any such action would involve a trespass upon the rights of both the whites and blacks too great to be thought of in our day.

Assuming that the only thing to do with the negroes is to shape them so that they may be fit for the place of citizens, the question is as to the steps which may be taken to attain this end. It is evident that it cannot quickly be done. Acting on the basis of our experience with immigrants from Europe, a majority of Congress concluded that all the negro needed to convert him from the slave to the truly free man was the ballot. We failed to see that between the primitive station of our race, two thousand years ago, and its present state there lay twenty centuries of toil and pain, spent in winning the state of mind of the citizen. We mocked the African with the gift of the franchise. We have now to begin where we should have begun thirty-five years ago, with measures that are proportionate to the need—with a system of education that may serve to develop the saving qualities of the race. What should this education be?

To most of us education begins with an alphabet and goes on to an indetermined limit of things that are to be had from books. The method is naturally esteemed, for we behold that the useful citizen comes forth from such teaching. Yet, logically, we might as well attribute the shape and quality of the body to the clothes it bears. The real education of our race, that which gives the most of its value to the trifle of instruction we give our children, is clearly a matter of race experience; of training in the generations of deeds since it began to pass from primitive savagery. First came the lessons in the art of continually laboring. Fortunately this lesson of labor the negro either brought with him, or learned so well in the generations of slavery that it is safely acquired. Next came the training in the occupations above the plane of simple agriculture—the industries of the forge, the loom, the ship and of military service and with it the habits of associated action. Along with these came the development of the commercial sense with the enlargements of view it gives, and from this the common sense of public affairs that makes a democracy possible. We assumed all this race training in the African when we cast him the ballot. Now that he has failed to profit by our folly, we begin to doubt whether there is, after all, the making of a citizen in him. A reasonable view of the facts leads us to conclude that he

can be made a valuable citizen, provided he has a fair share of real help in the task of becoming such.

The first need of the negro is the conviction that his salvation depends upon himself. So long as he is deluded by the hope that some great external power is to lift him to the social and economic level of the whites, there is no chance that he will come to depend on himself for advancement. From this point of view, at least, it is advantageous that the attention of this country is for the time turned away from them in a search for other, and less practicable endeavors, to lift lowly peoples to the Saxon's estate. The next is that the negroes be as rapidly as possible employed in varied craft work—work in which they may receive a larger training than the toil the fields afford. The simple yet valuable lessons of the soil-tiller they have had. For the greater number of their race, particularly those of the Guinea type, this grade of employment is as high as they may be expected to attain. Yet somewhere near one-third of the people of their color are fit for employments demanding more skill and, because of that skill, giving a better intellectual station. The mechanical employments of the day are ever gaining in their culture-giving powers. The complication of the machines which are used, and the mysterious nature of the powers which they apply, seem to make them more effective means of enlargement than the old simple tools. Those who have observed the process by which the horse-car driver of a decade ago has been converted into the motor-man of to-day have had a chance to see what the control of energies may do for them. I feel safe in saying, from the basis of personal experience with the negroes, that somewhere near one third of them are fit to be trained for mechanical employment of a fairly high grade. They will need more instruction than the average whites, but they will have a keen interest in their work, and are more likely than the whites to lead up their children in their own trades. For such employment the types which, for lack of a better name, I have termed the Zulu and the Semetic are clearly well fitted. Here and there in the South we find these people of the abler stocks already so employed.

There seems no reason to believe that there is at present enough race prejudice in the South to oppose any effective resistance to negroes entering on any such employment as that of the engineer. It is true that among the women operatives in spinning and weaving mills there has been such objection already found as to make it impossible to employ the negro and

white in the same rooms. It is, however, improbable that there would be any opposition to having the black women engaged in the industry, provided the personal association with the whites was not required. Whatever resistance it would be necessary to overcome in order to make the negro free to engineering employments would proceed from the poor white class or from Northern loom operators who brought to the South the obdurate hatred of the negro which is so strong in the regions where he is rarely seen. The old slave-holding class, and those who inherit their motives, will, I am convinced, welcome the effort to open such places to well-trained blacks. As an evidence of the state of mind of this ruling class, I may relate an experience of a year or two ago in one of the most remote corners of the extreme South:

I was lodged for some days in a small rustic inn whereto came, in the evening, a dozen men of the planter class to spin yarns, smoke and drink. They had all been Confederate soldiers—some of them were the very remnants of war. Willingly they allowed the talk to be led to the question as to the future of the black people. They showed their interest in all the forms of trade schooling that could be given them, and their contempt for the results of the literary education which they have received. Repeated reference was made to the great work that Booker Washington was doing at Tuskegee, and for it there was nothing but praise. One of the men dwelt with pleasure on the fact that in the nearest large town two negroes, trained at Tuskegee, were doing all the contract building, having ‘run out’ some cheap, ill-trained whites who had long been in the business. This talk was clearly not shaped for Northern ears, for the double reason that the Southern folk are not in the least moved to such deception, and also because I was with them as one of their own people. Very many such occasions for learning the temper of the ex-slaveholder class have convinced me that at present, and until the Southern conditions are assimilated to those of the North, there will be no difficulty in developing the technical skill of the blacks arising from the disinclination of the people when they are thus employed. It is true that the old slaveholder, with his care-taking humor towards the blacks, is passing away; but his motives are likely to be continued in his descendants at least for some generations.

There are at present in the South many thousand places for which it would be easy to train negroes—places which would give them a liberal

education of the kind most needed by their race. It is not too much to reckon that each year, in the development of the industries of that region, adds some thousand chances which can not well be filled from the native white people, but are likely to go to men brought from elsewhere. Every opportunity to establish a family supported by a skilled mechanic is of value. With even five per cent of the male negroes thus employed, the prospects of their future would be greatly benefited. The means for attaining this end are not difficult to find. What is needed is an extension of the system followed at Tuskegee, where youths are trained with the intent that they shall be made ready for high-grade manual labor, the general schooling being limited to what is necessary to ensure success in such practical work. A system of trade schools for negroes, sufficient to supply the present demand for skilled mechanics, is now the gravest need of the South.

It has been suggested that the troops which are required for the Federal service in tropical lands might well be recruited from the negroes. It has indeed been proposed that these soldiers should be permitted to take their families with them so that they might become permanently and contentedly established in Luzon and elsewhere in the colonies. There is no doubt but that the abler negroes, when properly officered, make excellent soldiers—at least as infantry men. The experience had with them during the Spanish War makes this point perfectly clear. It may also be reckoned that they would endure tropical climates better than the whites. It may further be said that the existence of a large and well respected force of blacks in the Federal army would unquestionably add to the social position of the negroes in the estimation of both races. Again, the return of these men to their homes, after their period of service, would be advantageous. Their training and experience would make them of much value to their people.

There are, however, certain signal disadvantages which would arise from the employment of negroes as soldiers. In the first place, it would tend to remove from the body of the folk the abler men—those to whom we should mainly look for the uplifting of their race. This evil, great in the case of all levies, would be most serious in this case; for the reason that, while with white troops the rank and file are not commonly by nature leaders of their society, they would be so with black recruits. If the choice could be made of the Guinea type, this loss would not be serious; but it certainly would fall to the more militant stocks—those to which we have to look for

advancement. In the next place, we must see that the negro does not need the training in passive obedience and mere order of life that the common soldier receives. He has had that already in quite sufficient measure. He now should have the lessons of individual responsibility—of control of his life from within—lessons that civil life alone can give. Therefore, the well-wisher of the race will be inclined to oppose this project of recruiting our armies from the negroes of the Southern States. If it is determined to enlist them it would be best to limit the age of the recruits to about twenty years, and the period of active service to five years, so that the men may be returned to civil life young enough to enter on ordinary employments.

At present it is most desirable that the negroes of the South should be induced to save money, for until that habit is formed, there is little chance of lifting them in the economic scale or of developing in them the business sense, which is one of the corner-stones of civilization. It is probable that more could be done in the way of correcting the faults and stimulating the latent capacities of the race by developing this motive than by any other means. It is difficult to suggest any effective system by which this end can be attained. The general conditions of the South make rural savings-banks impossible. The receipts, at least for many years, would be too small to render the business remunerative. The only practicable method appears to be that of a Federal system operated through the post-offices. The institution of such a system appears to be justified by the two conditions: the exceeding need of such a provision and the impossibility of doing the work except through the postal machinery of which the Federal Government holds a monopoly. It may be said that this method has proved successful under other governments, and that it has been for some time established in Canada. In our own country it is clearly demanded, in all rural communities, though nowhere else so gravely as in the Southern States.

In looking over the latent possibilities of the negro people, the observer can not fail to remark their keen delight in music. Statistics on this, as on other facts, are lacking; but from what I have been able to learn, it appears probable that a far greater proportion of the blacks are sensitive to musical effects than is the case with the white people. I have indeed never been able to find a black man who was so far lacking in this sensibility that he did not enjoy the songs of his people. It is not unlikely that close inquiry would show this to be a remarkable feature in this unexplored race. As yet little

effort has been made to determine the true measure of this capacity of the negro for music. It may be that they can not attain to the higher levels of the art; yet it is perfectly evident that their voices are exceptionally good, and that they have a keen native sense of time and tune. The most effective dance music I have ever heard has been made by negroes who could not read a note. When we consider how large a place music has in our life, it is a fair suggestion that this quality of the black nature might well be made the subject of experiment.

Those who look closely at the conditions of the negroes of the South are led to the belief that the existing separation in sympathy of the races is not likely long to continue. The greater number of the negroes instinctively crave a protective relation with the whites. It is the ancient disposition of the weak man to lean upon the strong which has in all ages and lands determined the relations of folk. At present the two peoples are held apart by the memories of slavery, rather than by any real personal dislike—the race prejudice which so commonly separates the Northern white from the negro. As this temporary barrier wears down, we may hope to find a new form of association arising—one in which the negroes will seek and find their friends among the trusted men of the superior race. I have seen marks of this new relation here and there, not many nor very clear, but fairly indicative of what may come about, provided the political excitement is allowed to subside and the people of the South, black and white, make their adjustments according to their motives and capacities, with no reference to the Federal power.

At first sight it will appear to most of the Northern people over-much to ask that the powers at Washington give up all efforts to deal with the needs of the negro folk—the so-called wards of the nation. Yet experience has shown the impracticability of the project of helping these negroes with the long arm of the Federal law. All that has been undertaken in this way has been fruitless or worse. The only chance for lifting the black man to the full status of the citizen is by leaving his future essentially in the hands of the masterful folk who alone can help him. We see that the ruling class in the South have a measure of interest in the status of the negro and an opportunity to benefit his state that can never belong to the people of the North. Although the country, as a whole, will, of course, suffer from the



failure to elevate the blacks, the burden will lie most heavily on those with whom they dwell.

The Southern whites have given evidence of political capacity of a high order. Even their blunder in the rebellion is in good part compensated for by the sagacity with which they accepted the results of the war and turned them to the best account they could. They are not likely to cower before the vast undertakings which the uplifting of the blacks will entail; as yet, they have not accepted the task as their own. They have indeed been brought to believe that their business was to defend their own class interests, as well as they might be able to, against the attacks of the negroes, aided by the Federal power. If they are forced to see that within the limits the Federal Constitution sets to action, the responsibility for the future of their several States is in the hands of those who control their politics, we may hope to find the political and economic skill which went to the development of the system of slavery given to the advancement of the Africans. While the work must needs be done by the men who are near to it, it should receive every possible aid and sympathy from those who, because they are far away, can not effectively control the matter. The cause is so large that it needs the help of all who wish it well.

It appears to me that the time has come for an effective union of endeavor on the part of those of North and South, ex-slaveholder and ex-abolitionist alike, who wish to see the negro have, not his rights in the common sense of the word (for mere rights are a pitiful share for a man), but rather a good human chance to climb the ladder of civilization, upon which our ancestors set him. The aims of these two ancient parties surely have for a common end the best that can be done for the negro people. It is just as much a mistake to suppose that the majority of the slaveholders in a malign spirit sought to oppress and torture the blacks, as to fancy that the abolitionists desired to set the negroes over their sometime masters; for history will probably write it down that the better men of these two parties were both dealing with the same very difficult problem: that their contentions grew from a failure on both sides to see the whole of the matter.

It is possible that something might be done to help towards effective work, looking to the end we have in view, through a society for the study of the African problem. Such an association, provided it included men who were guided by a true spirit of inquiry and had no political ends to win,

especially if it was in part made up of Southerners who had a large-minded view of the matter, could do much to guide action in profitable ways. In general, I am opposed to the increase in the number of societies; so that, if there be any in existence that could fairly undertake this task, I should prefer to see it set about the work. I am not aware, however, that there is any existing association which includes such questions in its field of inquiry.

It will be observed that the suggestions I have made concerning the immediate needs of the negro do not include any mention of the higher scholastic education. This is not because I disbelieve in such training for those blacks who, by their evident capacity, show that it fits them; but because it seems futile at the present time to waste efforts in giving these people an education for which they are in general by no means ready—which, if attained, does not afford them a way to a suitable station. The few youths of the race who really desire what is commonly called a college education, are reasonably certain to receive it in some one of the many schools where they are sure of a welcome and of all due help. Even in the case of those blacks who, by some rare chance, have inherited the proper foundations of the higher mental training, and are made ready for the so-called professions, I see but a very poor chance of advancement to any fit positions in this country. Even in the part of the North where one would expect these well-trained negroes would have a fair chance in life, it does not avail them. As physicians, lawyers, clergymen or engineers they can look forward to no future having a definite relation to their capacities. They can not expect to have any range of social opportunities, and their employment will have to be essentially with their own people.

The youth of negro blood might naturally expect to find in a community devoted to the maintenance of his rights at least a welcome to the external business society. He will, however, find that the people who would willingly sacrifice much to ensure him an equal place in matters political, allow their race prejudices or those of their associates to deny him fair play. It is a lamentable fact that this dislike to these men of the other aspect is far stronger in the North than in the South. In the parts of the North where negroes are rare, there is, it is true, a sense of duty by them that ensures their place before the law; but not enough personal contact with them to wear away the first offence of their diverse aspect. In most parts of the Southern States the black man is so constantly in view that the instinctive prejudice is

worn away—he is perhaps, in a somewhat contemptuous way, personally liked. The race prejudice takes the form of certain rules of intercourse, expressing about the feeling that separates the commissioned officers and the enlisted men of an army. There is an element of truth in the statement, attributed to Thomas Carlyle, that the Northern man said, “God d—d you, Sambo, be free;” and the Southerner, “God bless you, Sambo, be slave.” The result to Sambo is the same—a deprivation of opportunities in all the higher walks of life.

The only safe way up for the negro appears to lie in the industrial field, in mechanical employments, where his race may not weigh against him, and where head and hands may help one another to profit of mind and pocket—in business of varied kinds where he may get money, and with it the station that, in the common view, nothing else will afford him; in good work done for his race such as will give him the dignity in the eyes of all men that the master of Tuskegee has won. It is very much better for a negro youth, and for his race, that he should be a successful blacksmith, farmer or engineer, than a lawyer or physician, hindered and shunned, sorely burthened as he is sure to be by the cross that his fellowmen force him to bear. Therefore, unless they are willing to betake themselves to countries where the government is in the control of mixed peoples, thereby escaping the worst evils of race prejudice, it seems best for negroes not to seek the so-called learned professions, but to win their way on the lines where they will find less resistance—on ways quite fit for a man, even if not the highest.

It has been suggested that our colonies may afford a field for professionally educated negroes; but there, if they are to be ruled by the home government, it is likely that they will find a white caste in control. We may thus expect that the same essential disfranchisement will be found there as at home. Moreover, as before remarked, this project of sending to far lands the individual of ability who is needed at home, can not commend itself to those who feel the need which is with us, a need that calls for all the capacity we can hope to develop among the black people. It is clearly not a time to consider a proposition to export these abler youths of the black population.

Back of all our projects to bring the negroes of the South to the full station of citizens, to get rid of the contempt and the consequences of the contempt in which they are, as a race, so generally held, is the grave

question as to the practicability of framing a social and political system in which men of such diverse origin may have a substantially equal chance. It must be granted that in no modern state of high grade has this problem been fairly solved. The instances from the tropical colonies of Great Britain are not really apposite; but there seems no fundamental difficulty to contend with in order to attain this end. With cultivated people of their own race about them the better negro youth would not be deprived of that element of education. We have taken into our political family races scarcely less different in motives from our own than are the negroes, making no kind of objection to their sharing the commonwealth with us. In certain ways it is true that a nation loses strength where it fails to have its elements closely knit together. But it may be doubted whether these losses are not more than compensated for by the gains that arise from diversities such as would come from the introduction into our system of a body of folk with the capacities which our Africans are likely with thorough training to develop.

In this matter there are but two courses open to us—one of folly, the other of wisdom. We may leave the black people to work out their own salvation as best they may, to lie as a mass at the bottom of our society, except so far as the abler men who may arise among them help their struggling fellows. The result of this will be the perpetuation of all the existing evils. Or we may set to work, after the true manner of our folk, with the full knowledge that the task is very great, but that we have the strength to see it done. With this spirit we may accomplish the noblest work that men have ever undertaken in any nation.

To the people of the South we may fairly say: “These negroes were brought here by your forefathers, and thus tied to the land. In their training as slaves, they were given an opportunity to rise far above their primitive savagery. You have seen in serious trials how, as a race, they are trustworthy. They are now your fellow-citizens in name, but are in a condition to be a permanent menace to your commonwealth. Properly aided on their way upward, they may be of great value to your descendants.” To the people of the North we may plead for all the help they can give; for hardly less than the Southerners, their ancestors shared in the actions which brought the negroes to this country. They gave the blacks the semblance of citizenship by the process of emancipation. If the work stops there, it may be questioned whether it was a boon to the masses of the folk it made

nominally free. To be what it was meant to be, then, it needs more than enactments. There must be long continued and devoted labor, wisely directed.

A necessary part of the work of a true emancipation of the negro is a careful inquiry into the history and former status of the people. Such an inquiry, placed and kept in good hands, is a necessary preliminary to sagacious action. It may serve to unite the men of all parts of the country in a work that so nearly concerns us all. There is not, nor is there likely to arise, a situation that so calls for intelligent patriotism as this we are sorely neglecting. We may go far away and rear an empire with our armies; but if we leave these, our neighbors, without a fair chance to develop the good that is in them, we shall have lost our real opportunity for great deeds—mayhap we shall fix among us evils that in the end will drag us down.

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# THE PHYSICAL GEOGRAPHY OF THE LANDS.

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THE most important principles established in physical geography during the nineteenth century are that the description of the earth's surface features must be accompanied by explanation, and that the surface features must be correlated with their inhabitants. During the establishment of these evolutionary principles, exploration at home and abroad has greatly increased the store of recorded facts; the more civilized countries have been in large part measured and mapped; the coasts of the world have been charted; the less civilized continents have been penetrated to their centers. This harvest of fact has been an indispensable stimulus to the study of physical geography; yet it can not be doubted that the spirit which has given life to the letter of the subject is the principle of evolution—inorganic and organic. This is especially true of the geography of the lands.

The century has seen the measurement of higher peaks in the Himalayas than had been previously measured in the Andes. The Nile has been traced to its source in the lakes of equatorial Africa, verifying the traditions of the ancients; and the Kongo has been found to cross the equator twice on its way to the sea. Facts without number have been added to the previous sum of knowledge. But at the same time, it has been discovered that the valleys of mountain ranges are the work of erosion; that the product of valley erosion is often seen in extensive piedmont fluvial plains; that waterfalls are retrogressively worn away until they are reduced to the smooth grade of a maturely established river; and that interior basins are slowly filling with the waste that is washed in from their rims upon their floors. Here are explanatory generalizations, involving, yet going far beyond matter of direct observation. Such generalizations in geography

correspond to the recognition in astronomy that planetary movements exemplify the law of gravitation; they are the Newton as against the Kepler of the subject.

The sufficient justification of the demand that has now arisen for explanation and correlation in the study of land forms is found in the repeated experience that until an explanatory description of a region can be given, one may be sure that some of its significant elements pass unnoticed; and until the controls that it exerts on living forms are studied, one may be confident that its geographical value is but half measured. A sentence from Guyot's *Earth and Man* may here be taken as a guide: "To describe, without rising to the causes, or descending to the consequences, is no more science than merely and simply to relate a fact of which one has been a witness." There could hardly be devised a more concise and searching test of good work than this quotation suggests. The causes, in so far as the physical geography of the lands is concerned, have been learned chiefly through the study of geology; yet it does not by any means follow that all geologists are possessed of such knowledge of these causes as will constitute them geographers. The consequences have been learned through the study of evolutionary biology; yet a distinct addition to the usual discipline of biology is required in order to apprehend its geographical correlations. The limited space allowed to this article will require that further consideration of the consequences be excluded, in order to give due consideration to the causes.

One of the preparatory steps in the century's advance was taken by the German geographer, Ritter, who, near the beginning of the century, advocated a new principle that may be illustrated by the change in the definition of geography from "the description of the earth and its inhabitants" to "the study of the earth in relation to its inhabitants;" but advance beyond this beginning was for a long time obstructed by certain ancient beliefs. Theological preconceptions as to the age of the earth and the associated geological doctrine of catastrophism, although attacked by the rising school of uniformitarianism, were then dominant. They gave to the geographer a ready-made earth, on which the existing processes of change were unimportant. Furthermore, the belief in the separate creation of every organic species led to the doctrine of teleology, which maintained the predetermined fitness of the earth for its inhabitants, and of its inhabitants

for their lifework. All this had to be outgrown before geographers could understand the slow development of land forms and the progressive adaptation of all living beings to their environments. Yet the beginning that Ritter made was of great importance, and it would have led further had it not happened that for many decades professors of geography in Europe brought chiefly a historical training to their chairs, to the almost entire neglect of physical geography. In the last thirty years there has been a reaction from this condition in Germany and France, but Italy, with many professors of geography in her universities, still for the most part follows historical methods.

In the victory of the uniformitarians over the catastrophists began the fortunate alliance of geography with geology, which was long afterwards happily phrased by Mackinder: "Geology considers the past in the light of the present; geography considers the present in the light of the past." Instead of believing in cataclysmic upheavals and in overwhelming floods, Playfair and other exponents of the Huttonian school taught that mountains were slowly upheaved and slowly worn down. The simplicity of Playfair's argument finds excellent illustration in the often quoted passage regarding the origin of valleys: "Every river appears to consist of a main trunk, fed from a variety of branches, each running in a valley proportioned to its size, and all of them together forming a system of vallies, communicating with one another, and having such a nice adjustment of their declivities that none of them join the principal valley either on too high or too low a level; a circumstance which would be infinitely improbable if each of these vallies were not the work of the stream that flows in it." Descriptions of valleys should always recognize the share that rivers have had in eroding them, or else the "nice adjustment of their declivities" may pass unnoticed.

It should be noted, however, that to this day explanation is not always allowed an undisputed place in the treatment of the lands, however fully it is accepted as appropriate to the presentation of other divisions of physical geography. But the manner in which explanation is extending over a larger and larger part of the subject gives assurance that the geographers of the coming century will insist upon a uniformly rational treatment of all divisions of their science. The active phenomena of the earth's surface first secured explanation; it has long been considered essential to explain as well as to describe such phenomena as the winds of the air and the currents of the



ocean; indeed, this is now so habitual that many geographers who may object to the explanation of a peculiar kind of a valley as a trespass upon geology, will nevertheless demand an explanation of rainfall and tides, although these truly geographical subjects are manifestly shared with physics and astronomy. Land forms of very elementary character, like deltas, or of rapid production, like volcanoes, have had to give some account of themselves all through the century; but it was not for many years after the announcement of Playfair's law, that the erosion of valleys by the rivers that drain them came to be regarded as a subject appropriate to a geographical treatise. Only in the later years of the century has the fuller treatment of this beautiful subject been attempted; even now much of it remains to be developed in the century to come.

The treatment of physical geography will be much more even, to the great advantage of its students, when explanatory description is applied to all its parts. The alluvial fans at the base of arid mountains should be accounted for as well as the dunes of deserts. The fault cliffs of broken plateau blocks and the weathered cliffs of retreating escarpments deserve to be considered as carefully as the wave-cut cliffs of coasts; the essential differences of these forms are reached most easily through their explanation. The varied sculpturing of a mountain slope may, in time, come to be as well understood as is now the erosion of a simple valley in a low plain.

One of the most notable elements of the century's progress is the increasing breadth of view gained as explanatory descriptions are extended further and further over the geographical field. At first explanation was given to various individual features, item by item; now it is recognized that an appropriate place must be provided for all kinds of land forms in a comprehensive scheme of physiographic classification. Many instances of the earlier stage might be given, beginning with examples from the works of Humboldt, the acknowledged leader of scientific explorers in the opening decades of the century. His attempts, more or less completely successful, to explain the facts that he observed, as well as to correlate life with environment, may be traced all through his writings; but his 'Cosmos' (1845) did not reach a careful discussion of land forms, although it entered so far into an explanatory treatment as to consider the formation of mountain ranges.

Innumerable examples of isolated facts and special explanations, unrelated to a comprehensive scheme of physiographic classification, might be taken from the reports of exploring expeditions and of geological surveys; from books of travel and from geographical and geological journals with which the nineteenth century has filled so many library shelves; but lack of space will prevent mention of all sources, save a few treatises in which the accumulated knowledge of their time is summarized. Such a work as Mrs. Somerville's 'Physical Geography' (1848) gives in the early pages a brief general consideration of land forms, and then enters at once upon the areal description of the continents; later pages present a short outline of the features of rivers, and then the rivers of the world are taken up. This is as if a text-book of botany should pass rapidly over the structure and classification of plants, and devote most of its pages to the flora of different regions. Again, Klöden's compendious geography includes a volume on 'Physical Geography,' in which much material is gathered (3d ed., 1873); but the treatment is very uneven, as is natural in the absence of a good scheme of classification. Glaciers receive much attention, but valleys are rather curtly dismissed; deltas are elaborately described, but little space is given to other forms assumed by the waste of the land on the way to the sea. Ansted's 'Physical Geography' (5th ed., 1871) contains abundant fact, but much of it is a kind that is better presented on a map than in verbal form. Many pages are devoted to statistical statements, from which no student can gain inspiration for further study, for example: "The Danube receives a large number of tributaries, of which the most important are, on the right, the Isar, Inn, Raab, Drave, Save, Morave, and Isker. On the left are the Altmühl, Regen, Waag, Gran, Theiss, Temes, Aluta, Sereth, and Pruth. Many of these are large streams with other important tributaries. The Danube drains upwards of 300,000 square miles of country."

A decided advance over earlier books in the way of rational or explanatory treatment is found in the works of Peschel and Reclus; it is to the former that a reaction against the historical treatment of geography in Germany is largely due; while the latter is to be credited with an enlarged attention to the detail of land forms; but the books of neither of these authors recognize the systematic evolution of land forms. The same may be said of various other treatises which approach, but do not yet reach, the ideal that seems to be in sight. One of the chief responsibilities of the geographer—the description of landscape—can not be fully met by students

who accept the principles set forth in these books as their guides; for in spite of the increasing attention given to the lands in modern books, and in spite of the greater number of forms recognized, the combination of all forms in a well-organized whole is not yet accomplished.

It seems to have been against the empirical method of such books as Ansted's that Huxley protested in his 'Physiography,' urging its replacement by a more educative method. He wrote:

"I do not think that a description of the earth, which commences by telling a child that it is an oblate spheroid, moving around the sun in an elliptical orbit, and ends without giving him the slightest hint towards an understanding of the ordnance map of his own country, or any suggestion as to the meaning of the phenomena offered by the brook which runs through his village, or of the gravel pit whence the roads are mended, is calculated either to interest or to instruct.... Physiography has very little to do with this sort of Physical Geography. My hearers were not troubled with much about latitudes and longitudes, the heights of mountains, depths of seas, or the geographical distribution of kangaroos or *Compositae*.... I endeavored to give them ... a view of the 'place in nature' of a particular district of England—the basin of the Thames—and to leave upon their minds the impression that the muddy waters of our metropolitan river, the hills between which it flows, the breezes which blow over it, are not isolated phenomena, to be taken as understood because they are familiar. On the contrary, I endeavored to show that the application of the plainest and simplest processes of reasoning to any one of these phenomena suffices to show, lying behind it, a cause, which again suggests another; until, step by step, the conviction dawns upon the learner that, to attain to even an elementary conception of what goes on in his own parish, he must know something about the universe; that the pebble he kicks aside would not be what it is and where it is, unless a particular chapter of the earth's history, finished untold ages ago, had been exactly what it was.... Many highly valuable compendia of Physical Geography, for the use of scientific students of that subject, are extant; but in my judgment most of the elementary works I have seen begin at the wrong end, and too often terminate in an *ominum gatherum* of scraps of all sorts of undigested and unconnected information; thereby entirely destroying the educational value of that study which Kant

justly termed the ‘propædæutic of natural knowledge.’” (Preface to ‘Physiography,’ 1878).

Here we find clear recognition of the need of introducing a consideration of causes, just as was urged by Guyot; and furthermore a recognition of the need of linking together in their natural relations all the items which together constitute the content of the subject. It may, however, be contended that the attempt to combine in a single course of study the elementary principles of chemistry and physics, of geology and astronomy, along with those of physical geography, is not practicable from an educational point of view; such a combination will not secure either the clear knowledge or the strong discipline that can be derived from systematic courses in two or three of these subjects, presented separately. Text-books like Hinman’s ‘Eclectic Physical Geography’ and Mill’s ‘Realm of Nature,’ in both of which a broad range of other than geographical subjects is covered, do not seem to-day to be in so much favor as those books which attend more closely to the true content of our subject. Indeed, with respect to physical geography, considered from the scientific and educational point of view, a report on College Entrance Requirements, recently published by our National Educational Association,<sup>D</sup> presents the best definition and outline of the subject that has yet appeared. It advises the omission of irrelevant matter, however interesting such matter may be in itself. The principles of physics and the succession of geological formations with their fossils, the classification and distribution of plants and animals must be taught elsewhere; but much profit may be had from terrestrial phenomena by which the principles of physics are illustrated, and from the consequences of past geological changes in determining present geographical conditions, and especially from the physiographic controls by which the distribution of organic forms is determined.

<sup>D</sup> Proceedings, 1899, 780–792; also in the *Journal of School Geography*, September, 1898.

The general scheme under which all land forms may receive explanatory description must consider chiefly the movement and erosion of

the earth's crust. Deformation offers a part of the earth's crust to be worked upon. Various destructive processes of erosion work upon the offered mass, and the streams, with their transported waste, follow the depressions in the carved surface. So important is the element of erosion, and so leading is the part played by rivers in erosive work, that McGee would gather all land forms under a classification determined by their drainage systems.<sup>E</sup> Others have preferred a classification based, first on peculiarities of structure as determined by accumulation and deformation; and, secondly, on the progress of erosion; but in either scheme, the erosive work of rivers is so important that a sketch of the progress of the physical geography of the lands towards a systematic classification of its items may well follow the order in which valleys have been explained, branching off, as occasion may require, from the leading theme of rivers that flow under a normal humid climate to special conditions of erosion under an arid or a frigid climate. The progress which has made the physical geography of the lands what it is to-day is more the work of geologists than of geographers; and the chief reason for this is the indifference of many geographers to the physical side of their subject; an indifference that was undoubtedly favored by the cultivation of historical geography in continental Europe, and by the acceptance of the traveler or explorer as a full-fledged geographer in Great Britain. In the United States, it is only in the latter part of the century that the physical geography of the lands has gained a scientific standing, and the advantages that it now enjoys are geographical grafts upon a geological stock.

<sup>E</sup> *Nat. Geogr. Magazine*, i, 1889, 27–36.

The emancipation of geology from the doctrine of catastrophism was a necessary step before progress could be made towards an understanding of the lands. The slow movements of elevation and depression of certain coasts in historic time were of great importance in this connection. Studies of geological structures at last overcame the belief in the sudden and violent upheaval of mountain chains, which, under the able and authoritative advocacy of Elie de Beaumont, held a place even into the second half of the century. But even when it came to be understood that mountains and plateaus have been slowly upheaved, it still remained to be proved that the valleys and canyons by which they are drained were produced by erosion, and not by fractures and unequal movements of elevation. Advance was here made on two lines. Along one, a better understanding was gained of the forms producible by deformation alone; along the other, sea currents, floods and earthquake waves, to which the earlier observers trusted as a means of modifying the forms of uplift, were gradually replaced by the slow action of weather and water. Processes of deformation were found to act in a large way, producing massive forms without detail—broad plains and plateaus, extensive domes, straight cliffs and rolling corrugations; and thus it was learned that the varied and detailed forms of lofty mountain ranges and dissected plateaus must be ascribed almost entirely to the processes of erosion. But it should be noted that in exceptional instances land forms initiated by deformation, so recently as to have suffered as yet only insignificant sculpture, may exhibit much irregularity. The most striking example of this kind, an example of the very highest value in the systematic study of land forms, is that afforded by the diversely tilted lava blocks of Southern Oregon, as described by Russell.<sup>E</sup>

<sup>E</sup> 4th *Ann. Rep. U. S. Geol. Survey*, 1883.

Turning now to the second line of advance, it is noteworthy that so keen an observer as Lesley insisted, as late as 1856, that the peculiar topographical features of Pennsylvania, which he knew and described so well, could have been produced only by a great flood. But the principles of the uniformitarians were constantly gaining ground against these older ideas; and after the appearance in England of Scrope's studies in Central France and of Greenwood's polemic little work on 'Rain and Rivers' (1857), victory may be said to have been declared for the principles long before announced by Hutton and Playfair, which, since then, have obtained general acceptance and application.

Yet even the most ardent uniformitarians would, in the middle of the century, go no further than to admit that rain and rivers could roughen a region by carving valleys in it; no consideration was then given to the possibility that, with longer and longer time, the hills must be more and more consumed, the valleys must grow wider and wider open, until, however high and uneven the initial surface may have been, it must at last be reduced to a lowland of small relief. The surface of such a lowland would truncate the underground structures indifferently; but when such truncating surfaces were noticed (usually now at considerable altitudes above sea level, as if elevated after having been planed, and therefore more or less consumed by the erosion of a new system of valleys), they were called plains of marine denudation by Ramsay (1847), or plains of marine abrasion by Richthofen (1882). Today it is recognized that both subaërial erosion and marine abrasion are theoretically competent to produce lowlands of denudation; the real question here at issue concerns the criteria by which the work of either agency can be recognized in particular instances. In the middle of the century, not only every plain of denudation, but every line of escarpments was held by the marinists to be the work of sea waves; and it was not till after a sharp debate that the bluffs of the chalk downs which enclose the Weald of southeastern England were accepted as the product of ordinary atmospheric weathering, instead of as the work of the sea. Whitaker's admirable essay on 'Subaërial Denudation,' which may be regarded as having given the victory in this discussion to the subaërialists, was considered so heterodox that it was not acceptable for publication in the *Quarterly Journal* of the Geological Society, of London, but had to find a place in the more modest *Geological Magazine* (1867), whose pages it now honors. So signal indeed was this victory that, in later

years, the destructive work of the sea has been not infrequently underrated in the almost exclusive attention given to land sculpture by subaërial agencies. Truly, the sea does not erode valleys; it does not wear out narrow lowlands of irregular form between enclosing uplands, as was maintained by some of the most pronounced marinists in the middle of the century; but it certainly does attack continental borders in a most vigorous fashion, and many are the littoral forms that must be ascribed to its work, as may be learned from Richthofen's admirable 'Führer für Forschungsreisende' (1886). As this problem can not be further considered here, the reader may be at once referred to the most general discussion of the subject that has yet appeared, in an essay on 'Shoreline Topography' recently published by F. P. Gulliver.<sup>G</sup>

<sup>G</sup> *Proc. Amer. Acad.*, Boston, 1899, 152–258.

At about the time when the subaërial origin of valleys and escarpments was being established in England, the explorations and surveys of our western territories were undertaken, and a flood of physiographic light came from them. One of the earliest and most important of the many lessons of the West was that Playfair's law obtained even in the case of the Grand canyon of the Colorado, which was visited by the Ives expedition in 1858. Newberry, the geologist of the expedition, concluded that both the deep and fissure-like canyon and the broader valleys enclosed by cliff-like walls "belong to a vast system of erosion, and are wholly due to the action of water." Although he bore the possibility of fractures constantly in mind and examined the structure of the canyons with all possible care, he "everywhere found evidence of the exclusive action of water in their formation." This conclusion has, since then, been amply confirmed by Powell and Dutton, although these later observers might attribute a significant share of the recession of cliffs in arid regions to wind action. In a later decade, Heim demonstrated that the valleys of the Alps were not explicable as the result of mountain deformation, and that they found explanation only in river erosion. By such studies as these, of which many examples could be given, the competence of rivers to carve even the



deepest valleys has been fully established; yet so difficult is it to dislodge old-fashioned belief that Sir A. Geikie felt it necessary to devote two chapters in his admirable 'Scenery of Scotland' (1887) to prove that the bens of the Highlands were not so many individual upheavals, but that the glens were so many separate valleys of erosion; and as able an observer as Prestwich, a warm advocate of the erosion of ordinary valleys by their rivers, maintained (1886), with the results of our western surveys before him, that fissures were probably responsible for the origin of the deep and narrow canyons of the Colorado plateau.

The tumultuous forms of lofty mountains 'tossed up' as they seem to be when viewed from some commanding height, are, in by far the greater number of examples yet studied, undoubtedly the result of the slow erosion of the valleys between them; but it should not be forgotten that regions of very recent disturbances—as the earth counts time—may possess strong inequalities directly due to deformation. The tilted lava blocks of Oregon have already been mentioned. The bold forms of the St. Elias Alps, also described by Russell, are regarded by him as chiefly produced by the tilting of huge crustal blocks on which erosion has as yet done relatively little work. An altogether exceptional case is described by Dutton, who says that on the margin of one of the "high plateaus of Utah a huge block seems to have cracked off and rolled over, the beds opening with a V and forming a valley of grand dimensions." 'Rift valleys,' or trough-like depressions produced by the down-faulting of long, narrow, crustal blocks with respect to the bordering masses, are occasionally found, as in eastern Africa, where the 'Great Rift valley' has been described by Gregory. Trough-like depressions of similar origin, but much more affected by the degradation of their borders and the aggradation of their floors, are known to European geographers in the valleys of the Saône and of the middle Rhine. But no rift valley, no depression between the tilted lava blocks, resembles the branching valleys that are produced by the erosive action of running water.

Thus far, while much attention had been given to the work of rivers, little or no attention had been given to the arrangement of their courses. It seems to have been tacitly assumed that the courses of all streams were consequent upon the slope of the initial land surface. The explicit recognition of this origin, indicated by the provision of a special name, 'consequent streams,' was an important step in advance due to our western

geologists. The discovery soon followed that rivers have held their courses through mountain ridges that slowly rose across their path; the rivers, concentrating the drainage of a large headwater region upon a narrow line, cut down their channels as the land was raised. This idea first came into prominence through Powell's report on the Colorado River of the West (1875), in which he gave the name, 'antecedent,' to rivers of this class. He believed that the Green river, in its passage through the Uinta mountains, was to be explained as an antecedent stream. Much doubt has, however, been thrown upon this interpretation. Other accounts of antecedent rivers have been published, and to-day the Green is not so safe a type of antecedence as the Rhine below Bingen, the Meuse in the Ardennes, or several of the Himalayan rivers in the gorges that they have cut through the youngest marginal ridges of the range.

Rapidly following the establishment of these two important classes of valleys came the recognition of the very antithesis of antecedent rivers in those streams which have grown by headward erosion along belts of weak structure, without relation to the initial trough lines. To these the term 'subsequent' has been applied. It is frequently in association with streams of this class that drainage areas are rearranged by the migration of divides, and that the upper waters of one river are captured by the headward growth of another. This is accomplished by a most beautiful process of inorganic natural selection, which leads to a survival of the fittest and thus brings about a most intimate adjustment of form to structure, whereby the more resistant rock masses come to constitute the divides, and the less resistant are chosen for the excavation of valleys. Many workers have contributed to the solution of problems of this class; notably Heim, in his studies of the northern Alps (1876), and Löwl, who showed that, in folded mountain structures of great age, the original courses of streams might be greatly altered through the development of new lines of drainage (1882). A valuable summary of this subject is given by Philippon in his 'Studien über Wasserscheiden' (1886). The extraordinary depredations committed by the waxing Severn on the waning Thames have recently been set forth by Buckman. The turning of side branches from the slender trunk of the Meuse has been recognized in France. Many remarkable instances of stream captures have been found in the Appalachians, where the opportunity for the adjustment of streams to structures has been exceptionally good. Hayes and Campbell have, on the other hand, emphasized the importance of

drainage modifications independent of the growth of subsequent streams on weak structures, but governed by a slight tilting of the region, whereby some streams are accelerated and their opponents are retarded. It should be noted that the proof of the adjustment or rearrangement of drainage marks a victory for the uniformitarian school that is even more significant than that gained in the case of the antecedent rivers; for in one case a growing mountain range is subdued by the concentrated discharge of a large drainage area; but in the other case, the mountain slowly melts away under the attacks of the weather alone on the headwater slopes of the growing valleys.

The reason why all these studies of land carving are of importance to the geographer is that they greatly enlarge the number of type forms that he may use in descriptions, and that they recognize the natural correlations among various forms which must otherwise be set forth in successive itemized statements. The brief terminology learned in early school days, somewhat enlarged by a more mature variety of adjectives, is usually the stock of words with which the explorer tries to reproduce the features of the landscapes that he crosses, and as a result his descriptions are often unintelligible; the region has to be explored again before it can become known to those who do not see it. The longitudinal relief of certain well-dissected coastal plains, or the half-buried ranges of certain interior aggraded basins, may be taken as examples of forms which are easily brought home and familiarized by explanation, but which commonly remain remote and unknown under empirical description.

It may be urged that in many geological discussions from which geography has taken profit, consideration is given to form-producing processes rather than to the forms produced. This was natural enough while the subject was in the hands of geologists; but geographers should take heed that they do not preserve the geological habit. The past history of land forms and the action upon them of various processes by which existing forms have been developed, are pertinent to geography only in so far as they aid the observation and description of the forms of to-day.

Further illustration of the growing recognition of form as the chief object of the physiographic study of the lands is seen in the use of the term, 'geomorphology' by some American writers; but more important than the term is the principle which underlies it. This is the acceptance of theorizing

as an essential part of investigation in geography, just as in other sciences. All explanation involves theorizing. When theory is taken piecemeal and applied only to elementary problems, such as the origin of deltas, it does not excite unfavorable comment among geographers. But when the explanation of more complicated features is attempted, and when a comprehensive scheme of classification and treatment, in which theorizing is fully and frankly recognized, is evolved for all land forms, then the conservatives recoil, as if so bold a proposition would set them adrift on the dangerous sea of unrestrained imagination. They forget that the harbor of explanation can only be reached by crossing the seas of theory. They are willing to cruise, like the early navigators, the empirical explorers, only close along shore; not venturing to trust themselves out of sight of the land of existing fact; but they have not learned to embark upon the open ocean of investigation, trusting to the compass of logical deduction and the rudder of critical judgment to lead them to the desired haven of understanding of facts of the past.

One of the bolder explorers of the high seas of theory is Powell, who defined in the term 'baselevel' an idea that had long been more or less consciously present in the minds of geologists, and which has been since then of the greatest service to physiographers. Powell and his followers, especially Gilbert, Dutton and McGee, have consistently carried the consequences of subaërial erosion to their legitimate end in a featureless lowland, and have recognized the controlling influence of the baselevel during all the sequence of changes from the initial to the ultimate form. It is not here essential whether such a featureless lowland exists or ever has existed, but it is absolutely essential to follow the lead of deduction until all the consequences of the theory of erosion are found; and then to accept as true those theoretical deductions which successfully confront the appropriate facts of observation. Only in this way can the error of regarding geography as a purely observational natural science be corrected. Following the acceptance of the doctrine of baselevels came the method of reconstituting the original form initiated by deformation, as a means of more fully understanding the existing form; for only by beginning at the initial form can the systematic sequence of the changes wrought by destructive processes be fully traced and the existing form appreciated. This had often been done before in individual cases, but it now became a habit, an essential step in geomorphological study. Naturally enough, the terms of

organic growth, such as young, mature, old, revived, and so on, came to be applied to stages in the development of inorganic forms; and thus gradually the idea of the systematic physiographic development of land forms has taken shape. This idea is to-day the most serviceable and compact summation of all the work of the century on the physical geography of the lands. It recognizes the results of deformation in providing the broader initial forms on which details are to be carved. It gives special attention to the work of destructive processes on these forms, and especially to the orderly sequence of various stages of development, recognizing that certain features are associated with youth, and others with maturity and old age. It gives due consideration to the renewed movements of deformation that may occur at any stage in the cycle of change, whereby a new sequence of change is introduced. It gives appropriate place, not only to the forms produced by the ordinary erosive action of rain and rivers, but to the forms produced by ice and by wind action as well; and it co-ordinates the changes that are produced by the sea on the margin of the land with the changes that are produced by other agencies upon its surface. It considers not only the various forms assumed by the water of the land, such as torrents, rapids, falls and lakes, appropriately arranged in a river system as to time and place, but also the forms assumed by the waste of the land, which, like the water, is on its way to the sea. In a word, it lengthens our own life, so that we may, in imagination, picture the life of a geographical area as clearly as we now witness the life of a quick-growing plant, and thus as readily conceive and as little confuse the orderly development of the many parts of a land form, its divides, cliffs, slopes and water courses, as we now distinguish the cotyledons, stem, buds, leaves, flowers and fruit of a rapidly-maturing annual that produces all these forms in appropriate order and position in the brief course of a single summer.

The time is ripe for the introduction of these ideas. The spirit of evolution has been breathed by the students of the generation now mature all through their growing years and its application to all lines of study is demanded. It is true that the acceptance of inorganic as well as of organic evolution is often implied rather than outspoken; yet evolution is favorably regarded, as is proved by the eagerness with which even school boards and school teachers, conservatives among conservatives, hail the appearance of books in which the new spirit of geography is revealed. In the last years of the century, the school books most widely used in this country have made

great advance in the explanatory treatment of land forms. Tarr's *Physical Geographies* and Russell's monographic volumes on the 'Lakes,' 'Glaciers,' 'Volcanoes' and 'Rivers' of North America, all presenting land forms in an explanatory rather than an empirical manner, have been warmly welcomed in this country. Penck's *'Morphologie der Erdoberfläche'* (1894), although largely concerned with the historical development of the subject, presents all forms as the result of process. De Lapparent's *'Leçons de géographie physique'* (1886) treats land forms generically; and a second edition of the book is called for soon after the first. *'Earth Sculpture,'* by James Geikie (1899), and Marr's *'Scientific Study of Scenery'* (1900), carry modern ideas to British readers. There can be little doubt that the books of the coming century will extend the habit of explanation even further than it has yet reached.

This review of the advance of the century in the study of land forms, the habitations of all the higher forms of life, might have been concerned wholly with the concrete results of exploration, as was implied in an earlier paragraph. Travels in the Far East of the Old World, or in the Far West of the New, have yielded fact enough to fill volumes. But such a view of the century has been here replaced by another; not because the first is unimportant, for it is absolutely essential, but because the second includes the first and goes beyond it. Not the facts alone, but the principles that the facts exemplify, demand our attention. These principles, founded upon a multitude of observations, are the greater contribution of the closing to the opening century in the study of the Forms of the Land.

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# THE NEW YORK BOTANICAL GARDEN.

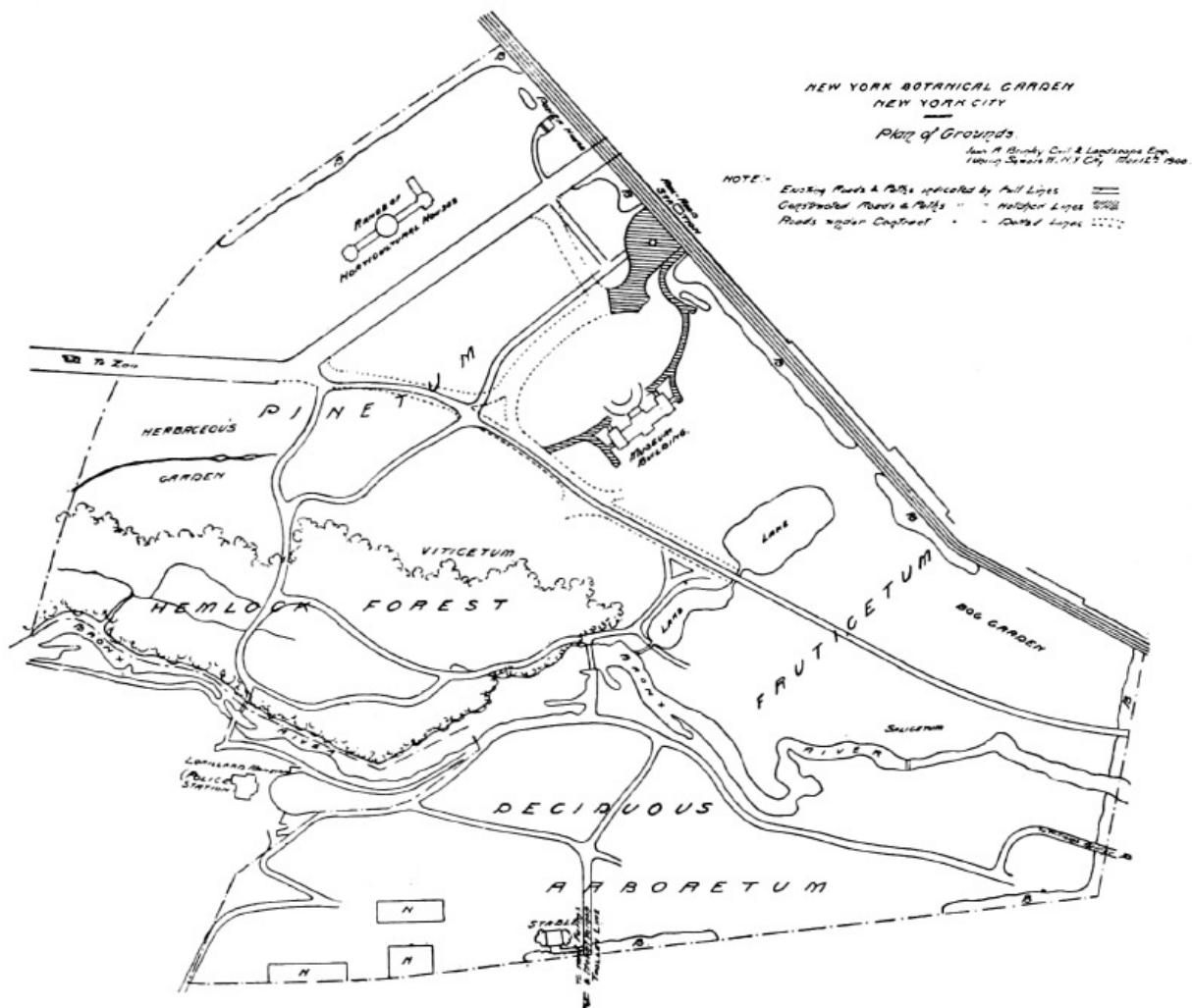
By DANIEL TREMBLY MACDOUGAL,  
DIRECTOR OF THE LABORATORIES.

A BOTANICAL garden is a museum of plants in the broadest sense of the term, and its chief purpose is to represent, by means of living specimens so far as possible, the principal types of the vegetation of the globe. It is obviously impossible to cultivate on any small area more than a few thousand of the quarter of a million of species in existence, and hence the plantations are supplemented by preserved specimens to illustrate the forms, which, by reasons of limitation of space, climate and soil, cannot be grown in the locality. In addition the species which formed the vegetation of the previous geological periods are represented by fossil specimens completing the history of the plant world so far as it is known, and yielding suggestions as to the descent of the present types.

Two general educational purposes are served by an institution of this character. Its collections are arranged to present information on the form, relationship, mode of life, habit and general biological character of the principal types of vegetation, in such manner as to be capable of comprehension by persons unacquainted with the technical aspects of the subject. Further interpretation of such facts may be made by means of books, journals, and lectures devoted entirely to this phase of the subject.

The material accumulated for the exploitation of popular knowledge of plants also affords an excellent basis for the induction of students into the more strictly scientific aspects of botany, and when supplemented by laboratories furnished with apparatus, microscopes, and other instruments of precision, the activities of these students may be carried beyond the frontiers of the subject in the investigation and discovery of new facts and phenomena. This extension of the boundaries of knowledge concerning the plant world may be carried on to advantage, only when a library is at hand,

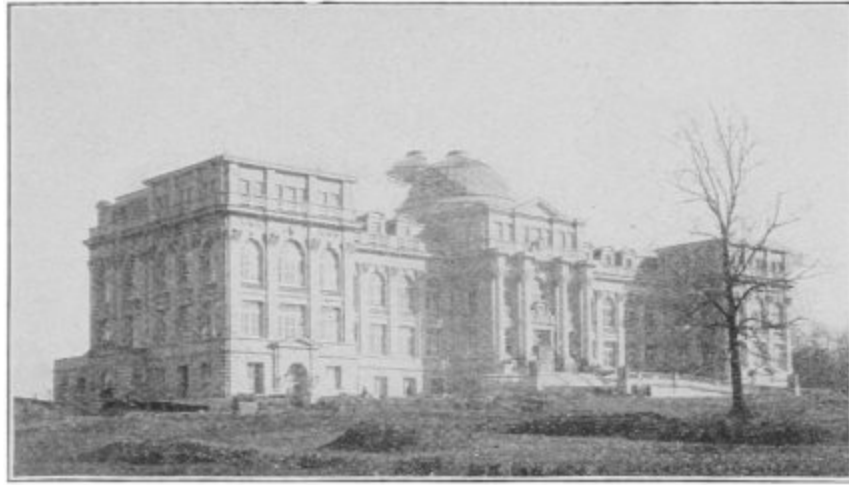
which contains all of the more important literature bearing upon the subject. The descriptions of the results of such researches should be made in publications devoted exclusively to this purpose, in accordance with the practice of all the more important botanical institutions in the world.





The Garden comprises two hundred and fifty acres of land in Bronx Park, in the City of New York, which was set aside for that purpose by the Department of Public Parks in 1895. A fireproof museum building of stone, brick and terra cotta, 308 by 110 feet, has been erected for the Garden by the city in the western part of the grounds, near the Bedford Park Station of the New York Central Railroad. The building has a basement floor and three stories, with a total floor space of nearly two acres, and a window area equal to half that of the floor area. The basement contains a lecture theater capable of seating seven hundred people, two large exhibition halls, preparation rooms, constant temperature laboratory, offices and storerooms. The first floor is devoted to a collection of economic plants, and the temporary installation of useful products in the way of foods, drugs, timbers, woods, fibers, gums, waxes, resins, oils, sugars, starches, poisons, utensils, etc., gives hints as to the great diversity of uses that may be made of vegetable products, together with an illustration of their method of preparation and their derivation.

The second floor is given over to an exhibit of types of all of the more important families and tribes of plants, from the simplest and most minute, to the highest and most complex. Specimens, models, fruits, seeds, drawings and photographs are used to bring the principal facts clearly before the observer. A set of swinging frames running parallel to the cases containing the types of the flora of the world, are used to display specimens of the plants found within a hundred miles of New York City. A number of special microscopes have been constructed for the purpose of forming a perfect exhibit, which will enable the visitor to see some of the more salient features in the minute structure of some of the plants in the cases.



**THE MUSEUM.**

The third floor contains the library, herbarium and laboratories. The library occupies a stack room extending to the rear of the middle of the building, two small storerooms and a large circular reading-room, under the illuminated dome. Here are assembled the botanical books of Columbia University, as well as those accumulated by the Garden, now numbering more than eight thousand volumes, with no reckoning of unbound separates and pamphlets. The collection of botanical periodicals is nearly complete, and the library is especially rich in literature concerning the mosses, ferns, and the flora of North and South America.

The main herbarium occupies a room in the east wing, eighty-five by forty-seven feet, and connected with it are storerooms and offices adequate to its administration. Windows on all sides of the main room and skylights give ample illumination. The number of mounted specimens on the shelves is not less than three quarters of a million, including the herbarium of Columbia University, which is deposited here in accordance with the agreement between the two institutions. The collection is especially rich in fungi, embracing the collections of Ellis and other eminent mycologists. A large amount of material of great historic value in connection with the work of Dr. John Torrey and the earlier botanical development of America is included. Accessions are being made to the herbarium at the rate of fifty to a hundred thousand specimens annually.

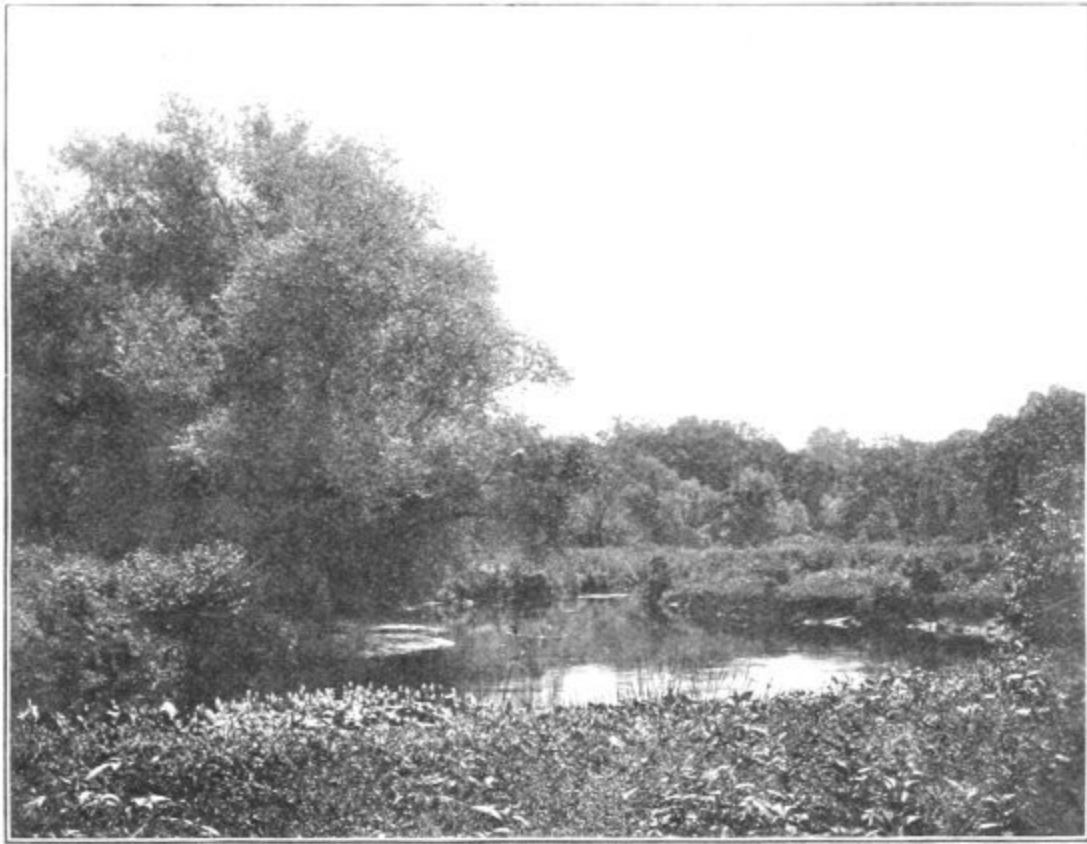


**IN THE FOREST.**

The laboratories consist of a series of rooms facing northward and westward, with special facilities for taxonomic, embryological and morphological investigations. Physiological and photographic darkrooms, the experiment room for living plants and chemical laboratories offer especially ample opportunities for the record and development of practically all phases of plant physiology. The laboratories, library and herbarium are open to the graduate students from Columbia University, in addition to those from other institutions of learning who may register directly at the Garden. The latter, in return, have the privileges of students at Columbia University.

A weekly convention of all of the workers in botany in New York City is held in the museum, at which the results of recent researches are given or an address is made by an invited speaker from out of the city.

The area of the Garden presents a very irregular topography, comprising, as it does, a half mile of the valley of the Bronx River, low marshes and swamps, artificial lakes, open glades, with heavy peaty soil, upland plains with gravelly sandy soil, granite ridges, and about seventy acres of natural forest. About forty acres of this forest consist of a dense grove of hemlocks, which has never been seriously disturbed by the hand of man. It is truly remarkable that the City of New York should include within its boundaries a primitive forest of this size, and this invaluable feature is to be preserved forever by a special contract between the Garden and the Department of Public Parks. Since a hemlock forest is a climactic formation, and is not replaced by any other growth unless cut down, it may be expected to endure through the present geological epoch, barring the accidents of flood, storm and fire. The great diversity of conditions offered by the natural features of the Garden gives it a very rich population of indigenous plants. A census of the ferns and seed-plants at the time the tract was converted to its present purpose showed nearly a thousand species.



**THE NORTH MEADOWS.**

The entire area has been handled most sympathetically by those in charge of the architectural features of the Garden. The buildings were erected in the more open western part of the grounds, which offered the least valuable landscape features, and the surface around them has been improved by plantings. The natural beauties of the tract have been most zealously guarded from disturbances of all kinds. The attractive panoramas of wild woodland and stream offered to the artist and lover of nature have been left absolutely untouched, but made more valuable by increased ease and safety of access.

A number of special biological groups of plants have been established in suitable places in various parts of the Garden. The trees are in the arboretum east of the Bronx on the side and summit of a long ridge; unassorted and reserve material of all kinds is kept in the nurseries on the eastern slope of the same ridge; the salicetum is established on the border of

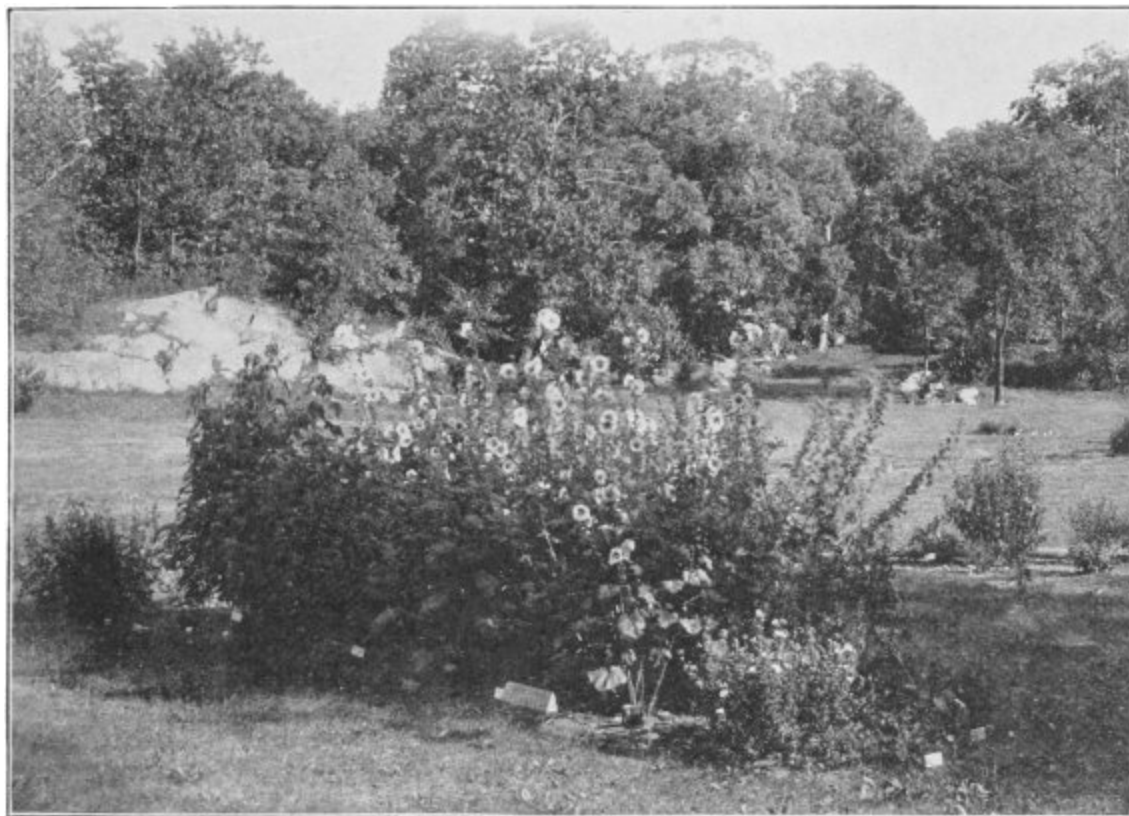
the marsh in the northern end of the Garden, giving the willows and poplars the conditions under which they grow best. The fruticetum occupies an adjoining upland plain underlaid with gravel to a depth of twenty feet, affording space for the cultivation of a large number of shrubs, while the conifers are located on slopes to the westward of the hemlock forest. The viticetum is along the western edge of the forest, and the trellises of logs and timbers, extending for a length of six hundred feet, give suitable support to the vines. The herbaceous plantation occupies an open glade to the westward of the forest, and lies between two granite ridges. It is traversed through the middle by a small stream widened at places into lagoons for aquatic forms. About twenty-two hundred species are now in cultivation in this plantation. The wide border plantations which are established along the boundaries also offer opportunities for the growth of a great variety of trees, herbs and shrubs.



**THE WATERFALL.**

The horticultural houses, also erected by the City for the Garden, are located in the western part of the grounds at some distance to the south of, and facing, the museum. A palm-house, with a total height of dome of ninety feet, is the central feature, from which lower ranges extend on either side, making a total length of front of five hundred and twelve feet. The horticultural houses, as well as the museum, are heated by steam furnished by a power house beside the railroad on the extreme edge of the Garden.

The collections of living plants in the plantations are arranged in the same system as the synoptic collection in the museum. Every plantation contains species of similar habit, and the horticultural houses are used for the cultivation of forms which may not endure the outdoor climate of this locality. Not only are the plants from warmer zones grown under glass, but when it is desired to develop native species out of their season, they may be forced and brought to full development and bloom in the winter.



**IN THE HERBACEOUS PLANTATION.**

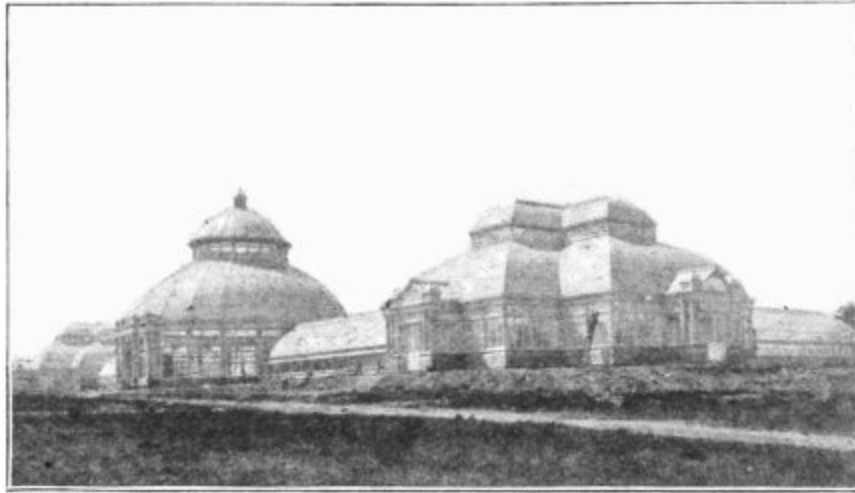
The construction of driveways and paths is being prosecuted by the Park Department with all available funds at their commands.

Public appreciation of the natural beauties of the Garden, and of the phases of botany illustrated by its collections has been most gratifying, as shown by the great and constantly increasing number of visitors. The series of popular lectures given in the museum on Saturday afternoons have been well attended. The Journal of the Garden, which serves as a means of communication with its members, brings to the notice of its readers interesting facts in botany, horticulture and forestry, and records a constantly swelling list of gifts of books, specimens and plants.

The library, herbarium and laboratories have been open for only a few months, yet twenty-two students have taken advantage of the facilities thus afforded during the collegiate year now closing. Investigations of importance have been carried forward by these students, by members of the staff, and by the members of the staff of Columbia University. The results of some of these investigations have been published in the Bulletin of the Garden, which also contains the official reports of the organization. Papers written by members of the staff or students are reprinted from the periodicals in which they appear as contributions, while a fourth series of Memoirs has been found necessary for the presentation of papers of great length.

Not the least important of the investigating functions of a garden consists in its participation in the exploration of remote or unknown parts of the world in an effort to obtain a better knowledge of the plant population of the earth. During the brief period of its activity the Garden has already carried out work of this character in the Rocky Mountains and in Porto Rico.





**HORTICULTURAL HOUSES.**

The ordinary work of the Garden is maintained by the income from its endowment fund, by the annual dues of its members (now numbering over eight hundred) and by an annual appropriation by the City. Its board of managers is authorized to hold and administer trust funds, and it is hoped by the aid of gifts or bequests for special or general purposes to expand its usefulness in directing investigation. Already it has been favored by a bequest of a considerable sum of money by the late ex-Chief Justice Charles P. Daly, which may be devoted to any purpose determined by the board of managers.

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# **GAS AND GAS METERS.**

BY HUBERT S. WYNKOOP, M. E.

**W**HAT is the matter with our illuminating gas? Why is its quality so poor? Why is it that our bills are creeping up, in spite of the fact that the rate per thousand cubic feet is going down? These are questions that periodically recur to the mind of every householder.

Just why the public has not been educated into a correct understanding of the gas situation is hard to say, unless it be that an inbred prejudice against believing the word of any corporation has led to an utter repudiation of such explanatory statements as may emanate from time to time from the gas office. And it must be admitted that many of the explanations are misleading, either through the intention of the superior officials or by reason of the ignorance of their subordinates.

Hardly has the chill of shortening days driven us indoors in the early twilight before complaints of poor gas become epidemic. Now, what is 'poor' gas? Is the gas deficient in light-giving constituents, or is it merely burned in such a manner as not to afford a satisfactory illumination?

The charter of Greater New York requires that the illuminating gas supplied throughout the city shall be of at least twenty candle power, or illuminating quality, or richness—that is to say, if we burn this gas in a standard burner at the standard pressure (or at as near this pressure as may be), so that the rate of consumption is five cubic feet an hour, the flame thus produced shall be equivalent to twenty standard sperm candles, each burning at the rate of one hundred and twenty grains of sperm per hour, and all bunched—if such a thing were possible. There can be hardly any doubt but that all the gas sent out from modern gas works fulfills the above requirement. Indeed, my own tests give results ranging from twenty-two to

twenty-eight candles, with an average of about twenty-four. Manifestly, the gas *sent out* is not 'poor.'

Nevertheless, the fact that the gas as manufactured is of the required candle power is no indication that the product as delivered to the consumer will give a similarly satisfactory test. Distribution of gas is attended with many perplexities, not the least of which is condensation. The illuminating hydrocarbons, or light-giving constituents held in suspension in the gas, are not so firmly fixed therein as to be unaffected by the size of the pipe, the character of the internal pipe surface, and barometric and thermometric variations. The transmission of gas causes, therefore, a loss of candle power ranging from a small fraction to several candles, although it is possible to conceive of conditions so extraordinarily favorable that the illuminating quality of the gas might be actually improved by distribution.

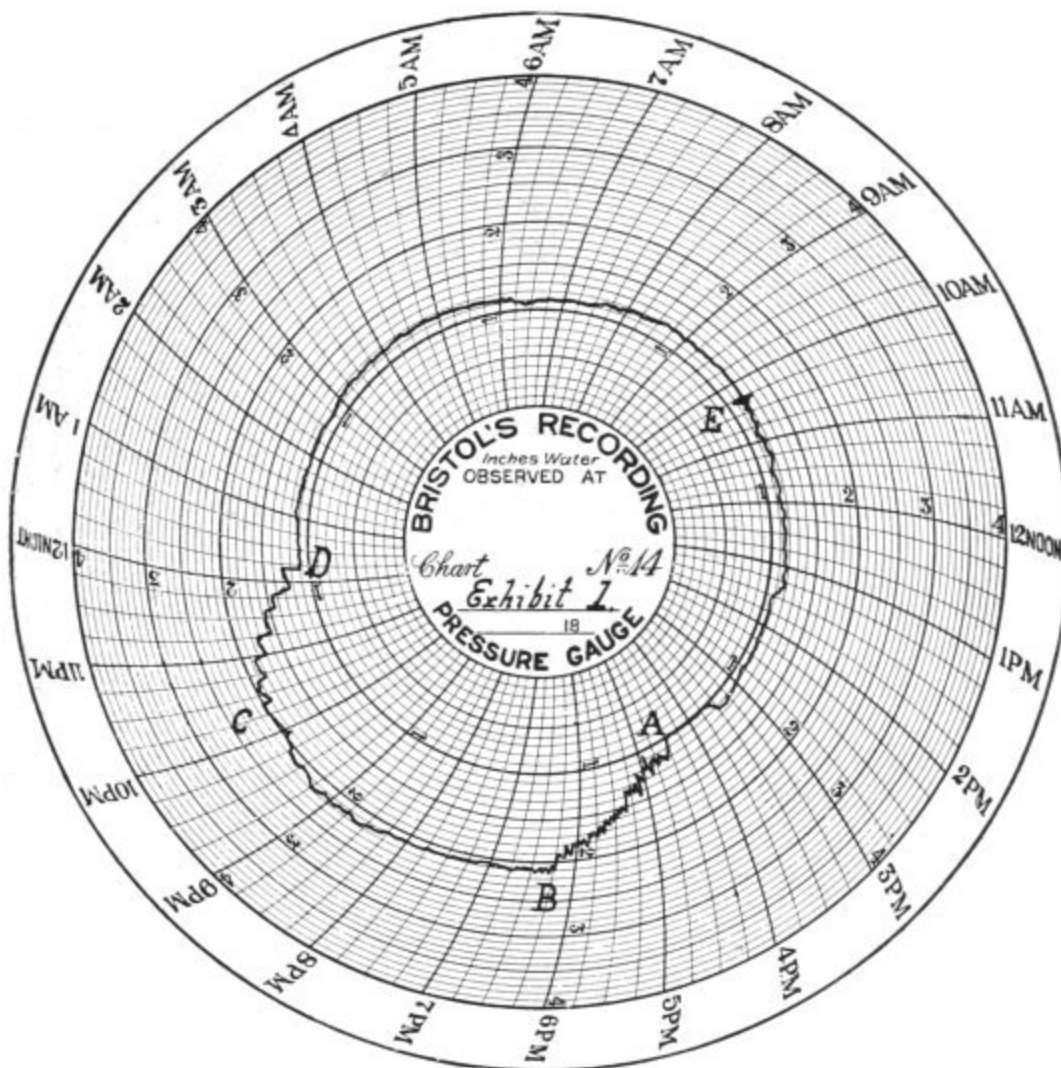
It will be readily understood from this explanation that tests made at the gas works, or even at points arbitrarily selected at a certain distance from these works, are hardly calculated to satisfy the consumer. For this reason I have preferred, in conducting these tests, to sacrifice to some degree the accuracy that obtains in laboratory experiments, in order to test gas samples *taken from the main directly in front of the complainant's own premises*. I argue that the consumer cares little or nothing as to whether the gas as manufactured complies with the law, or whether tests made at a point perhaps a mile away from the works show the required candle power; but that he does want to know what is the quality of the gas passing in at his service pipe. The method of collecting and transporting to a laboratory the gas samples enables one to say with positiveness that the gas at the point of complaint has an illuminating power of at least so many candles, and that it may be even one candle better than the tests indicate. The figures thus obtained range from twenty and a half to twenty-five. So, then, the gas *delivered to the consumer* is not 'poor.'

Hygienic reasons demand that the impurities in the gas shall not exceed a definite percentage. Whatever effect these impurities may have upon the candle power has been covered by the tests above explained, so that any further consideration of these impurities may be omitted here.

It is always a difficult matter to convince an indignant householder that the quality of the gas supplied to him is satisfactory. He knows perfectly

well that he is not getting the desired result, and no explanation, however elaborate, as to candle power will placate him, unless it be supplemented by a further statement detailing the cause of the trouble. When you are trying to draw water in the bathroom while the cook is filling the washtubs in the basement, do you say the water is 'poor'? Why, then, should you characterize the gas as 'poor,' when people nearer to the gas works than you are happen to be drawing heavily upon the common gas main? Imagine, if you please, a long gas main, with consumers tapping in at points throughout its entire length, and with a gas holder forcing the gas in at one end. Since there is a loss of pressure, caused by the transmission, it follows that the pressure will be higher at the gas holder than anywhere else along the line, the *difference* in pressure depending, roughly, upon the size and length of the pipe and upon the amount of gas flowing. Now, for any one customer the size and length of pipe will remain constant, but the flow of gas along the line will vary from hour to hour, consequently the pressure at his house may be expected to vary from hour to hour.

The unit of measurement of gas pressure is that pressure which will cause a difference of water level of one tenth of an inch in the two legs of a V-shaped tube when one end is connected with the gas main and the other end is left open to the outer air. Ten tenths, or one inch, is the standard, or normal pressure.



**EXHIBIT 1.**

Any appliance—even a gas-burner—operates to best advantage under certain well-defined conditions. Depart from these conditions, and the efficiency of the device is impaired to an extent depending largely upon the nature of the appliance under consideration. For example, burn an incandescent lamp at fifty per cent. above normal voltage and it breaks down; burn a gas jet at two hundred per cent. above normal pressure, and it still operates—how satisfactorily ‘deponent sayeth not.’ Now, the gas-burner is supposed to operate to best advantage at ten tenths of an inch. At this pressure the flame is neither so wavering as to be affected by every

chance draught, nor so rigid as to permit the gas to blow through without being properly consumed. Below the normal the flame decreases; above, the light is increased somewhat, but not by any means in proportion to the increase in the gas flow. Thus we see that the satisfactory employment of gas as an illuminant depends upon the maintenance of a pressure high enough to deliver the required amount of gas, but not so high as to cause wasteful consumption.

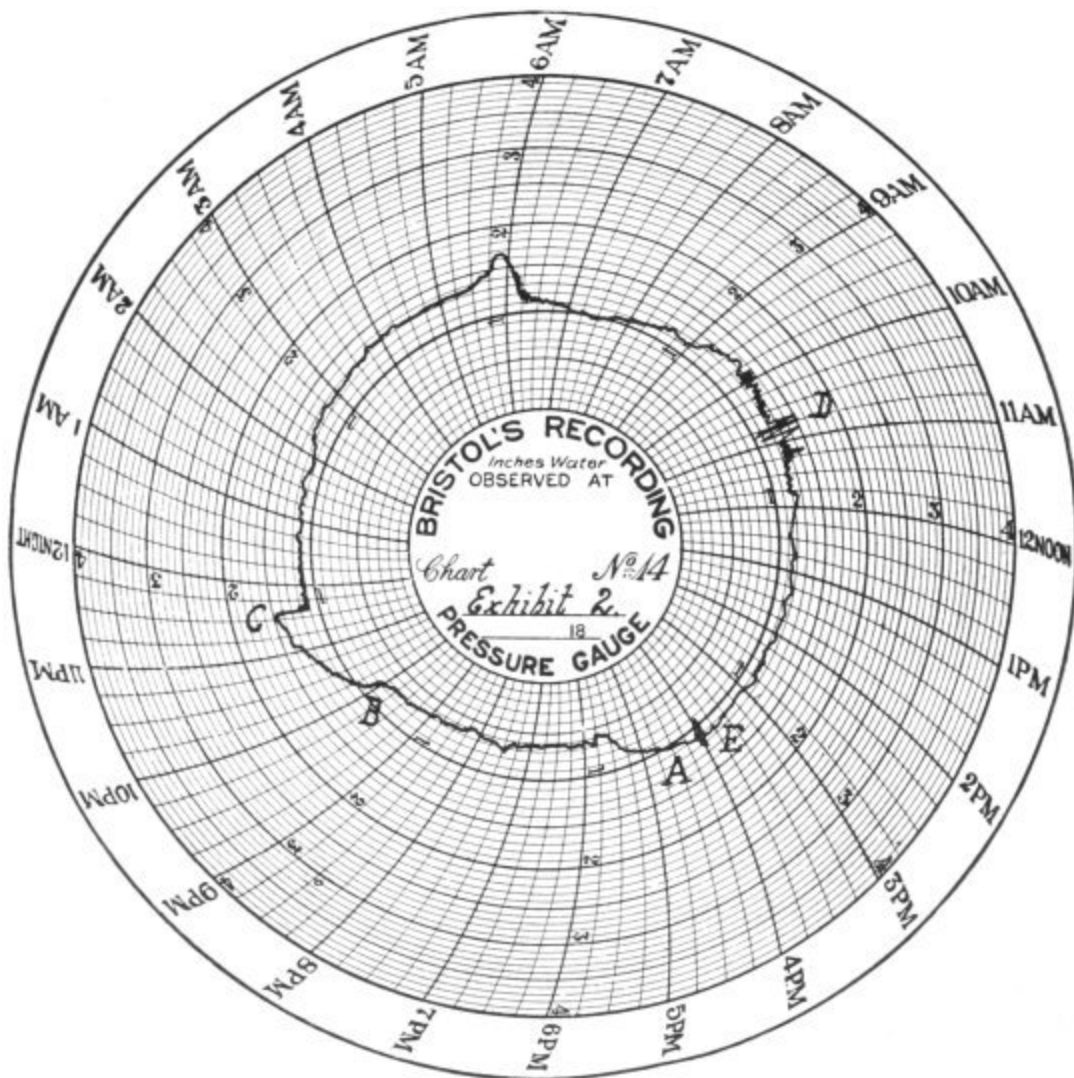
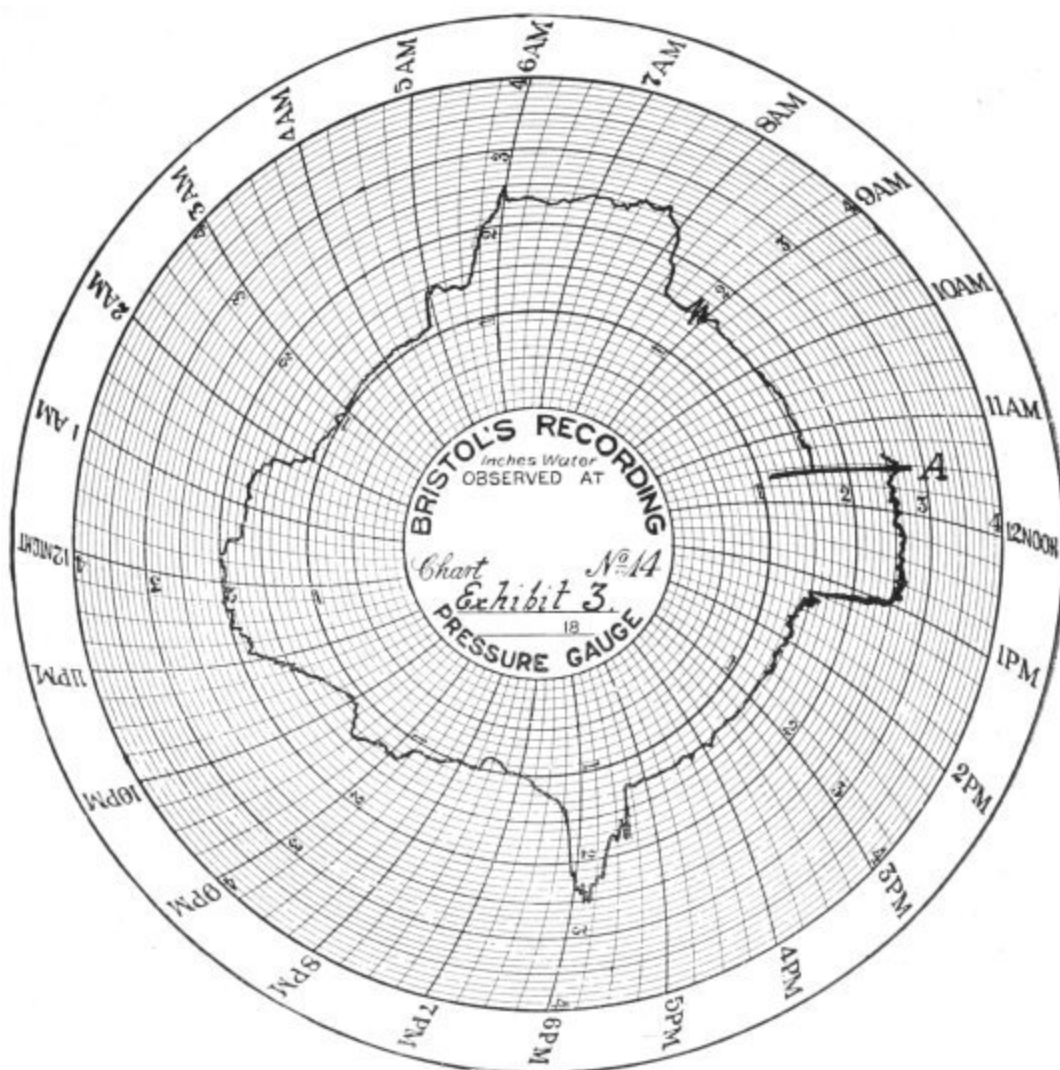


EXHIBIT 2.

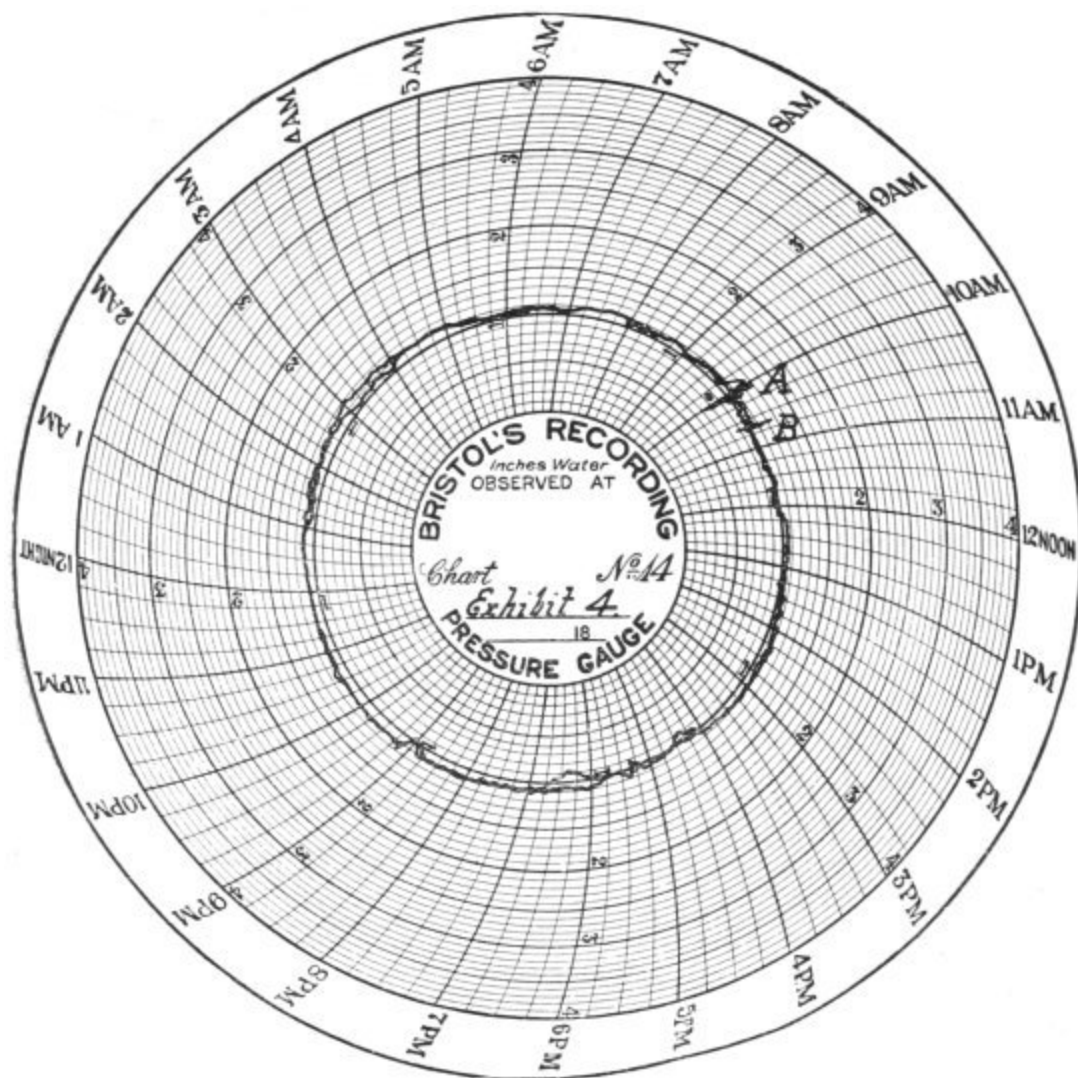
Turning back now to the gas main, let us consider the pressures actually existing. [Exhibit 1](#) is a photograph of a twenty-four-hour record of pressure at a point not far from the works. The radial lines represent time, and there is a line for each quarter of an hour. The circles represent pressure, there being one circle for each tenth of an inch. Starting at *E*, the point at which the record begins, and following the irregular line clockwise, one may readily determine the fluctuations of pressure and the time of their occurrence. Interpreting the diagram, we find that the pressure was slightly above the normal until 4.30 P. M. (*A*), when the works began to raise the pressure little by little, in order to compensate for the increased loss due to increased flow through the mains. At 6.15 P. M. (*B*), the works ceased increasing the pressure. While this increase lasted—from 6.15 P. M. (*B*) to 10.15 P. M. (*C*)—our friend near the works suffered under twenty-one tenths pressure, the gas blowing merrily through the tips and the meter conscientiously registering gas wasted as well as gas utilized. From 10.15 P. M. (*C*) the pressure falls by steps during the ensuing two hours, finally reaching eleven tenths just after midnight (*D*), which latter pressure is quite steadily maintained until the following forenoon. The service from bedtime to dinner time should have proved quite satisfactory. One would naturally expect to find this consumer complaining of high bills, however.



**EXHIBIT 3.**

Visiting the fellow at the distant end of the line, we find conditions widely at variance from those already considered. [Exhibit 2](#) tells a new story. The recording gauge was placed in service at 4 P. M. (E), and shortly afterward (A), the pressure began to fall. The jets grew dimmer and dimmer, while the Welsbach mantles became petticoats of red, with hems of white at the bottom. No wonder this man complains of 'poor' gas, while some learned friend, dropping in for an evening cigar, explains that there is 'air in the pipes.' The one consolatory reflection is that, at all events, the poor fellow had a good light to undress by (B to C).





**EXHIBIT 4.**

[Exhibits 3](#) and [4](#) come from my own residence. Together they form a ‘before-taking’ and ‘after-taking’ advertisement—not of medicine, but of a gas governor. The fact that I am located at a considerable distance—several miles—from the works, and am supplied through a main laid a number of years ago, when the territory was sparsely settled, enables me to present [Exhibit 3](#). Comment on this record is unnecessary. After securing this diagram I installed a governor and set it at eleven tenths. [Exhibit 4](#) shows what happened. I am now doing for myself, and at my own expense, that which the gas company fails to do for me. This governor, therefore, renders

me almost entirely independent of the gas company; and, in order to demonstrate more clearly to what degree this independence extends, the gauge has been allowed to run for forty-eight hours without changing the card, thus super-imposing the record of the second day upon that of the first. Note how closely the readings for the two days agree. The governor is a protection against excess of pressure only; if the street pressure falls below eleven tenths—the point at which my governor is set—automatic regulation ceases, and my gas simply becomes subject to practically the same variations as exist on the main. Happily, the latter condition is infrequently realized in our neighborhood. No argument is needed to prove how successfully a governing device of this nature can cope with the trouble indicated by [Exhibit 1](#), or how utterly inadequate it is to afford relief from the evil depicted in [Exhibit 2](#). Increased pressure is the only remedy for the latter.

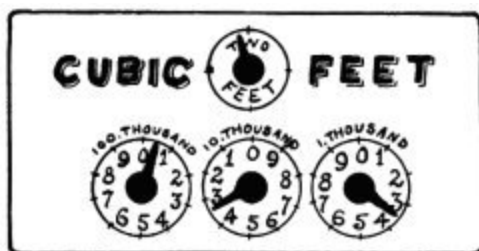
The gas company does not recommend the use of these house-to-house governors—presumably because such a recommendation would be in effect an admission that the service as now maintained by the company is not satisfactory. Indeed, the less enlightened officials—and it is these, unfortunately, with whom the consumer has generally to deal—positively and unreasoningly condemn all such regulating devices. In spite of this, there exist to-day several gas-reduction companies, whose sole occupation consists in exploiting various gas-pressure-regulating appliances, which are rented to consumers for a certain percentage of the monthly saving in the gas bills which their use effects.

It would appear to be a self-evident proposition that when one pays for gas delivered at his meter he is entitled to receive that gas under such a pressure as will afford the most satisfactory service. This pressure is found to be one inch. Making due allowance for reasonable fluctuations of a few tenths above the normal, any further departure from the standard may be taken as a sure indication of a disinclination on the part of the company to meet the expense of new pipes and regulating apparatus. The time is not far distant when the public will demand, not cheaper gas nor better gas, but a more satisfactory service. But before condemning the gas company one must look to his house piping. The company's responsibility ends just inside the meter, and from that point the consumer must provide satisfactory appliances, giving the same attention to the gas pipes as he gives to the

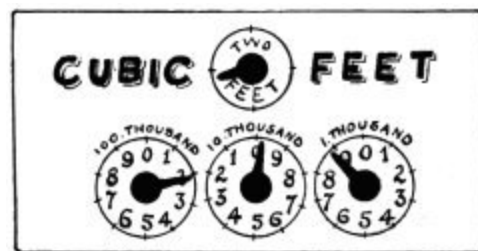
plumbing. This is seldom done and the company is frequently blamed for the neglect of the householder.

The gas engineer, steering between the Scylla of 'poor' gas and the Charybdis of excessive pressures, finds himself still 'dangerous in the rapids' of financial expenditure. At present he is doing the best he can with the money doled out to him by the management.

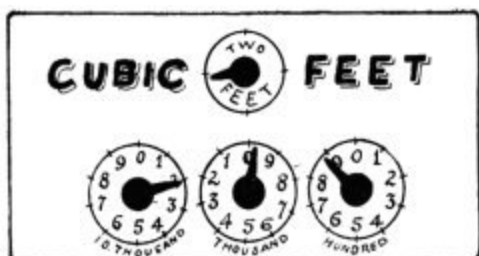
It will be observed that up to the present point the gas meter itself has played no part in the discussion. The meter, although greatly maligned, is in reality an eminently satisfactory piece of mechanism. Concerning this apparatus many erroneous notions prevail. One of these is that a householder may burn thousands of feet of gas without cost to himself, provided he keeps the company in blissful ignorance of the employment of gas for heating purposes upon his premises. The demonstration of the falsity of this idea lies within the reach of any one who will take the trouble to read his own meter on those days on which the company's indexer pays his monthly visits.



**FIG. 1.—READS 3,300 CUBIC FEET.**



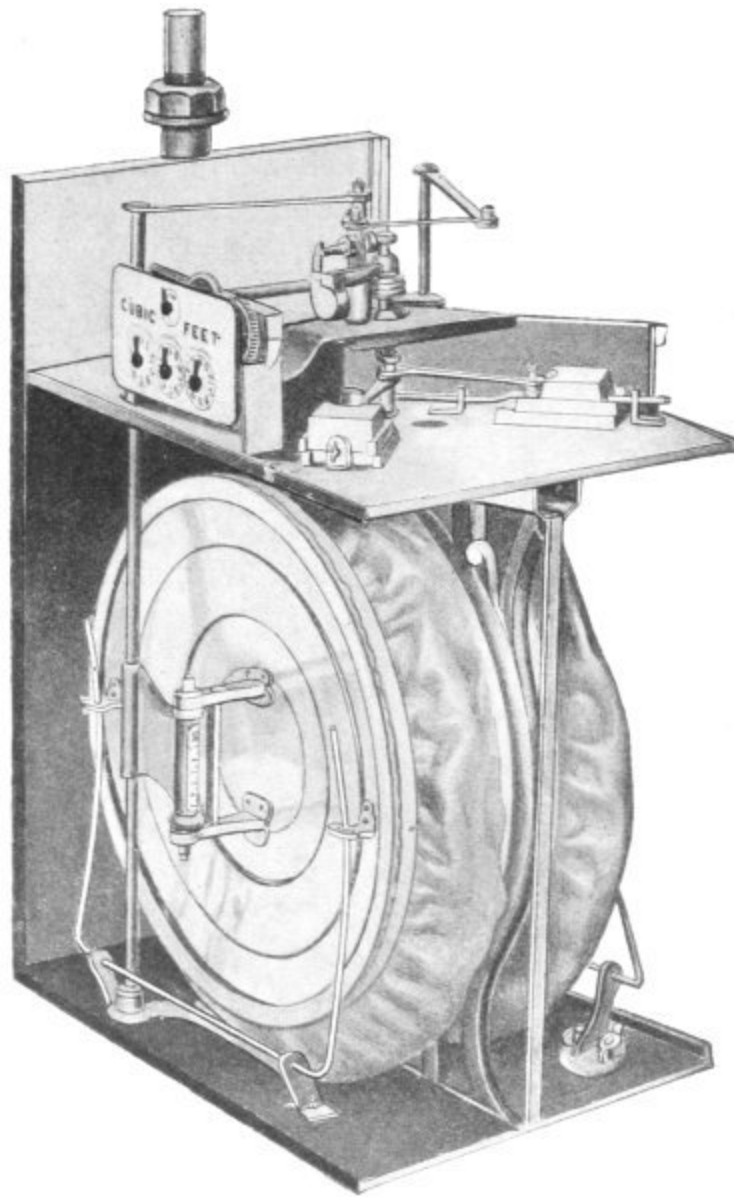
**FIG. 2.—READS 19,800 CUBIC FEET.** The apparent reading is 29,800. The gearing of the indexing mechanism is not especially delicate, and it frequently happens that the dial of one denomination begins to record before the hand of the next lower denomination has made a complete revolution.



**FIG. 3.—READS 19,800 CUBIC FEET.**

[Figs. 1](#), [2](#) and [3](#) represent different states of the index usually employed on the three, five and ten light meters, the sizes commonly found in our dwellings. The smaller dial, placed centrally above the other, is known as the 'proving dial,' and, being used merely for testing purposes, is not considered in reading the gas consumption. Although the index dials vary in nomenclature as well as in number, it is generally safe to consider that if the name is placed *above* the dial a complete revolution of the pointer is required to register the amount of gas indicated by the name; whereas if the name is placed *below* the dial each numbered division of the dial represents the amount corresponding to the name. If doubt still exists as to the value of each division of the lowest or right-hand dial, remember that no meter index is designed to read less than one hundred cubic feet for each division of the circle.

After one has indexed his own meter for a month or two he is in a position to begin checking the bills presented. The 'present state of meter' and the 'previous state of meter' are always specified, and the mere subtraction of the former from the latter gives the consumption. This is not invariably the case, however. After a meter has registered its maximum reading—100,000 in the smaller sizes—it passes over the zero point and begins to build up a new record. This happens at intervals as long as the apparatus is kept in service. Before me lies a bill giving the 'present state' as 1,700 and the 'previous state' as 96,300. Since the meter was continuously employed, it must have registered up to 100,000, so that it registered 3,700 cubic feet on the old score before recording 1,700 cubic feet on the new. Consequently, adding 1,700 to the difference between 'previous state' and the highest possible reading gives 5,400 cubic feet—the amount consumed during the month. By reading one's own meter the detection of any error on the part of the indexer or of the clerical force at the gas office becomes possible. Errors of this nature are of rare occurrence, as those who have adopted this plan of checking gas bills will testify. The responsibility for excessive bills is thus taken from the gas employees and thrown entirely upon the gas-registering mechanism itself. Those people, then, who chuckle furtively over the fact that the gas company has not 'caught on' to the surreptitious use of gas ranges are either the fortunate possessors of 'slow' meters or are deluding themselves as to the amount of gas which they actually consume.



**FIG. 4.—INTERIOR OF COMMON GAS METER.**

[Fig. 4](#) is a photograph of the common dry meter, with the front, back, top and left side removed. It is called a 'dry' meter to distinguish it from those meters, having little vogue in this country, which employ a liquid in place of a valve motion. The apparatus shown consists of a case divided into three compartments by a horizontal partition one fourth of the way down from the top, and by a vertical partition centrally placed and extending upward from the bottom of the casing to the horizontal partition.



The upper compartment contains the registering mechanism and a small valve chamber, the latter corresponding to the steam chest of an engine. In each of the lower compartments is a metal disk attached to the central partition by well-oiled flexible leathers, each disk, leather and the partition forming a bellows. As in a locomotive, the meter really consists of two separate mechanisms, set to operate out of phase and avoid dead centers.

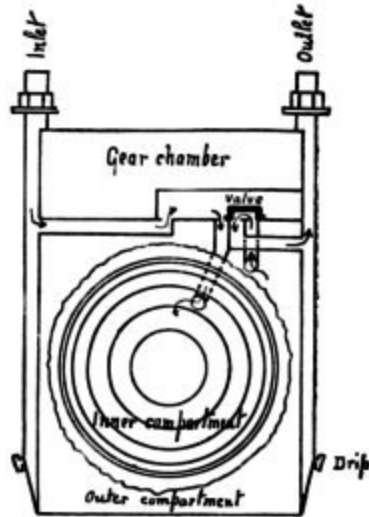


FIG. 5.

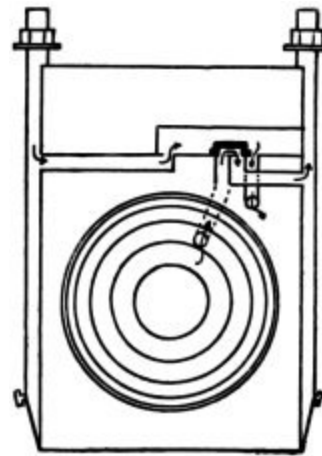


FIG. 6.

Considering one mechanism only, recourse may be had to a diagrammatic representation of the action ([Fig. 5](#)). Gas entering the inlet passes into the valve chamber. Here an ordinary D-slide-valve closes two of the openings, leaving a third through which the gas may flow into the bellows or inner compartment. The bellows expands, gradually filling the outer compartment, and forcing the gas out under the valve into the outlet pipe, as indicated by the arrows. When the bellows is fully distended the valve shifts into the position shown in [Fig. 6](#), admitting the inflowing gas to the outer compartment and collapsing the bellows, whose contents are forced into the outlet pipe by the paths traced by the arrows.

Thus, it will be observed, the meter is a volume measurer pure and simple, measuring cubic feet with as much deliberation as is required to deal water out of a cask by means of a pint dipper. Its percentage of error is the same at all pressures and under all loads within its capacity, and it

measures cubic feet of gas regardless of whether that gas be expanded or compressed.

And so we are obliged to realize, as another fallacy is exposed, that the meter does not spin around most energetically under the higher pressures, cheerfully and accommodatingly serving its masters by adding a mythical cubic foot or two to the count at each revolution.

It remains, then, to consider the error of the meter. The custom is, in New York at least, not to set a meter that registers fast—that registers a greater volume of gas than actually passes through it. If it is found to be slow, however, and not more than three per cent., it is allowed to go out. As a result, the meter, when first placed, always favors the consumer, sometimes to the extent of recording only ninety-seven feet of gas for each one hundred feet actually passed. Owing to the aging of the mechanism and the drying out of the leathers, there exists a tendency to increase the registry for each cubic foot passed. In this way a slow meter may become a fast meter after a period of active service. From the meager data at my disposal, it would appear that every meter should be called in for a thorough overhauling and readjustment at periodic intervals of from three to five years.

Assuming that there are several million gas meters in Greater New York alone, it is but natural to expect that out of this vast number, in spite of any reasonable care that may have been exercised in their adjustment originally, many will be found subsequently to be defective—some because of mechanical injury, some through sheer old age. Unfortunately, it is not possible as yet to obtain a convincingly large array of figures; but in the Borough of Brooklyn, where there are in service nearly a quarter of a million meters, and where complaints against them have been studiously encouraged by the authorities, one hundred and eighty-seven meters have been carefully tested. Here are the results:

21 correct

|                                                                                                          |                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 114 fast, average 3 per cent<br>(recording 103 cubic feet<br>for each 100 cubic feet<br>actually passed) | $\left\{ \begin{array}{l} 3 \text{ more than 10 per cent.} \\ 42 \text{ between 3 and 10 per cent.} \\ 69 \text{ less than 3 per cent.} \\ \hline 114 \end{array} \right.$ |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                  |   |                               |
|----------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------|
| 52 slow, average $2\frac{1}{4}$ per cent<br>(recording $97\frac{3}{4}$ cubic feet<br>for each 100 cubic feet<br>actually passed) | < | 0 more than 10 per cent.      |
|                                                                                                                                  |   | 13 between 3 and 10 per cent. |
|                                                                                                                                  |   | 39 less than 3 per cent.      |
|                                                                                                                                  |   | <hr/> 52                      |

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When one remembers that these one hundred and eighty-seven meters are presumably the worst of their kind, having been put in evidence by a naturally suspicious public, it is but fair to assume that the figures overrate rather than underestimate the errors of the average gas meter. Quoting from *The Progressive Age*, a journal devoted largely to the interests of the gas industry: "The meters made to-day will remain a long while in service before they begin to register incorrectly, and when we consider the dampness, extremes of temperature and hard usage they receive as they are transferred from cellar to attic, from among the dust, cobwebs and litter of a basement closet to the corner shelf of some coal cellar, to be the playground of rats, spiders and cockroaches, to be drenched in summer by sweating or leaky water pipes and wear a venerable beard of icicles in winter—to be, in fact, the worst-used machine about a gas plant—we can not fail but express surprise that it registers at all correctly."

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# THE SUN'S DESTINATION.

BY PROFESSOR HAROLD JACOBY,  
COLUMBIA UNIVERSITY.

THREE generations of men have come and gone since the Marquis de Laplace stood before the Academy of France and gave his demonstration of the permanent stability of our solar system. There was one significant fault in Newton's superbly simple conception of an eternal law governing the world in which we live. The labors of mathematicians following him had shown that the planets must trace out paths in space whose form could be determined in advance with unerring certainty by the aid of Newton's law of gravitation. But they proved just as conclusively that these planetary orbits, as they are called, could not maintain indefinitely the same shapes or positions. Slow indeed might be the changes they were destined to undergo; slow, but sure, with that sureness belonging to celestial science alone. And so men asked: Has this magnificent solar system been built upon a scale so grand, been put in operation subject to a law sublime in its very simplicity, only to change and change until at length it shall lose every semblance of its former self, and end perhaps in chaos or extinction?

Laplace was able to answer confidently: No. Nor was his answer couched in the enthusiastic language of unbalanced theorists who work by the aid of imagination alone. Based upon the irrefragable logic of correct mathematical reasoning, and clad in the sober garb of mathematical formulæ, his results carried conviction to men of science the world over. So was it demonstrated that changes in our solar system are surely at work, and shall continue for nearly countless ages; yet just as surely will they be reversed at last, and the system will tend to return again to its original form and condition. The objection that the Newtonian law meant ultimate dissolution of the world was thus destroyed by Laplace. From that day

forward, the law of gravitation has been accepted as holding sway over all phenomena visible within our planetary world.

The intricacies of our own solar system being thus illumined, the restless activity of the human intellect was stimulated to search beyond for new problems and new mysteries. Even more fascinating than the movements of our sun and planets are all those questions that relate to the clustered stellar congeries hanging suspended within the deep blue vault of night. Does the same law of gravitation cast its magic spell over that hazy cloud of Pleiads, binding them, like ourselves, with bonds indissoluble? Who shall answer, yes or no? We can only say that astronomers have as yet but stepped upon the threshold of the universe, and fixed the telescope's great eye upon that which is within.

Let us then begin by reminding the reader what is meant by that Newtonian law of gravitation. It appears all things possess the remarkable property of attracting or pulling each other. Newton declared that all substances, solid, liquid or even gaseous, from the massive cliff of rock down to the invisible air—all matter can no more help pulling than it can help existing. His law further formulates certain conditions governing the manner in which this gravitational attraction is exerted; but these are mere matters of detail; interest centers about the mysterious fact of attraction itself. How can one thing pull another with no connecting link through which the pull can act? Just here we touch the point that has never yet been explained. Nature withholds from science her ultimate secrets. They that have pondered longest, that have descended farthest of all men into the clear well of knowledge, have done so but to sound the depths beyond, never touching bottom.

This inability of ours to give a good physical explanation of gravitation has led numerous paradoxers to doubt or even deny that there is any such thing. But fortunately we have a simple laboratory experiment that helps us. Unexplained it may ever remain, but that there can be attraction between physical objects connected by no visible link is proved by the behavior of an ordinary magnet. Place a small piece of steel or iron near a magnetized bar, and it will at once be so strongly attracted that it will actually fly to the magnet. Any one who has seen this simple experiment can never again deny the possibility at least of the law of attraction as stated by Newton. Its possibility once admitted, the fact that it can predict the motions of all the

planets, even shown to the minutest details, transforms the possibility of its birth into a certainty as strong as any human certainty can ever be.

But this demonstration of Newton's law is limited strictly to the solar system itself. We may indeed reason by analogy, and take for granted that a law which holds within our immediate neighborhood is extremely likely to be true also of the entire visible universe. But men of science are loath to reason thus; and hence the fascination of researches in cosmic astronomy. Analogy points out the path. The astronomer is not slow to follow; but he seeks ever to establish upon incontrovertible evidence those truths which at first only his daring imagination had led him to half suspect. If we are to extend the law of gravitation to the utmost, we must be careful to consider the law itself in its most complete form. A heavenly body like the sun is often said to govern the motions of its family of planets; but such a statement is not strictly accurate. The governing body is no despot; 'tis an abject slave of law and order, as much as the tiniest of attendant planets. The action of gravitation is mutual, and no cosmic body can attract another without being itself in turn subject to that other's gravitational action. If there were in our solar system but two bodies, sun and planet, we should find each one pursuing a path in space under the influence of the other's attraction. These two paths or orbits would be oval, and if the sun and planet were equally massive, the orbits would be exactly alike, both in shape and size. But if the sun were far larger than the planet, the orbits would still be similar in form, but the one traversed by the larger body would be small. For it is not reasonable to expect a little planet to keep the big sun moving with a velocity as great as that derived by itself from the attraction of the larger orb. Whenever the preponderance of the larger body is extremely great, its orbit will be correspondingly insignificant in size. This is in fact the case with our own sun. So massive is it in comparison with the planets, that the orbit is too small to reveal its actual existence without the aid of our most refined instruments. The path traced out by the sun's center would not fill a space as large as the sun's own bulk. Nevertheless, true orbital motion is there.

So we may conclude that as a necessary consequence of the law of gravitation every object within the solar system is in motion. To say that planets revolve about the sun is to neglect as unimportant the small orbit of the sun itself. This may be sufficiently accurate for ordinary purposes; but it

is unquestionably necessary to neglect no factor, however small, if we propose to extend our reasoning to a consideration of the stellar universe. For we shall then have to deal with systems in which the planets are of a size comparable with the sun; and in such systems all the orbits will also be of comparatively equal importance.

Mathematical analysis has derived another fact from discussion of the law of gravitation which perhaps transcends in simple grandeur everything we have as yet mentioned. It matters not how great may be the number of massive orbs threading their countless interlacing curved paths in space, there yet must be in every cosmic system one single point immovable. This point is called the Center of Gravity. If it should so happen that in the beginning of things, some particle of matter were situated at this center, then would that atom ever remain unmoved and imperturbable throughout all the successive vicissitudes of cosmic evolution. It is doubtful whether the mind of man can form a conception of anything grander than such an immovable atom within the mysterious intricacies of cosmic motion.

But in general, we can not suppose that the centers of gravity in the various stellar systems are really occupied by actual physical bodies. The center may be a mere mathematical point in space, situated among the several bodies composing the system, but nevertheless endowed with the same remarkable property of relative immobility.

Having thus defined the center of gravity in its relation to the constituent parts of any cosmic system, we can pass easily to its characteristic properties in connection with the inter-relation of stellar systems with one another. It can be proved mathematically that our solar system will pull upon distant stars just as though the sun and all the planets were concentrated into one vast sphere having its center in the center of gravity of the whole. It is this property of the center of gravity which makes it preëminently important in cosmic researches. For, while we know that center to be at rest relatively to all the planets in the system, it may, nevertheless, in its quality as a sort of concentrated essence of them all, be moving swiftly through space under the pull of distant stars. In that case, the attendant bodies will go with it—but they will pursue their evolutions within the system, all unconscious that the center of gravity is carrying them on a far wider circuit.

What is the nature of that circuit? This question has been for many years the subject of earnest study by the clearest minds among astronomers. The greatest difficulty in the way is the comparatively brief period during which men have been able to make astronomical observations of precision. Space and time are two conceptions that transcend the powers of definition possessed by any man. But we can at least form a notion of how vast is the extent of time, if we remember that the period covered by man's written records is registered but as a single moment upon the great revolving dial of heaven's dome. One hundred and fifty years have elapsed since James Bradley built the foundations of sidereal astronomy upon his masterly series of star-observations at the Royal Observatory of Greenwich, in England. Yet so slowly do the movements of the stars unroll themselves upon the firmament, that even to this day no one of them has been seen by men to trace out more than an infinitesimal fraction of its destined path through the voids of space.

Travelers upon a railroad can not tell at any given moment whether they are moving in a straight line, or whether the train is turning upon some curve of huge size. The St. Gothard railway has several so-called 'corkscrew' tunnels, within which the rails make a complete turn in a spiral, the train finally emerging from the tunnel at a point almost vertically over the entrance. In this way the train is lifted to a higher level. Passengers are wont to amuse themselves while in these tunnels by watching the needle of an ordinary pocket compass. This needle, of course, always points to the north; and as the train turns upon its curve, the needle will make a complete revolution. But the passenger could not know without the compass that the train was not moving in a perfectly straight line. Just so we passengers on the earth are unaware of the kind of path we are traversing, until, like the compass, the astronomer's instruments shall reveal to us the truth.

But as we have seen, astronomical observations of precision have not as yet extended through a period of time corresponding to the few minutes during which the St. Gothard traveler watches the compass. We are still in the dark, and do not know as yet whether mankind shall last long enough upon the earth to see the compass needle make its revolution. We are compelled to believe that the motion in space of our sun is progressing upon a curved path; but so far as precise observations allow us to speak, we can but say that we have as yet moved through an infinitesimal element only of

that mighty curve. However, we know the point upon the sky towards which this tiny element of our path is directed, and we have an approximate knowledge of the speed at which we move.

More than a century ago Sir William Herschel was able to fix roughly what we call the Apex of the sun's way in space, or the point among the stars towards which that way is for the moment directed. We say for the moment, but we mean that moment of which Bradley saw the beginning in 1750, and upon whose end no man of those now living shall ever look. Herschel found that a comparison of old stellar observations seemed to indicate that the stars in a certain part of the sky were opening out, as it were, and that the constellations in the opposite part of the heavens seemed to be drawing in, or becoming smaller. There can be but one reasonable explanation of this. We must be moving towards that part of the sky where the stars are separating. Just so a man watching a regiment of soldiers approaching, will see at first only a confused body of men. But as they come nearer the individual soldiers will seem to separate, until at length each one is seen distinct from all the others.

Herschel fixed the position of the apex at a point in the constellation Hercules. The most recent investigations of Newcomb, published only a few months ago, have, on the whole, verified Herschel's conclusions. With the intuitive power of rare genius, Herschel had been able to sift truth out of error. The observational data at his disposal would now be called rude, but they disclosed to the scrutiny of his acute understanding the germ of truth that was in them. Later investigators have increased the precision of our knowledge, until we can now say that the present direction of the solar motion is known within very narrow limits. A tiny circle might be drawn on the sky, to which an astronomer might point his hand and say: Yonder little circle contains the goal towards which the sun and planets are hastening to-day. Even the speed of this motion has been subjected to measurement, and found to be about ten miles per second.

The objective point and the rate of motion thus stated, exact science holds her peace. Here genuine knowledge stops; and we can proceed further only by the aid of that imagination which men of science need to curb at every moment. But let no one think that the sun will ever reach the so-called apex. To do so would mean cosmic motion upon a straight line, while every consideration of celestial mechanics points to motion upon a curve.

When shall we turn sufficiently upon that curve to detect its bending? 'Tis a problem we must leave as a rich heritage to later generations that are to follow us. The visionary theorist's notion of a great central sun, controlling our own sun's way in space, must be dismissed as far too daring. But for such a central sun we may substitute a central center of gravity belonging to a great system of which our sun is but an insignificant member. Then we reach a conception that has lost nothing in the grandeur of its simplicity, and is yet in accord with the probabilities of sober mechanical science. We cease to be a lonely world, and stretch out the bonds of a common relationship to yonder stars within the firmament.



# A BIOGRAPHICAL SKETCH OF AN INFANT<sup>H</sup>.

BY CHARLES DARWIN.

<sup>H</sup> Reprinted from *Mind*, July, 1877.

[Child-study has recently become a most active department of psychology. It is the serious pursuit of men of science and the fad of women's clubs; a late accession to the magazines devoted to it comes from Japan. In spite of this wide-spread zeal, few of the followers of child-study have ever heard of one of the most valuable contributions to it. And in spite of the eminence of the author, Darwin's observations on the mental growth of his child are practically unknown to most zoölogists and psychologists.

It is a witness to the breadth of Darwin's interests that he should have been among the few men who anticipated by a generation or more what is now so wide a movement in psychology. His retention of his notes for thirty-seven years before publishing them is thoroughly characteristic. In this respect there is a notable difference between Darwin and the present-day enthusiasts for child-study.]

**M**• Taine's very interesting account of the mental development of an infant, translated in the last number of *Mind* (p. 252), has led me to look over a diary which I kept thirty-seven years ago with respect to one of my own infants. I had excellent opportunities for close observation, and wrote down at once whatever was observed. My chief object was expression, and my notes were used in my book on this subject; but as I attended to some other points, my observations may possibly possess some little interest in comparison with those by M. Taine, and others which



hereafter no doubt will be made. I feel sure, from what I have seen with my own infants, that the period of development of the several faculties will be found to differ considerably in different infants.

During the first seven days various reflex actions, namely sneezing, hickuping, yawning, stretching, and, of course, sucking and screaming, were well performed by my infant. On the seventh day, I touched the naked sole of his foot with a bit of paper, and he jerked it away, curling at the same time his toes, like a much older child when tickled. The perfection of these reflex movements shows that the extreme imperfection of the voluntary ones is not due to the state of the muscles or of the co-ordinating centers, but to that of the seat of the will. At this time, though so early, it seemed clear to me that a warm, soft hand applied to his face excited a wish to suck. This must be considered as a reflex or an instinctive action, for it is impossible to believe that experience and association with the touch of his mother's breast could so soon have come into play. During the first fortnight he often started on hearing any sudden sound, and blinked his eyes. The same fact was observed with some of my other infants within the first fortnight. Once, when he was 66 days old, I happened to sneeze, and he started violently, frowned, looked frightened, and cried rather badly; for an hour afterwards he was in a state which would be called nervous in an older person, for every slight noise made him start. A few days before this same date, he first started at an object suddenly seen; but for a long time afterwards sounds made him start and wink his eyes much more frequently than did sight; thus, when 114 days old, I shook a pasteboard box with comfits in it near his face and he started, whilst the same box when empty or any other object shaken as near or much nearer to his face produced no effect. We may infer from these several facts that the winking of the eyes, which manifestly serves to protect them, had not been acquired through experience. Although so sensitive to sound in a general way, he was not able even when 124 days old, easily to recognize whence a sound proceeded so as to direct his eyes to the source.

With respect to vision—his eyes were fixed on a candle as early as the 9th day, and up to the 45th day nothing else seemed thus to fix them; but on the 49th day his attention was attracted by a bright-colored tassel, as was shown by his eyes becoming fixed and the movements of his arms ceasing. It was surprising how slowly he acquired the power of following with his

eyes an object if swinging at all rapidly; for he could not do this well when seven and a half months old. At the age of 32 days he perceived his mother's bosom when three or four inches from it, as was shown by the protrusion of his lips and his eyes becoming fixed; but I much doubt whether this had any connection with vision; he certainly had not touched the bosom. Whether he was guided through smell or the sensation of warmth or through association with the position in which he was held, I do not at all know.

The movements of his limbs and body were for a long time vague and purposeless, and usually performed in a jerking manner; but there was one exception to this rule, namely, that from a very early period, certainly long before he was 40 days old, he could move his hands to his own mouth. When 77 days old, he took the sucking bottle (with which he was partly fed) in his right hand, whether he was held on the left or right arm of his nurse, and he would not take it in his left hand until a week later, although I tried to make him do so; so that the right hand was a week in advance of the left. Yet this infant afterwards proved to be left-handed, the tendency being no doubt inherited—his grandfather, mother, and a brother having been or being left-handed. When between 80 and 90 days old, he drew all sorts of objects into his mouth, and in two or three weeks' time could do this with some skill; but he often first touched his nose with the object and then dragged it down into his mouth. After grasping my finger and drawing it to his mouth, his own hand prevented him from sucking it; but on the 114th day, after acting in this manner, he slipped his own hand down so that he could get the end of my finger into his mouth. This action was repeated several times, and evidently was not a chance but a rational one. The intentional movements of the hands and arms were thus much in advance of those of the body and legs; though the purposeless movements of the latter were from a very early period usually alternate, as in the act of walking. When four months old he often looked intently at his own hands and other objects close to him, and in doing so the eyes were turned much inwards, so that he often squinted frightfully. In a fortnight after this time (i. e., 132 days old), I observed that if an object was brought as near to his face as his own hands were, he tried to seize it, but often failed; and he did not try to do so in regard to more distant objects. I think there can be little doubt that the convergence of his eyes gave him the clue and excited him to move his arms. Although this infant thus began to use his hands at an early period, he

showed no special aptitude in this respect, for when he was two years and four months old, he held pencils, pens, and other objects far less neatly and efficiently than did his sister, who was then only fourteen months old, and who showed great inherent aptitude in handling anything.

ANGER.—It was difficult to decide at how early an age anger was felt; on his eighth day he frowned and wrinkled the skin round his eyes before a crying fit, but this may have been due to pain or distress, and not to anger. When about ten weeks old, he was given some rather cold milk, and he kept a slight frown on his forehead all the time that he was sucking, so that he looked like a grown-up person made cross from being compelled to do something which he did not like. When nearly four months old, and perhaps much earlier, there could be no doubt, from the manner in which the blood gushed into his whole face and scalp, that he easily got into a violent passion. A small cause sufficed; thus, when a little over seven months old, he screamed with rage because a lemon slipped away and he could not seize it with his hands. When eleven months old, if a wrong plaything was given him, he would push it away and beat it; I presume that the beating was an instinctive sign of anger, like the snapping of the jaws by a young crocodile just out of the egg, and not that he imagined he could hurt the plaything. When two years and three months old, he became a great adept at throwing books or sticks, etc., at any one who offended him; and so it was with some of my other sons. On the other hand, I could never see a trace of such aptitude in my infant daughters; and this makes me think that a tendency to throw objects is inherited by boys.

FEAR.—This feeling probably is one of the earliest which is experienced by infants, as shown by their starting at any sudden sound when only a few weeks old, followed by crying. Before the present one was four and a half months old, I had been accustomed to make close to him many strange and loud noises, which were all taken as excellent jokes, but at this period I one day made a loud snoring noise, which I had never done before; he instantly looked grave and then burst out crying. Two or three days afterwards I made, through forgetfulness, the same noise, with the same result. About the same time (viz., on the 137th day), I approached with my back towards him and then stood motionless; he looked very grave and much surprised, and would soon have cried, had I not turned round; then his face instantly relaxed into a smile. It is well known how intensely

older children suffer from vague and undefined fears, as from the dark, or in passing an obscure corner in a large hall, etc. I may give as an instance that I took the child in question, when two and one fourth years old, to the Zoölogical Gardens, and he enjoyed looking at all the animals which were like those that he knew, such as deer, antelope, etc., and all the birds, even the ostriches, but was much alarmed at the various larger animals in cages. He often said afterwards that he wished to go again, but not to see 'beasts in houses'; and we could in no manner account for this fear. May we not suspect that the vague but very real fears of children, which are quite independent of experience, are the inherited effects of real dangers and abject superstitions during ancient savage times? It is quite conformable with what we know of the transmission of formerly well-developed characters, that they should appear at an early period of life, and afterwards disappear.

PLEASURABLE SENSATIONS.—It may be presumed that infants feel pleasure whilst sucking, and the expression of their swimming eyes seem to show that this is the case. This infant smiled when 45 days, a second infant when 46 days old; and these were true smiles indicative of pleasure, for their eyes brightened and eyelids slightly closed. The smiles arose chiefly when looking at their mother and were therefore probably of mental origin; but this infant often smiled then, and for some time afterwards, from some inward pleasurable feeling, for nothing was happening which could have in any way excited or amused him. When 110 days old he was exceedingly amused by a pinafore being thrown over his face and then suddenly withdrawn; and so he was when I suddenly uncovered my own face and approached his. He then uttered a little noise which was an incipient laugh. Here surprise was the chief cause of the amusement, as is the case to a large extent with the wit of grown-up persons. I believe that for three or four weeks before the time when he was amused by a face being suddenly uncovered, he received a little pinch on his nose and cheeks as a good joke. I was at first surprised at humor being appreciated by an infant only a little above three months old, but we should remember how very early puppies and kittens begin to play. When four months old, he showed in an unmistakable manner that he liked to hear the pianoforte played; so that here apparently was the earliest sign of an æsthetic feeling, unless the attraction of bright colors, which was exhibited much earlier, may be so considered.

AFFECTION.—This probably arose very early in life, if we may judge by his smiling at those who had charge of him when under two months old; though I had no distinct evidence of his distinguishing and recognizing any one, until he was nearly four months old. When nearly five months old he plainly showed his wish to go to his nurse. But he did not spontaneously exhibit affection by overt acts until a little above a year old, namely, by kissing several times his nurse who had been absent for a short time. With respect to the allied feeling of sympathy, this was clearly shown at six months and eleven days by his melancholy face, with the corners of his mouth well depressed, when his nurse pretended to cry. Jealousy was plainly exhibited when I fondled a large doll, and when I weighed his infant sister, he being then fifteen and one half months old. Seeing how strong a feeling jealousy is in dogs, it would probably be exhibited by infants at an earlier age than that just specified, if they were tried in a fitting manner.

*Association of Ideas, Reason, Etc.*—The first action which exhibited, as far as I observed, a kind of practical reasoning, has already been noticed, namely, the slipping his hand down my finger so as to get the end of it into his mouth; and this happened on the 114th day. When four and a half months old, he repeatedly smiled at my image and his own in a mirror, and no doubt mistook them for real objects; but he showed sense in being evidently surprised at my voice coming from behind him. Like all infants, he much enjoyed thus looking at himself, and in less than two months perfectly understood that it was an image; for if I made quite silently any old grimace, he would suddenly turn round to look at me. He was, however, puzzled at the age of seven months, when being out of doors he saw me on the inside of a large plate-glass window, and seemed in doubt whether or not it was an image. Another of my infants, a little girl, when exactly a year old, was not nearly so acute, and seemed perplexed at the image of a person in a mirror approaching her from behind. The higher apes which I tried with a small looking-glass behaved differently; they placed their hands behind the glass, and in doing so showed their sense, but far from taking pleasure in looking at themselves, they got angry and would look no more.

When five months old, associated ideas arising independently of any instruction became fixed in his mind; thus as soon as his hat and cloak were put on, he was very cross if he was not immediately taken out of doors. When exactly seven months old, he made the great step of associating his

nurse with her name, so that if I called it out he would look round for her. Another infant used to amuse himself by shaking his head laterally; we praised and imitated him, saying, “Shake your head;” and when he was seven months old, he would sometimes do so on being told without any other guide. During the next four months the former infant associated many things and actions with words; thus when asked for a kiss he would protrude his lips and keep still—would shake his head and say in a scolding voice, “Ah,” to the coal-box or a little spilt water, etc., which he had been taught to consider as dirty. I may add that when a few days under nine months old, he associated his own name with his image in the looking-glass, and when called by name would turn towards the glass, even when at some distance from it. When a few days over nine months, he learned spontaneously that a hand or other object causing a shadow to fall on the wall in front of him was to be looked for behind. Whilst under a year old, it was sufficient to repeat two or three times at intervals any short sentence to fix firmly in his mind some associated idea. In the infant described by M. Taine, the age at which ideas readily became associated seems to have been considerably later, unless, indeed, the earlier cases were overlooked. The facility with which associated ideas due to instruction and others spontaneously arising were acquired, seemed to me by far the most strongly marked of all the distinctions between the mind of an infant and that of the cleverest full-grown dog that I have ever known. What a contrast does the mind of an infant present to that of the pike, described by Professor Mobius,<sup>I</sup> who, during three whole months dashed and stunned himself against a glass partition which separated him from some minnows; and when, after at last learning that he could not attack them with impunity, he was placed in the aquarium with these same minnows, then in a persistent and senseless manner he would not attack them!

<sup>I</sup> ‘Die Bewegungen der Thiere,’ etc., 1878, p. 11.

Curiosity, as M. Taine remarks, is displayed at an early age by infants, and is highly important in the development of their minds; but I made no special observation on this head. Imitation likewise comes into play. When

our infant was only four months old, I thought that he tried to imitate sounds; but I may have deceived myself, for I was not thoroughly convinced that he did so until he was ten months old. At the age of eleven and a half months, he could readily imitate all sorts of actions, such as shaking his head and saying “Ah” to any dirty object, or by carefully and slowly putting his forefinger in the middle of the palm of his other hand, to the childish rhyme of “Pat it and pat it and mark it with T.” It was amusing to behold his pleased expression after successfully performing any such accomplishment.

I do not know whether it is worth mentioning, as showing something about the strength of memory in a young child, that this one, when three years and twenty-three days old, on being shown an engraving of his grandfather, whom he had not seen for exactly six months, instantly recognized him and mentioned a whole string of events which had occurred whilst visiting him, and which certainly had never been mentioned in the interval.

MORAL SENSE.—The first sign of moral sense was noticed at the age of nearly thirteen months; I said, “Doddy (his nickname) won’t give poor papa a kiss,—naughty Doddy.” These words, without doubt, made him feel slightly uncomfortable; and at last, when I had returned to my chair, he protruded his lips as a sign that he was ready to kiss me; and he then shook his hand in an angry manner until I came and received his kiss. Nearly the same little scene recurred in a few days, and the reconciliation seemed to give him so much satisfaction that several times afterwards he pretended to be angry and slapped me, and then insisted on giving me a kiss. So that here we have a touch of the dramatic art, which is so strongly pronounced in most young children. About this time it became easy to work on his feelings and make him do whatever was wanted. When two years and three months old, he gave his last bit of gingerbread to his little sister, and then cried out with high self-approbation, “Oh, kind Doddy, kind Doddy.” Two months later he became extremely sensitive to ridicule, and was so suspicious that he often thought people who were laughing and talking together were laughing at him. A little later (two years and seven and a half months old) I met him coming out of the dining-room with his eyes unnaturally bright, and an odd, unnatural or affected manner, so that I went into the room to see who was there, and found that he had been taking pounded sugar, which he

had been told not to do. As he had never been in any way punished, his odd manner certainly was not due to fear, and I suppose it was pleasurable excitement struggling with conscience. A fortnight afterwards I met him coming out of the same room, and he was eyeing his pinafore, which he had carefully rolled up; and again his manner was so odd that I determined to see what was within his pinafore, notwithstanding that he said there was nothing, and repeatedly commanded me to “go away,” and I found it stained with pickle-juice; so that here was carefully planned deceit. As this child was educated solely by working on his good feelings, he soon became as truthful, open and tender as any one could desire.

UNCONSCIOUSNESS, SHYNESS.—No one can have attended to very young children without being struck at the unabashed manner in which they fixedly stare without blinking their eyes at a new face; an old person can look in this manner only at an animal or inanimate object. This, I believe, is the result of young children not thinking in the least about themselves, and therefore not being in the least shy, though they are sometimes afraid of strangers. I saw the first symptom of shyness in my child when nearly two years and three months old; this was shown towards myself, after an absence of ten days from home, chiefly by his eyes being kept slightly averted from mine; but he soon came and sat on my knee and kissed me, and all trace of shyness disappeared.

MEANS OF COMMUNICATION.—The noise of crying or rather of squalling, as no tears are shed for a long time, is of course uttered in an instinctive manner, but serves to show that there is suffering. After a time the sound differs according to the cause, such as hunger or pain. This was noticed when this infant was eleven weeks old, and I believe at an earlier age in another infant. Moreover, he appeared soon to learn to begin crying voluntarily, or to wrinkle his face in the manner proper to the occasion, so as to show that he wanted something. When 46 days old, he first made little noises without any meaning to please himself, and these soon became varied. An incipient laugh was observed on the 113th day, but much earlier in another infant. At this date I thought, as already remarked, that he began to try to imitate sounds, as he certainly did at a considerably later period. When five and a half months old, he uttered an articulate sound “da,” but without any meaning attached to it. When a little over a year old, he used gestures to explain his wishes; to give a simple instance, he picked up a bit



of paper, and, giving it to me, pointed to the fire, as he had often seen and liked to see paper burnt. At exactly the age of a year, he made the great step of inventing a word for food, namely, *mum*, but what led him to it I did not discover. And now, instead of beginning to cry when he was hungry, he used this word in a demonstrative manner or as a verb, implying "Give me food." This word, therefore, corresponds with *ham*, as used by M. Taine's infant at the later age of fourteen months. But he also used *mum* as a substantive of wide signification; thus he called sugar *shu-mum*, and a little later after he had learned the word 'black,' he called liquorice *black-shu-mum*—black-sugar-food.

I was particularly struck with the fact that when asking for food by the word *mum* he gave to it (I will copy the words written down at the time), "a most strongly marked interrogatory sound at the end." He also gave to "Ah," which he chiefly used at first when recognizing any person or his own image in a mirror, an exclamatory sound, such as we employ when surprised. I remark in my notes that the use of these intonations seemed to have arisen instinctively, and I regret that more observations were not made on this subject. I record, however, in my notes that at a rather later period, when between eighteen and twenty-one months old, he modulated his voice in refusing peremptorily to do anything by a defiant whine, so as to express, "That I won't;" and again his humph of assent expressed, "Yes, to be sure." M. Taine also insists strongly on the highly expressive tones of the sounds made by his infant before she had learned to speak. The interrogatory sound which my child gave to the word *mum* when asking for food is especially curious; for, if any one will use a single word or a short sentence in this manner, he will find that the musical pitch of his voice rises considerably at the close. I did not then see that this fact bears on the view which I have elsewhere maintained that before man used articulate language, he uttered notes in a true musical scale, as does the anthropoid ape *Hylobates*.

Finally, the wants of an infant are at first made intelligible by instinctive cries, which after a time are modified in part unconsciously, and in part, as I believe, voluntarily as a means of communication,—by the unconscious expression of the features—by gestures and in a marked manner by different intonations,—lastly by words of general nature invented by himself, then of a more precise nature imitated from those which he hears; and these are acquired at a wonderfully quick rate. An

infant understands to a certain extent, and as I believe, at a very early period, the meaning or feeling of those who tend him, by the expression of their features. There can hardly be a doubt about this with respect to smiling; and it seemed to me that the infant whose biography I have here given understood a compassionate expression at a little over five months old. When six months and eleven days old, he certainly showed sympathy with his nurse on her pretending to cry. When pleased after performing some new accomplishment, being then almost a year old, he evidently studied the expression of those around him. It was probably due to differences of expression and not merely of the form of the features that certain faces clearly pleased him much more than others, even at so early an age as a little over six months. Before he was a year old, he understood intonations and gestures, as well as several words and short sentences. He understood one word, namely, his nurse's name, exactly five months before he invented his first word, *mum*; and this is what might have been expected, as we know that the lower animals easily learn to understand spoken words.

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## **CORRESPONDENCE.**

### ***COMPARATIVE LONGEVITY AND GREATNESS.***

Whether or not great men are favored by an increase of years above those allotted to more ordinary mortals has long been a question of interest, and has acquired a special importance in connection with the study of the natural history of men of genius, and the discussions of the possible relation of greatness to degeneracy and to insanity. Questions of this type can only be decided on the basis of extensive and carefully collected data, which unfortunately it is difficult and at times impossible to collect or to find. It is therefore natural that such evidence as seems to exist and to carry with it some degree of logical force should be brought forward in proof of a claim which on general principles is both pleasing and plausible. Of this type is the problem of the relation between longevity and greatness, and of this type is the evidence now and then brought forward to substantiate the belief that great men are, as regards longevity, an unusually favored class.

The most recent presentation of the topic (by Mr. Thayer in the Forum, February, 1900) collects a list of some five hundred prominent men and women of the nineteenth century and finds that these persons lived on an average sixty-eight years and eight months; that is, nearly thirty years longer than the population as a whole. And on the basis of this conclusion the writer combats the notion that nineteenth-century men of genius or of eminence exhibit signs of degeneracy, because longevity and the ability to do sustained work for a large number of years is in itself a sign of unusual vitality and vigor. As these conclusions are apt to be extensively quoted, and as they seem to me founded upon a serious fallacy, I shall attempt to present as simply as possible the nature of the desired evidence which alone could prove that great men are longer lived than others, and to show that the evidence thus far presented is inadequate to support the conclusion which

has been drawn. Mr. Thayer is not the first one to present the average age at death of a number of eminent persons as evidence of unusual longevity. In an article which was reprinted in the Popular Science Monthly for May, 1884, the average age at death of 1,741 astronomers was given, and found to be sixty-four years and three months; and on the basis of this fact the author claimed that astronomers enjoyed unusual longevity. In a brief contribution published in Science, October 1, 1886 (and republished in Nature, November 4, 1886), I called attention to the fallacy inherent in such conclusions, and also presented some new contributions to the question of the longevity of great men. The materials of that article I shall utilize in the present discussion.

To reach the kernel of the matter at once, the reader must note that the fallacy consists in neglecting to consider that in dealing with astronomers or with great men, or with persons of eminence of the nineteenth century, one is dealing with a group which is already carefully selected, *and the selection of which inevitably involves the attainment of a certain age*. The result is that we are not dealing with average persons as regards longevity, but with persons who in the very nature of things have already reached a certain period of maturity. No one can become a poet, or a novelist, or a painter, or a philosopher, or a commander or a statesman unless he lives at least a sufficient number of years to acquire the development of an adult, and to have the opportunity of developing his abilities and distinguishing himself. If great men were great from their infancy, and if we had the means of ascertaining this fact, then, and only then, would the method used be correct.

It is ordinarily stated that the average duration of life is somewhere between thirty-three and forty years, and Mr. Thayer considers that in the present century it has moved forward towards the latter figure. What this means is that if we were to keep a record of the age at death of all Americans who are to be born within the first ten years of the coming century, we should find that their average age at death would be some thirty odd years. But this number can by no means be used as a standard with which to compare the average age at death of men of distinction, or indeed of any other class of men selected according to a standard which involves on their part the attainment of mature years. If we were investigating the longevity of twins, or of persons with supernumerary toes, or indeed of

persons possessing any quality which one could detect in new-born infants, and if we could determine the average life-period of this class of persons and find that it markedly exceeded the average of the entire community, we should be entitled to conclude that twins, or persons who have supernumerary toes, are blessed with a greater longevity than the average man. But so long as men who are to acquire distinction bear no traces upon them of this power until they exhibit their powers and actually gain distinction, it is obvious that we are concerned with their longevity only from that moment when they have entered, or have become promising candidates for that class of selected individuals whose longevity we are investigating. Proceeding on this basis, I tried to determine the age at which, on the average, men of genius had accomplished a work sufficient to entitle them to be so denominated. This investigation was instigated by Mr. C. S. Peirce, then in charge of courses in logic at the Johns Hopkins University. Under his leadership a small company, of whom I was one, proposed to study certain traits of great men, and for this purpose we tried to select the three hundred greatest men of all times. The work was never carried on to completion, so that the final selection of the names, and particularly their use in the present connection, must rest on my sole responsibility. I mention these facts mainly to indicate the general representative character of the list which I used. I take from my previously published article the following essential facts: Omitting all doubtful names, about two hundred and fifty names remain, presenting a list which most persons would agree to be fairly representative of the greatest men of all times. Of these again I selected *at random* those about whom it was easiest to fix the age at which they had done work which would entitle them to a place on this list, or work which almost inevitably led to such distinction. It is a date about midway between the first important work and the greatest work. The average of over sixty such ages is thirty-seven years; which means that, on the average, a man must be thirty-seven years old in order to be a candidate for a place on this list. The real question, then, is, How does the longevity of this select class of thirty-seven-year-old men compare with that of more ordinary individuals? The answer is given by the expectation of life at thirty-seven years, which is twenty-nine years, making the average age at death sixty-six years. And this is precisely the age at death of these sixty great men; showing that, as a class (for these sixty may be considered a fair sample) great men are not distinguished by longevity from other men.

It will thus be seen that my own conclusion is entirely opposed to that of Mr. Thayer. But this opposition rests not upon a difference of data, but upon a difference of logic. To my mind the enumeration of ages at death of any number of great men cannot prove unusual longevity unless we take into consideration and can determine the number of years which, on the average, a person must have lived in order to become a candidate for the class under consideration. The comparison with the average age (that is, the period of about thirty-five or more years) is not only false; it is essentially absurd; for it would become possible only if we had among poets, and painters, and musicians, and historians, and scientists, and generals a goodly number who succumbed to the diseases of early infancy, or to some of the ills that juvenile flesh is heir to.

It may be well to illustrate at this point just what conclusions may be drawn from the data which Mr. Thayer and other writers have presented. The first conclusion is that it takes a considerable length of time to become eminent—on the whole a very natural and comprehensible statement. And with regard to the astronomers previously mentioned it is even possible to go farther; for these astronomers have been divided into four degrees of eminence, and it is found that astronomers of the first rank are longer-lived than those of the second, and these in turn are longer-lived than those of the third class, and these in turn are longer-lived than those of the fourth class. Therefore, the author concludes, the greatest astronomers have been most favored with length of years, and adds, as practical advice, “Be an astronomer and live long.” Now, of course, the true conclusion is that it takes longer to accomplish work which will entitle one to pre-eminence amongst astronomers than to do work which will only achieve moderate distinction. And the practical conclusion would read, “Live long enough to become great as an astronomer and you will probably, with the ordinary expectation of life, have a good chance of completing your three score and ten.” In the same way Mr. Thayer’s list of nineteenth-century celebrities might fairly be said to suggest the conclusion that in the present century one must already have labored for a goodly number of years before one’s name would be selected by a student of the longevity of great men. So far, then, these facts have an interesting interpretation.

It may also be worth while to note that if all the men whose longevity is to be compared are of a comparable class (that is, comparable with regard

to the attainment of years which they assume), then the longevity of different groups of celebrities may be compared with one another. Thus it is possible to compare the longevity of musicians with that of scientists (of about equal eminence), and according to Mr. Thayer's lists the scientists lived ten years longer than the musicians. The same conclusion appears in my own study, in which the scientists appear amongst the longest-lived, and the musicians amongst the shortest-lived men of genius. This conclusion must not be pressed too far, but in a general way it certainly is a bit of evidence worthy of consideration as proving that distinguished scientists live longer than distinguished musicians. It would be wrong to draw rigid conclusions from comparisons of small groups, and therefore it is better to contrast the average age at death of the various men studied in as large and as general classes as possible; e. g., as men of thought, men of feeling and men of action. All of the studies with which I am acquainted point to the conclusion that men of thought live longer than those who achieve distinction through unusual qualities of their emotional natures.

We may now approach the question, whether or not it is possible to prove that the men of distinction of the nineteenth century are longer-lived or shorter-lived than their every-day contemporaries. It would be possible to do this had we statistics of the age at death of the various professions; and again, had we these deaths classed according to the distinction which the individuals attained. In addition to this it would be necessary to ascertain (with some rough approximation, as I have attempted to do with regard to the greatest men of all times) the age at which they had accomplished sufficient work to entitle them to be enrolled in their special class. To take concrete instances, let us suppose that we wish to investigate the longevity of American lawyers. Now to be a lawyer in name only requires the candidate to have lived twenty-one years, and the average number of years which the average person of twenty-one years of age will continue to live is about forty; so that the mere fact that a man is a lawyer would bring his average age at death up to sixty-one years. I find in Mulhall's Dictionary of Statistics the statement made that the lawyers of Frankfurt die at the average age of fifty-four years, while merchants live to be fifty-seven years old. I know nothing about the authority of these figures, and am using them for illustration only. Assuming all the data to be correct (and twenty-one seems not too high an age for this purpose), this would seem to suggest that the ordinary lawyer of Frankfurt is not favored with abundance of years. In

passing, it is interesting to note that these Frankfurt statistics of lawyers and merchants and other classes show a uniformly lower age at death than those of the more eminent representatives of their professions. This is just what we should expect; for to be included in the one group one must have lived only long enough to prepare and establish one's self as a lawyer or as a merchant; while for the other group one must in addition have had opportunity to cultivate one's ability to a riper fruitage, and in a keen, and often long competition gain public recognition. It thus follows that the average longevity of the most distinguished lawyers will be greater than that of ordinary lawyers, because it takes longer to enter the more select class. But this argument, like many others, should not be pressed too far; innate ability may accomplish in a brief period what for more moderate powers is the work of many years. Nonetheless, in the study of comparative longevity it is the average that is significant; and it is the fluctuation of the average that we aim to discover. Thus, in the investigation of the longevity of an unwholesome occupation, such as would be accepted by a life insurance company only at special rates, we should expect to find the age at death of such individuals less than that of other classes involving an equal period of apprenticeship; but, of course, not less than that of the 'population as a whole.' And, to continue with the main argument, if we wish to investigate the longevity of shoemakers we should again have to decide upon some age at which on the average a person has probably already acquired the dexterity requisite to be a shoemaker. Even if we fix this so low as ten years, at which time the expectation of life is forty-eight years, it would bring the average age at death of shoemakers to fifty-eight years. It has thus become extremely obvious that if we compared these ages at death with the average life-period it would be just as easy to prove that lawyers and shoemakers and merchants enjoy exceptional longevity, as to prove that great men do. The average longevity is low because of the very large infant mortality, which enters into the composition of this average. When once the first ten years of life are passed the further expectation of life increases quite slowly. Roughly speaking, for every ten years between ten and fifty years the added expectation of life is but three years for each decade. We therefore see that in the very nature of things no one class of adults can possibly live as much as thirty years longer than 'the population as a whole.' The differences with which we are dealing are differences of a finer order, of a small number of years, and being slight differences, must be



substantiated by a relatively large number of cases; the cases, moreover, must be collected in a wholly unobjectionable manner; that is, in a manner in which the principle of selection bears no influence upon the longevity. To my knowledge adequate statistics which exhibit the relative longevity of different classes do not exist, and they certainly do not exist with regard to great men. We may therefore conclude that the facts which have thus far been collected are not opposed to the conclusion that great men enjoy favorable longevity, but they certainly have not established or contributed to the establishment of this fact. While it is not impossible to collect material which may serve as corroborative evidence of the longevity of great men, it seems probable that we must be content with evidence of a far inferior character.

Although I regard Mr. Thayer's argument concerning longevity as entirely fallacious, I find myself in sympathy with his main contention. It seems to me that much of the evidence which has been brought forward to assimilate greatness with degeneracy is of questionable value and that the logical force of such evidence has been very much overrated. That genius and insanity are related is probably capable not of demonstration, but of a moderate degree of substantiation; but this evidence must be both judiciously collected and judiciously interpreted. It cannot be presented in a popular form without subjecting it to the danger of serious and harmful misrepresentation. In the same way the question of degeneracy and its bearing upon modern life has been frequently misstated, so that statements of protests such as Mr. Thayer offers are both opportune and likely to have a wholesome effect. But the present concern is only with the relation of longevity to greatness as an indication of the absence of degeneracy. That long life is inconsistent with a general degeneracy may be admitted; but that great men exhibit this quality to any unusual degree has certainly not been proven.

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***SCHOOL REFORM.***

School teachers and educational reformers undoubtedly take themselves and their ideas too seriously. Accordingly one rejoices to see an eminent man put his own affairs aside for a moment and discuss educational theories in a humorous vein. Even ridicule should be welcomed if it can relieve the sombre earnestness of the educational platform and press. Professor Münsterberg, in the *Atlantic Monthly* for May, has done pedagogy this service by subjecting the elective system and professional training for high-school teachers to considerable good-natured ridicule. His article is so readable that one is led to suppose that it was written to be read, not to be believed. Moreover, Professor Münsterberg's eminence as a psychologist should not be taken as a sign that he thinks he knows aught of education. He has himself warned us against the illusion that psychology can derive truth about teaching, or that the psychologist can inform the teacher or anything of value. It may be that the wholesome matters of fact, as well as the brilliant imaginative criticism of this article are only play. The very strenuousness of the teacher's nature, however, will probably lead him to try to extract some new gospel of reform from Professor Münsterberg's lightest pleasantry; consequently it seems wise to consider the article as a serious argument and provide a possible antidote for it.

Professor Münsterberg contends that it is unwise to give high-school teachers special professional education apart from knowledge of the subjects which they are to teach; that it is folly to replace a prescribed course of study by an elective system; that the salvation of our schools depends upon the scholarship of the teachers and the attitude of parents. As the reformers agree heartily with this last claim (unless it is made an exclusive aim), and as its meaning is so vague that almost anything can be urged as a corollary to it, it may be dismissed. The first two contentions are about concrete matters of educational practice which need to be thought over. If professional preparation is a waste of time, there is every reason why we should omit it; if a prescribed course of study is better for the boys and girls, we can conscientiously lessen the expense and labor of administration in many schools.

The argument on the first point is, briefly, that Professor Münsterberg's teachers were good teachers and that they had no notion of even the vocabulary of educational theories. But obviously that may not have been the secret of their success. A majority of the high-school teachers in New

England have had no professional training, yet no one has observed that they are superior to those of their class who have. The argument is really a bare assertion of an unverified guess. It is the hap-hazard opinion of an eminent psychologist who perchance is trying to furnish evidence of his previous theory that psychology does not give one knowledge about teaching. It is worth while to note here a certain interesting aspect of human nature. Training in one sphere of intellectual activity need not bring ability in other spheres. The habit and power of observation or reasoning acquired in connection with chemistry need not make a man a good observer or reasoner in politics or philology. So we should not be surprised that a man eminent for his scientific habits as a psychologist should, on a question in another field, offer imaginative hypotheses without an attempt to verify them, or to collect pertinent evidence or to eliminate factors outside those he discusses. We may be allowed to feel sorry. If a scientist wishes to really clear up the question of the value of professional training, why does he not find representatives of the classes, 'teachers with professional training' and 'teachers their equals in other respects, who have replaced the effort after professional training by equal effort after further scholarship,' and compare the work of the two classes? If other factors enter to disturb such an investigation, why not carefully look at the facts to ascertain their influence? Until he does so his dicta will stand as mere opinions. It would be a blessing if scientific men would use the weight of their reputations, not to bolster up their after-dinner opinions about things in general, but to teach the public scientific methods of studying them.

Apart from the danger of offering pedagogy an unproved opinion as a fact, it seems poor economy to leave a question in such shape that only the opinion of another eminent man on the opposite side is required to destroy the result you have attained. Precisely this has occurred in the case of Professor Münsterberg's contributions to educational discussion two years ago. Another eminent man, Professor Dewey, has recently squarely denied what Professor Münsterberg affirmed. It only remains for some equally eminent German professor to rise and declare that his teachers were bad and that they had no professional training, or that his teachers were good and had it, and Professor Münsterberg's effect is neutralized.

Professor Münsterberg's argument against the elective system is more complex. He regards the elective system as partly a concession to the

obvious need of fitting young people earlier for their occupations in life and partly an attempt to use the likes and dislikes of children as a guide to what is good for them. This is a very narrow view. The elective system has been in part the result of the progress of science and the consequent conviction that the scientific study of things and human affairs should be a part of one's education. The elective system furnished a compromise by which such studies found a place in the college and school curricula. If the student is left to choose among them, instead of having a new prescribed course made out on the basis of modern views of life's needs, it is partly because they are more easily introduced and retained as electives and partly because there is no agreement as to which studies will be the best to prescribe.

The idea that reformers desire to have a course containing studies good for children and studies not good for them and to trust the scholars' likes and dislikes to guide them to the former, is absurd. Whether they are right in assuming that what is best for one boy may not be best for another, that his teachers and parents can help him to pick out a course of study better for him than any inflexible course prescribed for all can be, is a question of importance, but one which Professor Münsterberg does not try to answer. Instead, he tells us about his gratitude to his parents and teachers for never letting him neglect his steady toil at prescribed Greek for the pursuits which he himself elected out of school, such as electrical engineering, botany, novel-writing, reading Arabic, writing books on the prehistoric anthropology of West Prussia, etc., etc. Now, this confession about his early life absolves us from paying any further attention to his experience as a lesson to our high-school youths. The youth Münsterberg and the average high-school student do not belong in the same class. For he was evidently an eminent boy as he is an eminent man. We must admit, however, that the rigorous discipline afforded by the prescribed Latin and Greek is evidenced in the present stern moral sense of the professor, who is willing to abandon his chosen and favorite pursuit, laboratory experimentation, and at the call of duty give himself to the hated but necessary tasks of writing philosophical disquisitions, political discussions and articles on school reform.



## SCIENTIFIC LITERATURE.

### *CHEMISTRY.*

The general interest which has been aroused the last few years in physical chemistry is reflected in the number of books which have appeared in this department. Some of these dwell more upon the older physical chemistry, devoting but relatively little space to the later developments, while others are chiefly concerned with the newer phases of the subject. Perhaps the most satisfactory book which has appeared along this line is Walker's 'Introduction to Physical Chemistry' (Macmillan). No attempt is made to exhaust the field but the subject is well covered. Especially commendable is the clearness of the book, which will render it useful to students. The non-mathematical treatment of the subject will also commend it to many who use it as an introduction to physical chemistry. A book of narrower scope is Dr. H. C. Jones's 'Theory of Electrolytic Dissociation and Some of Its Applications,' from the press of the same publishers. The author gives first a short review of the development of physical chemistry up to the days of van't Hoff, and then surveys the origin of the theory of electrolytic dissociation, its proofs and some of its applications. While making no pretense to cover the whole field of physical chemistry, the author furnishes a very readable account of the most important of the later generalizations. It is a book which should be read especially by those chemists and physicists who are working in other fields, that they may gain a fair view of the electrolytic dissociation theory written by one thoroughly competent for his task. Biologists, too, will find the latter part of the book, treating of the applications of the theory to animal and plant life, of especial interest. Dr. Jones, with S. H. King, has also translated Biltz on 'Practical Methods of Determining Molecular Weights.' This is a successful attempt to gather together the best of the different methods of real value, and it is very satisfactorily carried out, presenting a good guide book for students.

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In the production of text-books of general chemistry, there seems to be a little lull, very few books having appeared in recent months. The first part of what promises to be a somewhat original work on inorganic chemistry, by Dr. Sperber, has appeared. After the introduction on general chemical laws, the elements of the seventh group (chlorine, etc.), are first considered, and then their hydrogen compounds; the sixth group (oxygen, etc.) and its hydrogen compounds; fifth group (nitrogen, etc.), etc. The method used is purely inductive, each subject being introduced by experiments from which the underlying principles are developed.

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A third edition of Elliott and Ferguson's 'Qualitative Analysis' has appeared which is a considerable improvement upon the previous editions. The principal merit of this book, is in the opinion of many its greatest drawback. In clearness and minuteness of directions it is hardly equalled by any manual of qualitative analysis, and thus it is a particularly easy book for the instructor to use, especially with a large class. But this, on the other hand, cannot fail to encourage mere mechanical work on the part of the student and to discourage independence. With large classes, however, it is a difficult problem how best to cultivate individuality of work.

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A little manual of 'Analysis of White Paints,' by G. H. Ellis, will prove of value to chemists to whom now and then paint samples are brought for analysis. It is a collection of notes by a chemist who has had much experience along this line.

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In the field of applied chemistry quite a number of books have come out lately, the most useful of which is probably the seventh volume of 'The Mineral Industry.' The field of mineral resources and industries of the world is very thoroughly surveyed, and the volume is brought as closely down to

date as possible. In this respect it has a great advantage over the corresponding publication of the United States Government. Among the subjects which are treated very thoroughly in the present volume are calcium carbide, fire brick and paving brick, coal mining methods and their economic bearing, progress in the metallurgy of copper and of gold, notes on the progress of iron and steel metallurgy (by Henry M. Howe), sulfuric acid, progress in ore dressing (by Robert H. Richards). It is a book necessary to the teacher, of great value to the economist and of much interest to the general reader. The second edition of McMillan's 'Electro-metallurgy' is a considerable improvement on the former edition, and is brought well down to date. The greater part of the book is devoted to the electro-deposition of metals, and is thorough and satisfactory. It is, however, unfortunate that the treatment of electro-metallurgical ore-extraction should be very inadequate, this whole subject, together with electro-refining, being confined to a single chapter of thirty pages.

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Lange's 'Chemische-technische Untersuchungsmethoden' is passing through its fourth edition, of which the second volume is just out. This treats of metals and metallic salts, fertilizers, fodders, explosives, matches, gas manufacture, ammonia and coal tar and inorganic colors. The book aims at exhaustive treatment, and while some subjects are in parts weak, as is naturally the case where there are many different authors, it is as a whole the best work in its field.

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A book in a new line is H. and H. Ingle's 'Chemistry of Fire and Fire Prevention' (Spon and Chamberlain). The book takes its origin from lectures delivered to an audience of insurance men. After three chapters on the history and theory of combustion, various industries more or less connected with fire are taken up; coal gas, dust explosions, fuel, illuminants, explosives, oils, volatile solvents, paints and varnish making, textile manufactures, spontaneous combustion, are some of the subjects treated. The last chapter is a quite useful one on fire prevention and extinction. The book contains much useful information and should prove of



very considerable value outside of the rather limited audience to which it is addressed.

## **ZOOLOGY.**

The past few months have witnessed the publication of many important works on zoölogical subjects, and among these it may not be amiss to note first Kingsley's 'Text-Book of Vertebrate Zoölogy,' since it adopts a new method, that of showing the bearing of embryology upon the morphology of vertebrates, and in turn, of morphology upon their classification. Its object is stated to be to "supplement both lectures and laboratory work, and to place in concise form the more important facts and generalizations concerning the vertebrates," and the author has succeeded in crowding a large amount of information into the 439 pages of the work. The illustrations are numerous, and for the most part very good, comprising some figures that have appeared in other text-books and some that are the outcome of Dr. Kingsley's own work. It is to be noted that in place of many of the standard European forms that have done morphological duty for years, we have such American types as *Acanthias*, *Necturus*, *Amblystoma* and *Sceloporus*, a change for which we are duly grateful.

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Parker and Haswell's admirable 'Manual of Zoölogy' has been revised and adapted for the use of American schools and colleges. It aims to give an outline of the structure and morphology of certain typical members of the various classes of animals and also briefly discusses such zoölogical questions as evolution, descent and distribution. An 'Elementary Course of Practical Zoölogy,' by T. J. Parker and W. N. Parker, has been issued somewhat on the lines of Huxley and Martin's 'Biology,' aiming to give a rather detailed account of the structure of a few types instead of glancing at the animal kingdom as a whole.

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Books on birds, and especially those devoted to the popularizing of ornithology, continue to be numerous, and among them may be mentioned Keeler's 'Bird Notes Afield,' which introduces us in a pleasant way to the better-known birds of California, a subject of which Mr. Keeler is well qualified to treat. Less attractive from a literary standpoint, but more important from a practical point of view, is Lange's 'Our Native Birds: How to Protect Them and Attract Them to Our Homes,' which discusses the various causes for the decrease of birds, and suggests methods by which this may be prevented. Of a totally different character is Shelley's 'Birds of Africa,' now in process of publication, the first part of Vol. II having recently appeared. While many undescribed forms may be expected from Africa in the future, this work brings the subject down to date. 'The Birds of South Africa' are described in one compact volume by Arthur C. Stark, and the Australian Museum is now issuing a new edition of 'A Catalogue of Nests and Eggs of the Birds of Australia,' by Alfred J. North, the original having long been out of print. It is to be hoped that the first volume of the new hand-list of birds, 'Nomenclator Avium tum fossilium tum viventium,' by R. Bowdler Sharpe, which was published last fall, may soon be followed by others, as the completed list will be a boon to all working ornithologists. Finally, it may not be known to all our readers that last year Newton's 'Dictionary of Birds' was issued in one volume at a reduced price.

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The second and final part of 'Insects,' of the Cambridge Natural History, by David Sharp, gives us one of the most important, if not *the* most important work on entomology that has appeared for a long time, the two volumes forming a condensed encyclopædia of entomology that will be needed by all working entomologists. Another useful work on entomology is Carpenter's 'Insects, Their Structure and Life,' that portion devoted to the 'life' of insects being the best, particularly the chapter on 'Insects and Their Surroundings.' Of a strictly popular nature is Scudder's 'Every-day Butterflies,' which deals in a charming way with some sixty species of eastern North America.

## ***BOTANY.***

The beginning of the year has been marked by the appearance of the usual number of elementary and popular books dealing with some phase of botany. Among these Professor Barnes's 'Outline of Plant Life' (H. Holt & Co.) is a simplified edition of a high school text of a year earlier. Only the gross anatomy of the plant is considered and the ordinary routine of beginning with the simpler forms and advancing to ones of successively more complex structure is followed, and the principles of reproduction and physiology are presented. The student is given an insight into the adaptive processes of the plant by a study of the special forms which live in the water, dry soil, deserts, and other special conditions.

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'Lessons in Botany,' by Professor Atkinson (H. Holt & Co.) is a similar edition of a high school text designed to meet the needs of students in half-year courses. The student is led to an interest in the plant by a consideration of seedlings and buds, then launched in a course dealing with types of varying morphological constitution with attention to physiology and morphology. The taxonomy of some of the more important families of seed plants is discussed in a special section. The author pays tribute to the present leaning toward ecology by chapters on seed distribution, the struggle for the occupancy of land, zonal distribution, soil formation in

rocky regions and moors, plant communities, and adaptations of plants to climate.

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In 'Nature and Work of Plants' (Macmillan) Dr. MacDougal approaches the subject of botany by a study of the functions of the plant, of the things which it must do to live and adapt itself to its surroundings. Such an introduction to the subject from the physiological point of view is a radical innovation in the matter of elementary texts. A second departure from the practice of current texts is the omission of illustrations, in order that the attention of the student may not be distracted from the plant at work by a picture of something it has done. The technique is simple and the book seems well-fitted to awaken enthusiastic interest and lead the student further into the subject. Chapters are devoted to such subjects as: composition and purposes of plants, the manner in which the different kinds of work are divided among the members of the body, the way in which new plants arise, and the relations of plants to each other.

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Miss Alice Lounsberry's 'Guide to the Trees' (Stokes & Co.) is an example of a type of popular books in botany indispensable to the amateur, and of great value to the working botanist. Nearly two hundred species, including shrubs, have been described. "Among them are all those most prominent in northeastern America, and a few distinctive or rare species from the South and West. Several also that are not indigenous but which have become identified with the tree life of this country are presented." The author has grouped forms of similar habit together in such manner that sections are devoted to: Trees preferring to grow in moist soil, lowlands and meadows; trees preferring to grow near water, in swamps, and running streams; trees preferring to grow in rich soil, in forests and thickets, and trees preferring to grow in light, dry soil and upland places. The general notes of information appended to the technical descriptions add much to the reading value of the book, which is beautifully illustrated by sixty-four colored plates, after paintings by Mrs. Rowan, and a hundred sketches in black and white.

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The amount of interest centered in the preservation of the forests of the national domain, and the establishment of forestry in the courses of several educational institutions, makes Mr. Bruncken's 'North American Forests and Forestry' (Putnam & Sons) most timely. The author discusses the sociological aspects of forestry, and the distribution of forests in North America. It is of interest to note that the forest is treated as a living plant formation subject to many vicissitudes in the struggle for existence with neighboring societies of plants, particularly with the bog and prairie. The fate of the forest in front of the advancing pioneer is well delineated, and forest finance management and protection are most sensibly considered. Perhaps no other work offers the citizen such a rational presentation of all aspects of the numerous questions involved in forestry as the one under discussion.

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Sachs's 'Physiology of Plants' has long been a classic among botanists because of the immense amount of new results which were brought out in its pages, marking the dawn of a new epoch in the history of botanical investigation. A large share of its conclusions have become invalidated by the general advance of the subject, however, and the next most notable work, Pfeffer's 'Plant Physiology,' is one which is bound to exert even a more lasting influence in the guidance and furtherance of research. The first volume issued, dealing with the metabolism and sources of energy in plants, is cyclopedic in its completeness of review of investigations in this phase of the physiology without cumbering its bibliographical lists with titles of unimportant papers. In general, subjects yet under controversy are set forth with judicial fairness, and the author has made himself familiar with the work of Russian, English and American botanists in a manner not practiced by some of his contemporaries. The translation of this work by Dr. Ewart (Clarendon Press) has given opportunity for the correction of any slight omissions in the bibliography, and the completed book must be regarded as of the greatest value not only to the botanist but to the animal physiologist who would cover the domain of that illusive subject known as "general physiology."

## ***ANTHROPOLOGY.***

Probably the most striking sign of the increasing interest in the study of primitive man is the organization of well-equipped expeditions for the investigation of prehistoric remains and particular groups of existing savages. Of the latter class, the Cambridge expedition to Torres Straits, under the leadership of Professor A. C. Haddon, has returned to England, and various preliminary reports of the results of its work have already appeared. A new departure in the scheme of work of this expedition was the introduction of psychological observations under experimental conditions among the natives. The tests which were made were necessarily simple, but covered a fairly wide field. They included tests of visual acuteness, color vision and color blindness, acuteness and range of hearing, appreciation of tones and differences of rhythm, tactile acuteness and localization, estimation of weights, simple reaction-times to visual and auditory stimuli, estimation of intervals of time, memory and a number of tests of a more general character.

The detailed results have not yet appeared, but it is evident that there is much of interest to be expected. For example, of about two hundred and fifty individuals of different tribes tested for color blindness, not a case was found, except on one island, where three out of eight subjects suffered from ordinary red-green blindness. Reaction-times are said to be shorter than among the uneducated classes of European peoples, but no figures have as yet appeared. A fact, important if true, is the reported lack of suggestibility among the natives of the region. This is directly opposed to the general observations of most ethnographers and seems hardly probable. On all points the detailed reports are needed.

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On this side of the world public attention has been called particularly to the admirable plans of the Jesup North Pacific Expedition, which has been at work for the past three years on the northwest coast of America and the opposite coast of Asia. During the year just past the first published accounts of its results have begun to appear in a series of handsome monographs from the American Museum of Natural History in New York. Professor Franz Boas, the director of the expedition, furnishes the first two memoirs,

one on 'Facial Paintings of the Indians of British Columbia' and the other on the 'Mythology of the Bella Coola Indians.' The first named is of importance because of its bearing on the evolution of decorative designs. The Indians of the northwest coast differ from most other primitive groups in the matter of decoration by their failure to develop geometric designs and their tendency to retain realistic portrayals with certain characteristic modifications. In the adaptation of the decorations to the human face the problem has been difficult and a large number of examples are given showing the method of solution. The memoir on Bella Coola mythology is the first account of the complex conceptions of these Indians which can lay any claim to completeness. The Bella Coola conception of the universe is interesting. They believe in five worlds, one above the other, of which the middle one is the earth. Above this are spanned two heavens and beneath, two underworlds. In the upper heaven resides the supreme deity, who interferes little with the affairs of men; in the lower heaven dwell the Sun and all the other deities who are more intimately connected with mankind. The first underworld is inhabited by ghosts who may rise to the first heaven and be sent again to earth, and in the second underworld dwell the ghosts of those who have died a second death; from this there is no return. Other memoirs in the series are 'The Archæology of Lytton, B. C.,' by Harlan I. Smith, descriptive of the work of the expedition in that line; 'The Thompson Indians of British Columbia,' by James Teit, which is an exhaustive ethnographical account of that tribe, and 'The Basketry Designs of the Salish Indians,' by Livingston Farrand, in which is shown the development of geometric designs from realistic forms among the Indians of the Salish stock, a development which contrasts sharply with that of the neighboring stocks described by Boas.

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With the results of field-work pouring in and the constant modifications of theory brought about thereby, it becomes a task practically impossible to write a general 'Anthropology' which will not be out-of-date before it issues from the press. Nevertheless, from time to time the attempt is made and one of the latest ventures is 'Man, Past and Present,' by Mr. A. H. Keane (University Press, Cambridge). It is a general classification and description of the races of man which is open to the same objections as to

validity of classification as can be offered to any work on the subject at the present stage of knowledge. At the same time it contains much information in compact form, is not technical and will doubtless be useful. Of similar scope is 'The Races of Man,' by J. Deniker, which has just appeared in English form (Scribner's Contemp. Sci. Series). This work, also compact, is somewhat more technical than Keane's and also more accurate. It contains an appendix, with brief tables of measurements and indices adapted for quick reference.

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Of more special studies, unquestionably the most important work of the year is Messrs. Spencer and Gillen's 'The Native Tribes of Central Australia' (Macmillan). This extraordinarily minute account of the customs of the tribes with which it deals has already begun to attract the attention which it deserves. The problems upon which it throws light are numerous, but probably that of most general interest is Totemism, with its many social and religious bearings. The origin of this well-known savage custom has been a puzzle and heretofore not even a plausible suggestion has been made toward its solution. Messrs. Spencer and Gillen's account of the totemic ceremonies of the Arunta tribe, however, points irresistibly toward a definite economic origin, an attempt to preserve and increase the totemic animals and objects for the good of the tribe. The underlying relation between the clansman and his totem, as well as the social relations between the members of a clan, with the rules regarding marriage and the resultant modification of the family organization, are all analyzed with quite exceptional skill and in this and other fields the book is destined to become a classic.

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## THE PROGRESS OF SCIENCE.

Dr. Wolcott Gibbs presented his resignation from the presidency of the National Academy of Sciences at the recent Washington meeting, and the occasion permits the publication of his portrait and a few words in reference to his great contributions to science. Born in New York City in 1822, Dr. Gibbs graduated from Columbia College fifty-nine years ago. He studied abroad under Liebig, the founder of the first chemical laboratory, occupied a chair in the College of the City of New York, and was for twenty-four years Rumford professor at Harvard University. He became professor emeritus in 1887, and established a private laboratory at Newport, where he has continued his researches. Dr. Gibbs is one of the great chemists of the world. He is the only American honorary member of the German Chemical Society. Among other important ideas, his suggestion that the electrolytic deposition of copper be used as a means of quantitative analysis is one which has grown to a remarkable extent. There are now a number of volumes devoted solely to the amplification of this idea, which has been applied to numerous substances. Many other methods of quantitative analysis have been improved and simplified under his guidance, but perhaps his greatest work is his extended experimental study of complex salts, especially the cobaltamine compounds, and a great number of singularly complicated bodies, containing some of the rarer elements. Most of these substances are of no practical value, but they are of great theoretical interest, because they are only partially explained by the present theories of molecular structure. While the resignation of Dr. Gibbs from the presidency of the Academy is doubly regretted because it is owing to the fact that his health no longer permits the strain of the office, chemical science will profit all the more from his exclusive devotion to research.

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The meetings of the National Academy of Sciences held annually at Washington during the third week of April, pass without the general attention that they deserve. For the Academy meets not only to listen to special scientific papers, but also as the official scientific adviser of the Government. As knowledge increases in range and exactness, it is evident that expert advice becomes more and more necessary, both for the enactment of legislation and for carrying it into effect. It may, indeed, be fairly claimed that the advisory or expert department of the Government should rank coördinate with its legislative, executive and judicial branches. The National Academy has on occasion been called to investigate scientific questions—thus it has recently presented a report to the Department of the Interior on a policy for the forested lands of the United States—but it has been of less service in this direction than was intended by the act of incorporation or than sound policy dictates. This limitation to the usefulness of the Academy seems to depend in part on the small membership, and the fact that it consists of the most eminent rather than the most efficient men of science of the country. The Academy has less than one hundred members, only one fourth as many as the Royal Society. Professor Jastrow shows in the present number of this journal that men of science do not become eminent until rather late in life, and the members of the Academy are apt to be somewhat lacking in initiative. University professors are now selected chiefly from younger men of promise, who are expected not only to attain scientific eminence, but also to possess executive ability and to exert personal influence. The National Academy needs a membership of this character, and has fortunately to some extent obtained it within recent years. Thus the members elected at the present meeting are Prof. James E. Keeler, director of the Lick Observatory; Prof. Franz Boas, of Columbia University and the American Museum of Natural History; Prof. Henry F. Osborn, also of Columbia University and the American Museum, and Prof. Samuel L. Penfield, of Yale University.

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There is perhaps no objection to regarding the National Academy of Sciences as a *quasi* hereditary upper house, whose functions are largely conservative, while the active duties on behalf of science devolve on a more democratic body—The American Association for the Advancement of

Science. This association meets at Columbia University, New York City, during the last week of the present month, and with it some fifteen special societies devoted to different sciences. The association celebrated its fiftieth anniversary in Boston two years ago, when about half of its nearly two thousand members were present, and there is every reason to hope that the New York meeting will be as largely attended. The members will be welcomed by Governor Roosevelt and President Low, and after listening to addresses by the vice-presidents, will divide into nine sections, before which special papers will be presented. The address of the retiring president, Mr. G. K. Gilbert, of the United States Geological Survey, will be given at the American Museum of Natural History on Tuesday evening, while the president, Prof. R. S. Woodward, of Columbia University, will preside at the general sessions. The American Association has during its long history performed a useful service in bringing men of science together and in attracting the attention of the general public to scientific work, but in some respects it has been less influential than its sister associations in Great Britain, Germany and France. This has been in some measure due to the large area of the country and the heat of the summer, making it difficult for men of science to come together, but it probably represents chiefly a certain lack of organization of science in America. With the growth of university centers and of scientific work under the Government, the number of men of science has greatly increased, while with the establishment of special societies and journals their means of intercommunication have improved. There is every reason for the support of an association which can represent the whole body of scientific men and forward the scientific movements that are of such importance to the country. The membership of the association is of two classes, fellows and members. The former are selected from those who are actively engaged in advancing science, while all those who are interested in science are eligible for membership. Those who would like to have their names proposed for membership may address the local secretary of the New York meeting, Prof. J. McKeen Cattell, Columbia University, or the permanent secretary, Dr. L. O. Howard, Department of Agriculture, Washington, D. C.

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A very ambitious project is on the stocks for the foundation of an 'International Association for the Advancement of Science, Arts and Education.' It will be remembered that there was last year an interchange of visits between the British Association meeting at Dover and the French Association meeting at Boulogne. Arrangements were then made resulting in the appointment of general committees for Great Britain and France, and it was decided to hold an international assembly at Paris during the Exposition. Prof. Patrick Geddes, secretary of the British Group, has since visited the United States, and a general committee has been formed with Dr. W. T. Harris, United States Commissioner of Education, and Prof. R. S. Woodward, president-elect of the American Association, as vice-presidents. M. Bourgeois, late French Minister of Education, is the general president, and M. Gréard, rector of the University of Paris, is president of the French Group. The plans for the Assembly this summer are based directly on the Paris Exposition. It is proposed to establish headquarters on the grounds of the Exposition, in the buildings of the University of Paris and at other places, where those interested in the scientific aspects of the Exposition and in the scientific and educational congresses may meet and receive information and guidance. Special visits to the Exposition and other excursions, special lectures and entertainments, special summaries of the work of the congresses, etc., are promised. The Association is not, however, limited to the Paris Exposition, but proposes a permanent organization for the holding of assemblies and the organization of relations between men of science of different nations. Those interested in the Paris Assembly may secure further information from Mr. Ely, secretary of the American Group, 23 East Forty-fourth street, New York City.

\* \* \* \* \*

The Government of the United States does more to develop the resources of the country and advance science than any other nation. On these objects the sum of over \$8,000,000 is spent annually and over 5,000 officers are employed. Yet in one direction it has fallen far behind the great European nations. Our Department of Agriculture, our Geological Survey and many other agencies surpass in range and efficiency the similar institutions elsewhere, but the applications of physics and chemistry to the arts have not enjoyed equal advantages. The Physikalische-Technische

Reichsanstalt, the national physical laboratory of the German Empire, established under the direction of von Helmholtz, is conducted at an annual cost of \$80,000, and there is in addition a German bureau of weights and measures on which the sum of \$36,000 is annually expended. For similar purposes Great Britain spends annually \$62,000, Austria, \$46,000, and Russia, \$17,500, whereas, our office of Standard Weights and Measures receives the meager appropriation of \$10,400. We are very glad to learn that the Secretary of the Treasury has submitted an amendment to the pending sundry civil bill, creating in place of the present office a National Standardizing Bureau. According to the amendment the functions of the bureau shall consist in the custody of the standards; the comparison of the standards used in scientific investigations, engineering, manufacturing, commerce and educational institutions with the standards adopted or recognized by the Government; the construction when necessary of standards, their multiples and subdivisions; the testing and calibration of standard measuring apparatus; the solution of problems which arise in connection with standards; the determination of physical constants and the properties of materials when such data are of great importance to scientific or manufacturing interests and are not to be obtained of sufficient accuracy elsewhere. Provision is also made for the erection of a laboratory and its equipment, and for the employment of an adequate staff, with a director, whose salary shall be \$6,000 per annum.

\* \* \* \* \*

It is satisfactory that the Secretary of the Treasury should recommend a reasonable salary for the director of the proposed bureau. Men of science are, as a rule, but poorly paid, and the officers in the scientific departments of the Government receive in many cases salaries that are a small part of what they could earn as physicians or lawyers. There is, of course, danger that if salaries are large, the offices will be sought by 'practical' politicians, and it is probably the part of wisdom to offer the best facilities for research rather than large salaries. Still, if the scientific man has the salary of a clerk, he will be ranked in the same class by legislators and executive officers. The small salaries offered at Washington also lead to the continual loss of those whose services are of the greatest value to the Government. Thus, the recent call to the presidency of the Massachusetts Institute of Technology of

Dr. Henry S. Pritchett, Superintendent of the United States Coast and Geodetic Survey, is a serious blow to the bureau and to science at Washington. Dr. Pritchett's scientific attainments and executive ability will find ample scope at the Massachusetts Institute of Technology, where he worthily succeeds Presidents Rogers, Runkle, Walker and Crafts. But he was also greatly needed in the Coast and Geodetic Survey, where, after the excellent administration of Dr. T. C. Mendenhall, there had been an unfortunate interregnum of three years. During the past three years, however, the work of the Survey has been placed on an excellent basis by Dr. Pritchett, and there is every reason to believe that the ground gained will not be lost.

\* \* \* \* \*

The transition of Dr. Pritchett from the professorship of mathematics and astronomy in Washington University to the superintendency of the United States Coast and Geodetic Survey and now to the presidency of the Massachusetts Institute, calls attention to the fact that the only promotion possible to men of science or university professors is an executive position. The type of the German *Gelehrte*, still current in literature and on the stage, is not common in America. The modern methods of advancing science—the laboratory, the observatory, the museum, the expedition, with their complex equipment—demand administrative ability of a high order. Science has been able to supply presidents, not only to the great technical schools, but also to Harvard, Johns Hopkins, Stanford and other universities. Still, it is unfortunate that the man of science can not look forward to promotion in the direction of his own work. He becomes a college professor or the like at a comparatively early age with a moderate salary. He has now as a motive the increase of his reputation, rather likely to degenerate into vanity, and the nobler motive of contributing to the advance of science and of civilization. But these motives appeal differently to different men—in any case, they bake no bread and educate no children. The average salary of scientific men can not be greatly increased; there must be a certain relation between supply and demand, and the average earnings of other professional men are also small. But the lawyer may look forward to becoming a judge, the physician to a large city practice, the clergyman to a bishopric, etc. In Germany a university professor may look forward to being called to Berlin,

to becoming a *Hofrat*, a *Geheimrat* and a 'von.' It seems that we need in each American university one or two chairs with very large endowments, the occupation of which would be a special honor.

\* \* \* \* \*

The French Academy of Sciences and French Science have lost two of their most distinguished representatives in the deaths of Joseph Bertrand and of Alphonse Milne-Edwards. Bertrand was born in 1823, and was somewhat of a prodigy when a boy, having published a paper on the theory of electricity when but sixteen years old, and being the author of numerous mathematical papers before he was twenty-one. His original contributions to mathematics and mathematical physics are of great importance, and he was the author of standard works on algebra, on arithmetic and on the calculus. As permanent secretary of the Paris Academy of Sciences he was continually engaged in administrative work, preparing obituary notices, acting as judge in the annual awards of its prizes, etc. He also contributed a large number of biographies and other articles to non-technical journals. Milne-Edwards, born in 1835, was a son of the eminent zoölogist, Henri Milne-Edwards, and the grandson of Bryan Edwards, the historian and member of the British Parliament. Milne-Edwards published important researches in paleontology and in zoölogy, especially in relation to birds, and was at the time of his death professor of zoölogy at Paris and director of the Jardin des Plantes.

\* \* \* \* \*

In the deaths of the Duke of Argyll and Prof. St. George Mivart, Great Britain loses two men of a type more common there than in the United States. Argyll was a man of great wealth, whose interests in science were only secondary, but who did much directly and indirectly for its advancement. His work, 'The Reign of Law,' published some twenty-five years ago, has been widely read, and he is the author of many books and articles concerned with the natural sciences. Mivart, although trained as a barrister, became perhaps a professional man of science, but he never occupied a regular university position. He published numerous contributions to comparative anatomy and zoölogy, but is perhaps best known for books and articles on general scientific subjects. Just before his

death, it will be remembered, he was excommunicated from the Roman Catholic Church owing to articles which were supposed not to be in conformity with its tenets. Both Argyll and Mivart represented an attitude towards the doctrine of evolution which may be regarded as now practically extinct.

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Two lectures have been recently delivered by Prof. James Dewar at the Royal Institution on the subject of liquid and solid hydrogen. These lectures have been illustrated by experiments and have attracted the attention of the most distinguished chemists and physicists of England. It is easy to understand such interest in the subject when we consider that even Clerk Maxwell thought it improbable that hydrogen would ever be liquified, and yet Dewar was able to exhibit not only liquid, but solid, hydrogen to his audience. Briefly recapitulated, the steps in the condensation of what were formerly called the permanent gases are these: in 1878 Cailletet, in Paris, and Pictet, at Geneva, by suddenly expanding gases which had been compressed to a high degree and cooled to a low temperature, succeeded in obtaining these gases in the shape of a mist or of a transitory liquid jet. In 1884 Wroblewski and Olszewski at Crakow obtained oxygen and nitrogen as static liquids. By expanding hydrogen from a compression of 190 atmospheres in a vessel cooled by liquid air evaporating under diminished pressure, this gas was obtained as a mist or momentary froth, though it was affirmed by Olszewski that he observed the liquid hydrogen in colorless drops and as a liquid running down the sides of the tube. In May, 1898, Dewar obtained hydrogen as a static liquid by allowing compressed hydrogen, cooled in a bath of boiling air, to escape rapidly at a jet, the liquid hydrogen being collected in a doubly isolated vacuum vessel. This liquid hydrogen is a colorless liquid, with a specific gravity of 0.07 or less than one sixth the weight of liquid marsh gas, the lightest liquid hitherto known. This is better realized by saying that while one gram of water has the volume of one cubic centimeter, one gram of liquid hydrogen has a volume of over 14 c. c. The boiling point of hydrogen is  $-252^{\circ}$  C. or  $21^{\circ}$  above the absolute zero, and by boiling in a vacuum the temperature of  $15^{\circ}$  can be obtained. Very recently by slowly evaporating very perfectly isolated



liquid hydrogen, solid hydrogen was obtained by Dewar as a white mass of solidified form, of the lowest temperature ever obtained,  $-258^{\circ}$  C.

\* \* \* \* \*

Among the most suggestive results obtained through recent work in experimental embryology are those of Prof. Jacques Loeb, of the University of Chicago, on the chemical fertilization of the eggs of sea-urchins without participation of the male element. There has for some time been reason to suspect that cell-division, both in tissue cells and in the egg, is incited by chemical stimulus; and several observers before Loeb had shown that when unfertilized sea-urchin eggs are treated by the addition to the sea-water of various substances, such as strychnine or chlorides of sodium or magnesium, they may undergo some of the preliminary changes of development and may even segment. Loeb was able to induce complete and normal development by first bringing the eggs for about two hours into a mixture of sea-water and a weak solution of magnesium chloride, and then transferring them to normal sea-water. Eggs thus treated segmented and underwent a development which, though somewhat slower than usual, was otherwise normal and produced perfect larvæ. This effect can not properly be called fertilization in the ordinary sense of the word, but is rather to be regarded as artificially induced parthenogenesis. It points unmistakably, however, to the possibility, or rather probability, that in normal fertilization the spermatozoon incites the egg to development by bringing to it certain definite chemical substances; and Loeb gives reasons for the view that these substances are probably in the form of ions, concluding that these and not the nucleins are essential to the process of fertilization. A highly suggestive new field for work is opened by these experiments.

\* \* \* \* \*

Astronomers can not bring the phenomena they study into the laboratory or test the behavior of the heavenly bodies under artificial conditions. They have to be satisfied with such opportunities as nature gives, even though she bestows them as meagerly as she does solar eclipses. Consequently, the total eclipse of the forenoon of May 28th has been the object of much preparation. Most of the important astronomical observatories in this country will have parties stationed along the path of

the eclipse from New Orleans to Norfolk, Va. Many European parties will observe the eclipse in Portugal, Spain and North Africa. It has been pointed out by Prof. R. W. Wood and Mr. A. L. Rotch<sup>J</sup> that there are several important physical observations to be made apart from the astronomical observations on the sun's corona described by Professor Bigelow, in the May number of this magazine. Just before and after totality alternate bright and dark bands are observed sweeping across the country. It is hoped that by the coöperation of a number of observers more complete and exact data concerning this phenomena may be gathered and its explanation found. The changes in the wind noted during eclipses will also be observed to ascertain whether the sudden cooling of the atmosphere by the passage of the moon's shadow is a sufficient explanation of the so-called 'eclipse wind.' Those who know nothing about theories of the corona or of the 'eclipse wind' will be interested in the more obvious phenomena and in some cases, in the opportunity to take such a photograph as can not be duplicated in this country until 1918. The most favored ones are those who live in the fifty-mile belt of the total eclipse, but the sun will be seen nine tenths covered in the eastern and southern States, and will be six tenths covered to those in the least favorable locality of the United States, the extreme northwest. The proper methods of observing and photographing the corona were described in Professor Bigelow's article on the eclipse in the May number of the POPULAR SCIENCE MONTHLY.

<sup>J</sup> In 'Science,' Apr. 27th and May 11th.

## Transcribers' Notes

Punctuation, hyphenation, and spelling were made consistent when a predominant preference was found in this book; otherwise they were not changed.

Simple typographical errors were corrected; occasional unbalanced quotation marks retained.

Ambiguous hyphens at the ends of lines were retained.

Page [162](#): “ominum gatherum” likely is a misprint for “omnium gatherum”.

Page [167](#): “resistent” was printed that way twice in this article. In a different article, it was twice printed “resistant”.

Page [207](#): Unmatched closing quotation mark deleted after “from other men.”

Page [211](#): “corrolary” was printed that way.

Page [214](#): “calcium carbid” was printed that way.

Page [220](#): The first occurrence of “American Museum of Natural History” was printed without the “of”; the second with it.

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